

FAST ALMOST-UNIFORM SAMPLING OF RANDOM k -SAT SOLUTIONS

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ABSTRACT. We study approximately uniform sampling of satisfying assignments from random k -SAT formulas. For every sufficiently large k and density $0 < \alpha \leq 2^k/k^{16}$, we prove that, with high probability over the formula, there is a sampler whose output distribution is within total variation distance ε of the uniform distribution on satisfying assignments and whose expected running time is at most $(nk(\alpha + 1)/\varepsilon)^C$, for a universal constant C . Our algorithm improves the counting and sampling algorithms obtained by Chen, Lonkar, Wang, Yang, and Yin (STOC 2025) at the density $2^k/\text{poly}(k)$ with running time $(n/\varepsilon)^{\text{poly}(k, \alpha)}$. Our result achieves this density region for sampling with a polynomial degree independent of both the width and the density.

Our algorithm separates a high-degree core from the remaining variables, and combines a recursive sampler for the residual formulas with approximate block heat-bath updates on the core. We adapt the recursive insertion-chain framework of Jain, Mizgerd, and Pham (2026) from 2-trees to ordinary connected violation sets. Expansion and random literal signs yield uniform moment bounds for the resulting correlated lists across all residual formulas, allowing the signed-flow analysis to give a universal polynomial running-time degree. A polymer expansion and an exploration bound establish a polynomial spectral gap for the core dynamics.

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1. INTRODUCTION

Random constraint satisfaction problems provide a fundamental setting in which to study the relationship between the existence of solutions, their geometry, and the computational cost of finding them. A central example is random k -SAT: a formula Φ on n Boolean variables consists of $m = \lfloor \alpha n \rfloor$ independently generated clauses, each containing k random literals. The density α controls the number of constraints and the interactions among variables. As the density increases, random formulas undergo a transition from being satisfiable with high probability to being unsatisfiable with high probability.

Numerical experiments and statistical-physics analyses of this transition motivated the *satisfiability conjecture* [MPZ02, MMZ05]. It asserts that, for each fixed $k \geq 3$, there is a critical density $\alpha_{\text{sat}}(k)$ such that

$$\lim_{n \rightarrow \infty} \mathbb{P}(\Phi(k, n, \lfloor \alpha n \rfloor) \text{ is satisfiable}) = \begin{cases} 1, & \alpha < \alpha_{\text{sat}}(k), \\ 0, & \alpha > \alpha_{\text{sat}}(k). \end{cases}$$

Establishing the conjecture and determining the threshold became central challenges in the study of random constraint systems. Friedgut [Fri99] established a sharp transition, although this alone does not identify a limiting critical density. Using the second-moment method, Achlioptas and Peres [AP03] narrowed the satisfiability bounds to an additive gap of order k around $2^k \ln 2$. Coja-Oghlan and Panagiotou [COP16] subsequently obtained the predicted constant-order correction in the large- k asymptotics. Ding, Sly, and Sun [DSS22] then proved the existence and identified the value of the limiting threshold for each sufficiently large fixed k , confirming the predicted asymptotic value

$$\alpha_{\text{sat}}(k) = 2^k \ln 2 - \frac{1 + \ln 2}{2} + o_k(1).$$

Knowing where solutions exist leaves a further algorithmic question: how can they be found efficiently? The threshold proof does not supply a polynomial-time search algorithm throughout the satisfiable regime. Coja-Oghlan’s search algorithm [Coj10] finds a satisfying assignment at densities up to $(1 - o_k(1))2^k \ln k/k$, a $\Theta(\log k/k)$ factor below the satisfiability threshold but matching the so-called *clustering threshold*, where the emergence of well-separated clusters and frozen variables creates long-range constraints that obstruct local decisions [ACO08]. Rigorous failure results for belief propagation and basic survey propagation guided decimation [Coj17, Het16], together with lower bounds for low-degree polynomial algorithms [BH22], show that broad families of algorithms encounter barriers at or before a constant multiple of this threshold, motivating the conjecture that the polynomial-time search transition coincides asymptotically with the clustering threshold; this remains open for general polynomial-time algorithms. This gap also highlights the distinction between existence and algorithmic access, and motivates asking how much of the solution space can be accessed efficiently at high densities.

The statistical-physics picture also describes changes within the satisfiable phase: a dominant connected region gives way to many separated clusters, and near the satisfiability threshold a few clusters carry most of the solution weight [MMZ05, DSS22, CLW⁺25]. Figure 1 places these geometric transitions alongside the density scales.

Beyond finding a single solution, we study *approximately uniform sampling*: given Φ , generate a satisfying assignment whose distribution is within a prescribed total variation distance ε of the uniform law μ_Φ . Sampling asks for more than a certificate of satisfiability. A search algorithm may consistently select a small, atypical part of the solution space; a sampler must represent the relative weights of all its parts. Such access allows us to estimate marginal probabilities and other observables of a typical solution, connecting the algorithmic study of random formulas with their Gibbs distributions and partition functions. Sampling is also closely related to approximate counting, provided that the required conditional instances remain tractable. Thus the sampling problem tests

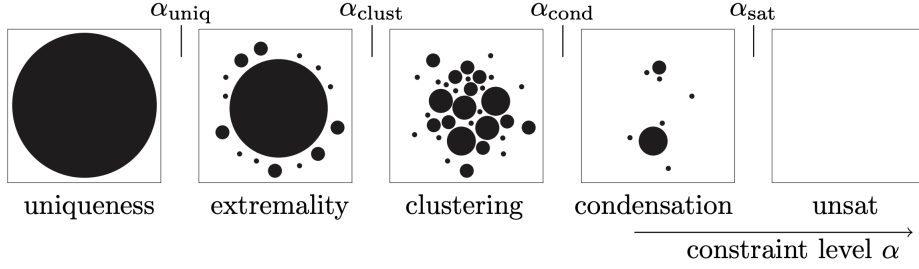


FIGURE 1. The conjectural phase diagram of random k -SAT, adapted from [KMR⁺07, DSS22].

whether the rich probabilistic information carried by a random formula can be made computationally accessible.

Progress in sampling has two dimensions: the range of densities covered and the dependence of the running time on the parameters. An algorithm taking time $(n/\varepsilon)^{\text{poly}(k,\alpha)}$ is polynomial-time for every fixed k, α , but its polynomial degree can increase with both parameters. This distinction is particularly important when the admissible density itself is exponential in k . Our objective is to combine the density scale $2^k/\text{poly}(k)$ with a running time whose polynomial degree is independent of the width and density.

Early work of Montanari and Shah [MS07] used orrelation decay to approximate the normalized logarithm of a partition function for a permissive, positive-temperature version of random k -SAT, at densities up to $(1 + o_k(1))2 \ln k/k$. This is the Gibbs uniqueness threshold and is exponentially below the satisfiability threshold. Galanis, Goldberg, Guo, and Yang [GGGY21] first extended relative approximation of the number of satisfying assignments to exponentially large densities. Their adaptation of Moitra’s linear-programming method works, for sufficiently large k , when $\alpha < 2^{k/300}/k^3$. A central innovation was to isolate high-degree variables and the variables heavily influenced by them, thereby recovering control of the remaining marginals. The resulting running time has a parameter-dependent polynomial exponent. He, Wu, and Yang [HWY23] substantially improved both the density range and the running time. Their recursive sampling algorithm reaches $2^{k/3}/\text{poly}(k)$ and runs in almost-linear time in n for fixed accuracy and parameters. It replaces the enumeration implicit in the linear-programming approach with recursive sampling, supported by a refined analysis of the random formula’s structure. Chen, Lonkar, Wang, Yang, and Yin [CLW⁺25] subsequently reached $2^k/\text{poly}(k)$ for both approximate counting and sampling, with running time $(n/\varepsilon)^{\text{poly}(k,\alpha)}$. Their analysis of recursive coupling uses denser witness structures in place of the usual 2-tree witnesses, exploiting expansion of random constraint systems. This established that the exponential density condition for sampling can match that of satisfiability and search, while leaving a polynomial factor in k between the respective density bounds.

These results rely crucially on the random structure of the formula. Approximate counting for general k -CNF formulas encounters hardness at variable degrees of order $2^{k/2}$, under the usual complexity assumptions [BGG⁺19]. The random model at density $2^k/\text{poly}(k)$ has average occurrence degree $k\alpha$, well beyond that exponential bound. The result of [CLW⁺25] therefore demonstrates the advantage of random structure for counting and sampling. The remaining question is whether this advantage can be obtained without a parameter-dependent exponent in the running time:

Can approximately uniform solutions of random k -SAT be sampled at density $2^k/\text{poly}(k)$ in time polynomial in $n, k, \alpha, \varepsilon^{-1}$, with a universal polynomial degree?

Recent counting and sampling versions of the Lovász local lemma suggest a possible route to a universal polynomial degree. Liu, Wang, Yin, Zhang, and Zhou [LWY⁺26] give a counting algorithm

under a symmetric condition of the form $p(D+1)^2 \leq c$, where p bounds a bad-event probability and D is the maximum degree of the dependency graph. Jain, Mizgerd, and Pham [JMP26] obtain both counting and sampling with a universal polynomial degree under an asymmetric condition on the square of that graph. These theorems do not directly settle the question above. Even the typical clause degree in random k -SAT is αk^2 ; substituting $p \approx 2^{-k}$ into the symmetric criterion leads only to an exponential density at $2^{k/2}$. Moreover, the maximum degree grows with n . Reaching the desired regime therefore requires both a different sufficient condition for the sampling mechanism and a separate treatment of high-degree variables.

1.1. Our result. We answer the preceding question affirmatively. We use the occurrence model: each clause chooses its k variable occurrences independently and uniformly from $[n]$, and each occurrence receives an independent fair sign. Repeated variables, tautological clauses, and different clause labels with identical contents are allowed.

Theorem 1.1. *There is a universal constant $C > 0$ and a randomized sampling algorithm ALG with the following properties. Fix an integer $k \geq 2^{20}$ and a real density $\alpha > 0$ with*

$$\alpha \leq \frac{2^k}{k^{16}}.$$

For sufficiently large n , with high probability over the random k -SAT formula $\Phi \sim \Phi(n, m, k)$ with $m = \lfloor \alpha n \rfloor$, for every $0 < \varepsilon < 1$, ALG returns a satisfying assignment X in time T almost surely with

$$\|\text{Law}(X) - \mu_\Phi\|_{\text{TV}} \leq \varepsilon, \quad \mathbb{E}[T] \leq \left(\frac{nk(\alpha + 1)}{\varepsilon} \right)^C.$$

Remark 1.2. The expected-time guarantee applies to good formulas and averages over the algorithm's randomness. It is easy to give an explicit bounded-time version of Theorem 1.1, following Markov's inequality. For example, by running the procedure of Theorem 1.1 with accuracy $\varepsilon/2$ and returning an arbitrary assignment if its running time exceeds

$$T_{\max} = \left\lceil \frac{2}{\varepsilon} \left(\frac{2nk(\alpha + 1)}{\varepsilon} \right)^C \right\rceil,$$

this gives a deterministic polynomial time cap on every input, while the output law is still within total variation distance ε of μ_Φ .

Remark 1.3. The cutoff 2^{20} , the power 16, and the running-time degree are conservative. We have not optimized these constants as much as possible.

Theorem 1.1 combines the density range $2^k/\text{poly}(k)$ with a universal polynomial running-time degree. Previous work achieves this density scale with a parameter-dependent polynomial degree [CLW⁺25], or almost-linear dependence on n over a smaller density range [HWY23]. Our result reaches the high-density scale while making the running time polynomial jointly in width, density, instance size, and inverse accuracy. This strengthens a separate polynomial-time guarantee for each fixed pair (k, α) to a single bound whose degree is independent of both parameters.

1.2. Technique overview. Our algorithm has two levels: a recursive sampler for the good variables conditional on an assignment to a high-degree core, and a block heat-bath chain targeting the marginal distribution on the core. The main tasks are to make the conditional sampler efficient throughout the high-density regime and to prove that the core chain mixes in polynomial time.

The core and residual formulas. Following the separation approach of [GGGY21, HWY23, CLW⁺25], we first collect the high-degree variables and those strongly affected by them into a core. With high probability, each core component has at most $\log_2 n$ variables, so its feasible assignments can be enumerated in polynomial time. Deleting core literals from non-core clauses gives a fixed good formula Ψ . Under a feasible core assignment, clauses satisfied by core literals disappear; the others retain their good literals. Thus every residual formula is a clause-deletion subformula of Ψ .

The recursive insertion chain for good variables. To give an efficient sampler for good variables, we adapt the recursive insertion-chain of [JMP26], inserting clauses one at a time and using auxiliary states to transfer probability from the old solution measure to the new one. We replace the 2-trees used in their construction by ordinary connected violation sets. For dependency degree D , the counting bound for rooted sets of size t improves from $O(D^2)^{t-1}$ to $O(D)^{t-1}$. The difficulty is that shared variables make violations dependent.

Whereas [JMP26] uses independent local samples for individual constraints, we generate each list from a single global product assignment. Expansion controls simultaneous violations of small sets, while for larger sets, averaging over the independent good-literal signs bounds their total contribution. These estimates control joint list moments and hold on one high-probability event for every residual formula, including adaptively chosen ones. Combined with the improved counting bound, they allow us to reach density $2^k/\text{poly}(k)$.

We also adapt the signed-flow and state-weight cost analysis of [JMP26] to these correlated lists. In particular, a list conditioned to contain a label (S, u) , specifying a connected set and its local assignment, can be expensive to generate. The joint moment bounds control the signed-flow energy, while weighting conditional list costs by the corresponding state weights bounds the simulation cost. Together, these estimates give an expected running time with a universal polynomial degree.

Rapid mixing of the core dynamics through entire regions. Small core components make individual updates affordable, but do not alone guarantee rapid mixing, since a component can still interact with many others and have strongly biased conditional weights. We update one core component at a time, weighting each feasible assignment by its number of satisfying extensions to the good variables. The good-variable sampler estimates these extension weights as products of conditional satisfaction probabilities, each at least $1/2$.

To prove rapid mixing, a polymer expansion represents the effect of the good variables as interactions among core components. A coupling of the conditional core measures bounds disagreements by connected regions in an auxiliary random hypergraph. We dominate this graph by one whose interactions are activated independently of the coupled samples, which lets us control both the spread of disagreements and the cost of changing assignments with very different extension weights.

The key is to bound each explored disagreement region as a whole. A structural estimate gives at most $O(k^2(\alpha + 1)|U|) + 4 \ln n$ clauses incident to any connected variable set U . Applying the incidence bound to a connected set covering the entire explored region, together with the proportion of high-degree variables in the core and the decay of interaction weights, implies an exponential-moment bound on the region's size and interaction cost. This yields a spectral gap of at least n^{-2} for the core heat-bath chain. The polymer expansion and random hypergraph are proof tools; actual updates estimate extension weights using the good-variable sampler. After the approximate core chain has mixed, we sample the good variables conditional on the resulting core assignment.

1.3. Organization. The remainder of the paper is organized as follows. Section 2 fixes the notation and reviews the probabilistic and analytic tools used throughout, including polymer models, the Lovász local lemma, and functional inequalities for Markov chains. Section 3 establishes the expansion and incidence properties of random formulas, constructs the high-degree core, and proves that feasible core assignments extend to satisfying assignments of the residual formula. Section 4 develops

the recursive insertion-chain and its signed-flow analysis, while Section 5 verifies its hypotheses for the good-variable formula and obtains an efficient sampler. Section 6 proves a polynomial spectral-gap bound for the heat-bath dynamics on the core using a polymer representation. Finally, Section 7 combines the good-variable sampler with approximate core updates to prove the main theorem. The appendix contains supplementary proofs for the core mixing analysis.

AI disclosure. ChatGPT suggested the main approach to proving Theorem 1.1, combining the rapid convergence of the heat-bath dynamics on the core with a modification of the recursive insertion chain in [JMP26] for the good formula. GPT was also used throughout the preparation of this manuscript for language editing and formatting. The authors have verified and refined all proofs and take full responsibility for the paper.

2. PRELIMINARIES

This section fixes notations and provides some necessary preliminaries.

2.1. Notations. For a natural number n , $[n]$ is the set $\{1, \dots, n\}$. The logarithm function is denoted by $\log$ and we use $\ln$ to denote the logarithm with natural base e . For an event $\mathcal{E}$, the indicator of $\mathcal{E}$ is denoted by $\mathbb{1}[\mathcal{E}]$.

For an assignment σ on U , we use $\sigma(v)$ to denote the value of σ on v . For a subset S of U , denote by $\sigma|_S$ the partial assignment $\sigma|_S$ of σ on S . To assign values, we use $S \leftarrow \sigma$ to describe the term that S is assigned the partial assignment σ . For the event A which is measurable by assignments, we use $\mathbf{1}_A$ to denote the indicator function of A , *i.e.*,

$$\mathbf{1}_A(\sigma) = \mathbb{1}[A \text{ holds under the assignment } \sigma].$$

Fix a probability distribution π on a state space Ω . For every test function $f : \Omega \rightarrow \mathbb{R}$, we define the expectation and the variance of f with respect to π as

$$\mathbb{E}_\pi[f] \triangleq \sum_{x \in \Omega} \pi(x) f(x), \quad \text{Var}_\pi(f) \triangleq \mathbb{E}_\pi[f^2] - \mathbb{E}_\pi[f]^2.$$

The total variation distance of two probability distributions μ and ν on a common state space Ω is defined as

$$\|\mu - \nu\|_{\text{TV}} \triangleq \frac{1}{2} \sum_{x \in \Omega} |\mu(x) - \nu(x)|. \quad (1)$$

A *coupling* of two probability distributions μ and ν is a joint distribution of a pair of random objects (X, Y) such that $X \sim \mu$ and $Y \sim \nu$. Equivalently, the first and second marginals of the joint distribution are μ and ν , respectively. The random objects X and Y need not be independent; their dependence may be chosen to facilitate a comparison while preserving both marginal laws.

We also extend (1) to zero-mean functions. For every function $f : \Omega \rightarrow \mathbb{R}$ such that $\sum_{x \in \Omega} f(x) = 0$, define $\|f\|_{\text{TV}} \triangleq \frac{1}{2} \sum_{x \in \Omega} |f(x)|$.

2.2. Random SAT formula. A random k -SAT instance $\Phi \sim \Phi(n, m, k)$ is a k -SAT formula on n variables $\mathcal{V} = \{v_1, \dots, v_n\}$ with m labeled clauses $\mathcal{C}$. Each clause C_a of label a selects k literals $\ell_{a,1}, \dots, \ell_{a,k}$ independently and uniformly at random from $\{v_1, \dots, v_n, \neg v_1, \dots, \neg v_n\}$. For simplicity, we abuse the use of the clause C_a and its label a . Let $\text{var}(\ell)$ be the variable name of a literal ℓ .

Let ν denote the uniform distribution over all assignments, regardless of whether they satisfy Φ . For a family $\mathcal{F}$ of constraints, we denote by $\text{Good}(\mathcal{F})$ the event that all constraints in $\mathcal{F}$ are satisfied. We use $\mu_{\mathcal{F}}(\cdot)$ to denote the distribution $\nu(\cdot \mid \text{Good}(\mathcal{F}))$.

For an event $\mathcal{E}$, $\text{vbl}(\mathcal{E})$ is the collection of all variables it depends on. For a variable $v \in \mathcal{V}$, the degree $\deg(v)$ is the number of clause labels containing v and for a variable subset $U \subseteq \mathcal{V}$, the number of its incident clause labels is denoted by $I(U)$, i.e.,

$$I(U) \triangleq |\{a \in \mathcal{C} : \text{vbl}(a) \cap U \neq \emptyset\}|. \quad (2)$$

The dependency and primal graphs induced by Φ are introduced to analyze the structure of a SAT formula. The dependency graph $\mathcal{H} = \mathcal{H}_\Phi$ has a vertex set $V(\mathcal{H}) = \mathcal{C}$. We add an edge in $\mathcal{H}$ for two clause labels if their representing clauses share at least a common variable. Meanwhile, the primal graph $\mathcal{G} = \mathcal{G}_\Phi$ of Φ has a vertex set $V(\mathcal{G}) = \mathcal{V}$ and joins two distinct variables if they occur in a common clause. For a variable subset U , we say U is connected if and only if $\mathcal{G}[U]$ is connected.

2.3. Abstract polymer models and the Kotecký–Preiss criterion. An abstract polymer model consists of a finite set $\mathcal{P}$ of polymers, a symmetric incompatibility relation $\not\sim$ on $\mathcal{P}$, and an activity $w_\gamma \in \mathbb{C}$ for each $\gamma \in \mathcal{P}$. The incompatibility relation is part of the model: $\gamma \not\sim \gamma'$ means that the polymers γ and γ' are incompatible and cannot occur together in a compatible family. We regard every polymer as incompatible with itself, so $\gamma \not\sim \gamma$ for every $\gamma \in \mathcal{P}$. A family $\mathcal{F} \subseteq \mathcal{P}$ is compatible if no two distinct members are incompatible. Its partition function is

$$\Xi(w) = \sum_{\mathcal{F} \text{ pairwise compatible}} \prod_{\gamma \in \mathcal{F}} w_\gamma, \quad (3)$$

where the empty family contributes one.

For an ordered tuple $\Gamma = (\gamma_1, \dots, \gamma_\ell) \in \mathcal{P}^\ell$, repetitions allowed, let $I(\Gamma)$ be its incompatibility graph on $[\ell]$: distinct i, j are adjacent exactly when $\gamma_i \not\sim \gamma_j$. Define its Ursell coefficient by

$$\varphi(\Gamma) = \sum_{\substack{F \subseteq E(I(\Gamma)) \\ ([\ell], F) \text{ connected}}} (-1)^{|F|}. \quad (4)$$

For $\ell = 1$, this coefficient is one. If $I(\Gamma)$ is disconnected, then $\varphi(\Gamma) = 0$. Define the formal cluster series

$$\mathcal{L}(w) \triangleq \sum_{\ell \geq 1} \frac{1}{\ell!} \sum_{(\gamma_1, \dots, \gamma_\ell) \in \mathcal{P}^\ell} \varphi(\gamma_1, \dots, \gamma_\ell) \prod_{i=1}^{\ell} w_{\gamma_i}. \quad (5)$$

The series in (5) is infinite because tuples may contain repeated polymers.

Theorem 2.1 (Kotecký–Preiss criterion [KP86]). *Let $u_\gamma \geq 0$ satisfy $|w_\gamma| \leq u_\gamma$ for every $\gamma \in \mathcal{P}$, and let $A : \mathcal{P} \rightarrow [0, \infty)$ be a penalty function. Suppose that*

$$\forall \gamma \in \mathcal{P}, \quad \sum_{\gamma' \not\sim \gamma} u_{\gamma'} e^{A(\gamma')} \leq A(\gamma). \quad (6)$$

Then the series $\mathcal{L}(w)$ in (5) converges absolutely and

$$\mathcal{L}(w) = \log \Xi(w).$$

Here $\log \Xi(w)$ denotes the analytic branch satisfying $\log \Xi(0) = 0$. Moreover, for every fixed polymer γ_0 ,

$$\sum_{\ell \geq 1} \frac{1}{(\ell-1)!} \sum_{\gamma_2, \dots, \gamma_\ell \in \mathcal{P}} |\varphi(\gamma_0, \gamma_2, \dots, \gamma_\ell)| \prod_{i=2}^{\ell} u_{\gamma_i} \leq e^{A(\gamma_0)}. \quad (7)$$

For $\ell = 1$, the inner tuple and product in (7) are empty, and their contribution is one.

For a fixed polymer γ , the left-hand side of (6) sums over all polymers directly incompatible with γ . The factor $e^{A(\gamma')}$ accounts for possible further extensions of a connected cluster from γ' . Thus (6) bounds the total weighted contribution of such extensions by $A(\gamma)$. The estimate (7) is called the pinned form of the criterion because the polymer γ_0 is fixed; it bounds the total absolute weight of all connected clusters containing this prescribed polymer.

2.4. Lovász local lemma. The Lovász local lemma, first established in [EL75], provides a sufficient condition for the satisfiability of a CNF formula. In this work, we use the following asymmetric form of the local lemma.

Theorem 2.2 (Asymmetric Lovász local lemma [HSS11]). *Given bad events $B_1, \dots, B_m$ with dependency neighborhoods $N(1), \dots, N(m)$, if there exist $\{x_a \in (0, 1)\}_{a=1}^m$ such that*

$$\forall a = 1, \dots, m, \quad \mathbb{P}(B_a) \leq x_a \prod_{c \in N(a)} (1 - x_c)$$

Then the probability avoiding all bad events is at least $\prod_{a=1}^m (1 - x_a) > 0$. Moreover, for any event $\mathcal{E}$, it holds that

$$\mathbb{P}\left(\mathcal{E} \mid \bigwedge_{a=1}^m \overline{B_a}\right) \leq \mathbb{P}(\mathcal{E}) \prod_{a \in N(\mathcal{E})} (1 - x_a)$$

where $N(\mathcal{E}) \triangleq \{a \in [m] \mid \text{vbl}(\mathcal{E}) \cap \text{vbl}(B_a) \neq \emptyset\}$.

2.5. The Efron–Stein inequality. We use the following conditional-variance form of the Efron–Stein inequality, also known as variance tensorization for product measures; see [BLB04, Theorems 4–5].

Lemma 2.3 (Efron–Stein inequality). *Let $r \geq 1$ and let $Z_1, \dots, Z_r$ be independent random variables, where Z_i takes values in a finite set $\mathcal{S}_i$. Write $Z = (Z_1, \dots, Z_r)$ and, for each $i \in [r]$, let $Z_{-i} = (Z_j)_{j \in [r] \setminus \{i\}}$ denote all coordinates except i . Then, for every $h : \prod_{i=1}^r \mathcal{S}_i \rightarrow \mathbb{R}$,*

$$\text{Var}(h(Z)) \leq \sum_{i=1}^r \mathbb{E}[\text{Var}(h(Z) \mid Z_{-i})]. \quad (8)$$

For each i , conditioning on Z_{-i} fixes every coordinate except Z_i . Thus the corresponding summand measures the average variance arising from the randomness of Z_i alone. The inequality bounds the total variance by the sum of these single-coordinate contributions. Independence of the coordinates is required, but the random variables $Z_1, \dots, Z_r$ need not have identical distributions.

2.6. Markov chains and functional inequalities. A Markov chain on a state space Ω consists of a transition kernel P on Ω with its stationary distribution π . In this paper, we only consider *ergodic* and *reversible* Markov chains (see details in [LP17]).

The *Dirichlet form* is usually used to study the convergence of Markov chains. For simplicity, we discuss more details of this relationship in Section 6 and only introduce basic concepts here. The Dirichlet form associated with P is the following functional defined as: for two test functions $f, g : \Omega \rightarrow \mathbb{R}$,

$$\mathcal{E}_P(f, g) \triangleq \frac{1}{2} \sum_{x, y \in \Omega} \pi(x) P(x, y) (f(x) - f(y)) (g(x) - g(y)).$$

The Rayleigh–Ritz variational characterization of the spectral gap (see [LP17, Lemma 13.7 and Remark 13.8]) gives¹

$$\text{gap}(P) \triangleq 1 - \lambda_2(P) = \inf_{\substack{f: \Omega \rightarrow \mathbb{R} \\ \text{Var}_\pi(f) > 0}} \frac{\mathcal{E}_P(f, f)}{\text{Var}_\pi(f)},$$

where $\lambda_2(P)$ is the second-largest eigenvalue of P .

3. STRUCTURAL PROPERTIES OF RANDOM SAT FORMULAE

Before the design of our sampler, we discuss the structure of random k -SAT formulae here. We follow the routine of high-degree separation in [HWY23, CLW⁺25]: given a chosen degree cut-off, we form a *core* of a random k -SAT formula by including variables of high degree or ‘significantly influenced’ by high-degree variables. With high probability, the core part of the original formula is composed of small components and therefore we can enumerate all feasible assignments on each component.

Before we state the structural properties, we clarify necessary parameters in our design and analysis here. Set

$$\begin{aligned} \lambda &= 1 + \alpha, \quad \Delta = 6k^7\lambda, \quad C_{\text{INC}} = 100k^2\lambda, \quad \overline{D} = 6k^8\lambda + 1, \\ n_0 &= e^{k-3}\lambda(k-4)^4, \quad q_0 = (4n_0)^{-1/3}, \quad r = \lfloor q_0 n \rfloor. \end{aligned} \tag{9}$$

3.1. Expansion of random SAT formulae. We apply the following expansion of a random k -SAT formula.

Lemma 3.1. *Suppose that $k \geq 30$ and $n^2 \geq 2n_0$. With probability at least $1 - \frac{2n_0}{n^2} - \frac{4}{3} \cdot 4^{-n^{1/3}}$, for every set S of at most r clauses, it holds that*

$$|\text{vbl}(S)| \geq (k-4)|S|. \tag{10}$$

Proof. Fix an integer $t \leq r$. Assume that there exists a subset S of exactly t clauses such that

$$|\text{vbl}(S)| < (k-4)|S| = (k-4)t.$$

Since $(k-4)t \leq (k-4)q_0 n < n$ by (9), it holds that all variables in S completely lie in some $U \subseteq [n]$ of size $(k-4)t$.

For a fixed subset S of clause labels with $|S| = t$ and a variable subset U of size $(k-4)t$, we bound the probability that all variables in S lie in U as

$$\mathbb{P}(\text{vbl}(S) \subseteq U) \leq \left(\frac{(k-4)t}{n} \right)^{kt}.$$

¹Lemma 13.7 uses nonzero mean-zero functions. For f with positive variance, set $g = f - \mathbb{E}_\pi[f]$. Then $\mathbb{E}_\pi[g] = 0$, $\mathcal{E}_P(g, g) = \mathcal{E}_P(f, f)$, and $\|g\|_{L^2(\pi)}^2 = \text{Var}_\pi(f)$. Thus centering yields the equivalent variance formulation displayed here; see Remark 13.8.

By the union bound together with $\binom{n}{k} \leq (en/k)^k$,

$$\begin{aligned}
\mathbb{P}(\exists S, |S| = t, |\text{vbl}(S)| < (k-4)|S|) &\leq \binom{m}{t} \binom{n}{(k-4)t} \left(\frac{(k-4)t}{n}\right)^{kt} \\
&\leq \left(\frac{em}{t}\right)^t \left(\frac{en}{(k-4)t}\right)^{(k-4)t} \frac{(k-4)^{kt} t^{kt}}{n^{kt}} \\
&\leq \left(e^{k-3} \cdot \frac{m}{n} (k-4)^4 \left(\frac{t}{n}\right)^3\right)^t \\
&\leq \left(n_0 \cdot \frac{t^3}{n^3}\right)^t.
\end{aligned}$$

For $t \leq n^{1/3}$, $n_0 t^3 / n^3 \leq n_0 / n^2 \leq 1/2$. Then

$$\sum_{t=1}^{n^{1/3}} \left(n_0 \cdot \frac{t^3}{n^3}\right)^t \leq \frac{n_0 / n^2}{1 - n_0 / n^2} \leq \frac{2n_0}{n^2}.$$

Otherwise, using $t/n \leq q_0$ and $n_0 \cdot q_0^3 = 1/4$,

$$\sum_{t=n^{1/3}}^n \left(n_0 \cdot \frac{t^3}{n^3}\right)^t \leq \sum_{t>n^{1/3}} 4^{-t} \leq \frac{4}{3} \cdot 4^{-n^{1/3}}.$$

Thus the two cases together conclude the lemma. $\square$

3.2. Estimate for incident clauses. We need the estimate for the number of incident clauses of a connected variable subset.

Lemma 3.2. *For every $k \geq 2$ and $m \geq 0$, with probability at least $1 - n^{-3}$, every connected variable subset U satisfies*

$$I(U) < C_{\text{INC}}|U| + 4 \ln n.$$

Proof. At first, we consider the case $|U| > 1$. For a connected variable subset U , pick a spanning tree of $\mathcal{G}[U]$. For every edge in this spanning tree, select a clause containing both endpoints of it. After discarding repeated clause labels, we obtain a clause certificate T such that

$$|T| \leq |U| - 1, \quad U \subseteq \text{vbl}(T).$$

Moreover, T is connected in $\mathcal{H}$.

Next, we count the number of connected clauses. Let N_t be the number of connected t -clause labels. For a collection of t clause labels, if they are connected, there is a spanning tree of size t . Root the tree at an arbitrary label. We reveal such a tree in the order that the parent precedes its children. Then conditional on everything already revealed, the probability that a new clause intersects its parent is at most $k \cdot k/n = k^2/n$. Consequently, by the Cayley's formula,

$$\begin{aligned}
\mathbb{E}[N_t] &\leq \binom{m}{t} t^{t-2} \left(\frac{k^2}{n}\right)^{t-1} \\
&\leq \frac{e^t m^t}{t^t} \cdot t^{t-2} \cdot \frac{k^{2(t-1)}}{n^{t-1}} \\
&\leq \frac{n}{k^2 t^2} (e \alpha k^2)^t \\
&\leq n (e \lambda k^2)^t
\end{aligned}$$

Now we expose a certificate T and count the remaining incidences. For a variable subset U whose certificate is T , let X be the number of remaining unexposed clauses which intersect U . Conditional on the reveal of T , X follows a binomial distribution, *i.e.*,

$$X \sim \text{Bin}(m - |T|, p) \quad \text{where } p = 1 - \left(1 - \frac{|U|}{n}\right)^k \leq \frac{k|U|}{n}.$$

Since $|T| \leq |U| - 1$, the event $\{I(U) \geq C_{\text{INC}}|U| + 4 \ln n\}$ implies

$$X \geq (C_{\text{INC}} - 1)|U| + 4 \ln n.$$

Note that $(C_{\text{INC}} - 1)|U| \geq e^2 \cdot \mathbb{E}[X | T]$ since $\mathbb{E}[X | T] \leq m \cdot k|U|/n \leq \alpha k|U|$. By the Chernoff bound,

$$\mathbb{P}(X \geq (C_{\text{INC}} - 1)|U| + 4 \ln n \mid T) \leq e^{-((C_{\text{INC}} - 1)|U| + 4 \ln n)}.$$

Set the event

$$\mathcal{E}_\ell \triangleq \{\exists U, |U| = \ell \wedge I(U) \geq (C_{\text{INC}} - 1)|U| + 4 \ln n\}.$$

By the union bound,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_\ell) &\leq \sum_{t=1}^{\min\{\ell-1, m\}} \mathbb{E}[N_t] \binom{kt}{\ell} e^{-(C_{\text{INC}} - 1)\ell + 4 \ln n} \\ &\leq \ell n (e\lambda k^2)^\ell (ek)^\ell e^{-(C_{\text{INC}} - 1)\ell - 4 \ln n} \\ &= n^{-3} \cdot \ell \exp(-\ell(C_{\text{INC}} - 1 - \ln(e^2 \lambda k^3))). \end{aligned}$$

By (9), for $k \geq 2$ and $\lambda \geq 1$,

$$C_{\text{INC}} - 1 - \ln(e^2 \lambda k^3) \geq 100k^2\lambda - 3 - 3 \ln k - \ln \lambda \geq 2.$$

Therefore,

$$\begin{aligned} \mathbb{P}\left(\bigcup_{\ell \geq 2} \mathcal{E}_\ell\right) &\leq n^{-3} \sum_{\ell \geq 2} \ell \cdot e^{-2\ell} \\ &= n^{-3} \cdot \frac{e^{-4}(2 - e^{-2})}{(1 - e^{-2})^2} \\ &\leq \frac{1}{20} \cdot n^{-3}. \end{aligned}$$

To conclude the lemma, it remains to consider the case $|U| = 1$, *i.e.*, $U = \{v\}$. Observe that $I(U)$ is dominated by a binomial distribution $\text{Bin}(m, k/n)$. By the Chernoff bound,

$$\mathbb{P}(I(U) \geq C_{\text{INC}} + 4 \ln n) \leq n^{-4} e^{-C_{\text{INC}}}.$$

By the union bound, the failure probability is at most

$$\frac{1}{20} \cdot n^{-3} + n \cdot e^{-C_{\text{INC}}} n^{-4} \leq n^{-3}.$$

Therefore we conclude the lemma. □

3.3. High-degree separation. For a subset $S \subseteq \mathcal{V}$, we construct the core induced by S as:

- (1) initialize the core as S ;
- (2) repeatedly select a remaining clause containing more than 8 distinct variables in the current core, and add all its variables to the core.

We use $\text{Core}(S)$ to denote the final set generated by S .

Consider the set collecting all variables of high degree:

$$\mathcal{V}^{\text{high}} \triangleq \{v \in \mathcal{V} \mid \deg(v) \geq \Delta\}.$$

The core of Φ is defined as $\mathcal{V}^{\text{bad}} = \text{Core}(\mathcal{V}^{\text{high}})$ and denote by $\mathcal{V}^{\text{good}} = \mathcal{V} \setminus \mathcal{V}^{\text{bad}}$ the collection of good variables.

A clause C_a of label a is called a *core* clause if its variables are completely contained in $\mathcal{V}^{\text{bad}}$ (equivalently, $\text{vbl}(a) \subseteq \mathcal{V}^{\text{bad}}$). We set $\mathcal{C}^{\text{good}}$ as the collection of all non-core clause labels, *i.e.*,

$$\mathcal{C}^{\text{good}} \triangleq \left\{a \in \mathcal{C} \mid \text{vbl}(a) \cap \mathcal{V}^{\text{good}} \neq \emptyset\right\}.$$

For every $a \in \mathcal{C}^{\text{good}}$, let C_a^{good} be the clause of C_a after violating all bad variables in it, *i.e.*,

$$C_a^{\text{good}} \triangleq \bigvee_{i \in [k] : \text{var}(\ell_{a,i}) \in \mathcal{V}^{\text{good}}} \ell_{a,i}.$$

and the good formula is set as

$$\Psi \triangleq \bigwedge_{a \in \mathcal{C}^{\text{good}}} C_a^{\text{good}}.$$

We first bound the size of $\mathcal{V}^{\text{high}}$.

Lemma 3.3. *For $k \geq 30$ and $h = \lceil q_0 n \rceil$, it holds that*

$$\mathbb{P}\left(|\mathcal{V}^{\text{high}}| \leq r\right) \geq 1 - e^{-h}.$$

Proof. It suffices to show that $\mathbb{P}(|\mathcal{V}^{\text{high}}| \geq h) \leq e^{-h}$. Fix a variable subset U of size h . Let X_U be the random variable to denote the number of literals whose variable names are in U . Then X_U is a binomial variable:

$$X_U \sim \text{Bin}\left(km, \frac{h}{n}\right) \quad \text{thus} \quad \mathbb{E}[X_U] = \frac{m}{n}kh \leq \lambda kh.$$

If $U \subseteq \mathcal{V}^{\text{high}}$, it must hold that $X_U \geq h\Delta$. The Chernoff bound provides that

$$\mathbb{P}(X \geq \Delta h) \leq \left(\frac{e\lambda kh}{\Delta h}\right)^{\Delta h} = \left(\frac{e}{6k^6}\right)^{\Delta h}$$

since $\Delta = 6k^7\lambda$. By the union bound,

$$\begin{aligned} \mathbb{P}\left(|\mathcal{V}^{\text{high}}| \geq h\right) &\leq \binom{n}{h} \left(\frac{e}{6k^6}\right)^{\Delta h} \\ &\leq \left(\frac{en}{h}\right)^h \left(\frac{e}{6k^6}\right)^{\Delta h} \\ &\leq \left(\frac{e}{q_0} \left(\frac{e}{6k^6}\right)^{\Delta}\right)^h. \end{aligned}$$

For $k \geq 30$, by choices of parameters (9),

$$\frac{e}{q_0} \left(\frac{e}{6k^6}\right)^{\Delta} \leq e^{-1}.$$

Hence we conclude the lemma. □

Another important property is the bounded growth of the core for every high-degree variable subset.

Lemma 3.4. *Assume $r \geq 1$, $|\mathcal{V}^{\text{high}}| \leq r$ and the expansion event of Lemma 3.1. For every $S \subseteq \mathcal{V}^{\text{high}}$, it holds that*

$$|\text{Core}(S)| \leq \frac{k-4}{5}|S| \leq \frac{k}{2}|S|.$$

Proof. By construction of $\text{Core}(S)$, every selected clause has at most $k-9$ new variables added to the core. Therefore, after any $t \leq r$ selections, by Lemma 3.1,

$$(k-4)t \leq |\text{vbl}(\text{selected clauses})| \leq |S| + (k-9)t$$

which implies that $5t \leq |S|$. It also implies that the process cannot reach $t = r$ otherwise it must hold $5r \leq |S| \leq r$. Therefore

$$|\text{Core}(S)| \leq |S| + (k-9)\frac{|S|}{5} = \frac{k-4}{5}|S|,$$

which concludes this lemma. $\square$

3.3.1. Small components in the bad core. A key property of $\mathcal{V}^{\text{bad}}$ is that with high probability, the components of it has small size. Let $B_1, \dots, B_s$ be the maximal connected components of $\mathcal{G}[\mathcal{V}^{\text{bad}}]$.

Lemma 3.5. *Assume the joint event of Lemmas 3.1 to 3.3. For any non-empty subset $J \subseteq [s]$ and $U = \bigcup_{i \in J} B_i$, it holds that*

$$|U| \leq \frac{k}{2} |U \cap \mathcal{V}^{\text{high}}|.$$

Moreover, each component is of size at most $\log_2 n$.

Proof. The key ingredient to prove the first inequality is the following identity:

$$\text{Core}\left(U \cap \mathcal{V}^{\text{high}}\right) = U \tag{11}$$

To show $\text{Core}\left(U \cap \mathcal{V}^{\text{high}}\right) \subseteq U$, suppose that there is a clause containing at least 9 variables in U . All its variables should be in $\mathcal{V}^{\text{bad}}$ by the construction of $\mathcal{V}^{\text{bad}}$ and they are connected in $\mathcal{G}[\mathcal{V}^{\text{bad}}]$. Note that each B_i is a maximal connected component for $i = 1, \dots, s$. These variables must lie in one component. Therefore they are in U since the clause meets U .

Now we prove that $U \subseteq \text{Core}\left(U \cap \mathcal{V}^{\text{high}}\right)$. Take a sequence of clause additions constructing $\mathcal{V}^{\text{bad}}$ from $\mathcal{V}^{\text{high}}$. Then we retain only the additions whose clauses lie in U . It is clear that this is a legal sequence to construct $\text{Core}\left(U \cap \mathcal{V}^{\text{high}}\right)$ from $U \cap \mathcal{V}^{\text{high}}$, and the sequence finally produces all of U since $U \subseteq \mathcal{V}^{\text{bad}}$. Hence $U \subseteq \text{Core}\left(U \cap \mathcal{V}^{\text{high}}\right)$.

Putting two directions together, we conclude (11). By Lemma 3.4,

$$|U| = \left| \text{Core}\left(U \cap \mathcal{V}^{\text{high}}\right) \right| \leq \frac{k}{2} |U \cap \mathcal{V}^{\text{high}}|$$

which is exactly the first conclusion.

To bound the size of each component, suppose that there is a component $B_i \in \mathcal{V}^{\text{bad}}$ such that $|B_i| > \log_2 n$. By the inequality above, it holds that

$$|B_i| \leq \frac{k}{2} |B_i \cap \mathcal{V}^{\text{high}}|.$$

Consider the number of clauses incident to $B_i \cap \mathcal{V}^{\text{high}}$. Since every variable in $B_i \cap \mathcal{V}^{\text{high}}$ has degree at least Δ ,

$$\Delta |B_i \cap \mathcal{V}^{\text{high}}| \leq \sum_{v \in B_i \cap \mathcal{V}^{\text{high}}} \deg(v) \leq kI(B_i).$$

The last inequality holds since every clause incident to B_i contains at most k distinct variables in $B_i \cap \mathcal{V}^{\text{high}}$. By Lemma 3.2,

$$\frac{2\Delta |B_i|}{k} \leq \Delta |B_i \cap \mathcal{V}^{\text{high}}| \leq kC_{\text{INC}}|B_i| + 4k \ln n.$$

By the assumption $|B_i| > \log_2 n$, $\ln n < |B_i| \cdot \ln 2$. Then we have

$$\frac{2\Delta}{k} |B_i| < k(C_{\text{INC}} + 4 \ln 2) |B_i|$$

giving that $1 < k^2(C_{\text{INC}} + 4 \ln 2)/(2\Delta)$. By (9),

$$\frac{k^2(C_{\text{INC}} + 4 \ln 2)}{2\Delta} = \frac{25}{3k^3} + \frac{\ln 2}{3k^5 \lambda} < 1$$

for $k \geq 30$ and $\lambda \geq 1$. This contradiction concludes that $|B_i| \leq \log_2 n$ for every $i = 1, \dots, s$. $\square$

Now we check the feasibility of the core formula with arbitrary literal signs, which provides a guarantee of validity for updates to the block dynamics in our sampling algorithm. As a first step, we show that, under the preceding structural properties, every connected component of core clauses is small.

Proposition 3.6. *Assume the joint event of Lemmas 3.1 to 3.3. For every integer $L \in [\log_2 n, r]$ and $i = 1, \dots, s$, there are at most L core clauses meeting B_i .*

Proof. Suppose that there exists B_i containing at least L core clauses. Select L of them to form S . By Lemma 3.1, since $L \leq r$, it holds that

$$|\text{vbl}(S)| \geq (k-4)L \geq L.$$

On the other hand, $\text{vbl}(S)$ must be fully contained in B_i by the assumption that B_i is maximal. This leads to contradiction that $|\text{vbl}(S)| \leq |B_i| < \log_2 n \leq L \leq |\text{vbl}(S)|$. $\square$

Now we are ready to place the following lemma.

Lemma 3.7. *Assume the joint event of Lemmas 3.1 to 3.3. Then the core formula is satisfiable with any choice of literal signs, and its feasible set is*

$$\Omega \triangleq \prod_{i=1}^s \Omega_i \neq \emptyset$$

where Ω_i consists of all feasible assignments to B_i satisfying all core clauses meeting that component.

Proof. For every $i = 1, \dots, s$, let $\mathcal{C}_i$ be the collection of core clauses meeting B_i . Construct a bipartite graph with:

- one left vertex for each clause label $a \in \mathcal{C}_i$;
- one right vertex for each variable in $v \in B_i$;
- an edge $a \sim v$ if and only if $v \in \text{vbl}(a)$.

For every $S \subseteq \mathcal{C}_i$, the neighbors of S are exactly $\text{vbl}(S)$. By Proposition 3.6, for every subset $S \subseteq \mathcal{C}_i$, $|S| \leq |\mathcal{C}_i| \leq r$, thus $|\text{vbl}(S)| \geq (k-4)|S|$. Thus by Hall's theorem, there is an injection $f_i : \mathcal{C}_i \rightarrow B_i$ such that $f_i(a)$ occurs in the clause C_a .

Now we construct a satisfying assignment based on the matching. Given any choice of literal signs, for every clause $a \in \mathcal{C}_i$,

- choose one occurrence of $f_i(a)$ in C_a ;
- set $f_i(a)$ making C_a true.

Since f_i is injective, different clauses prescribe values to different variables. Finally, we set all unmatched variables in B_i arbitrarily. Hence we conclude that $\Omega_i \neq \emptyset$ for every $i = 1, \dots, s$.

Since $B_1, \dots, B_s$ are disjoint and thus each core clause belongs to exactly one component, then it holds that every assignment in $\Omega \triangleq \prod_{i=1}^s \Omega_i$ is feasible for the full core formula. On the other hand, given a feasible assignment of the core formula, its restriction to B_i also satisfies $\mathcal{C}_i$ for each $i = 1, \dots, s$, meaning that it belongs to Ω . Thus we conclude that the feasible set of the core formula is exactly Ω . $\square$

3.3.2. Extension from the core formula to good variables. Now we consider the good formula Ψ . Let $\mathcal{H}_\Psi$ be the dependency graph of Ψ and $\text{vbl}^{\text{good}}(\cdot)$ denote the variables in the good formula Ψ .

Proposition 3.8. *Assume the joint event of Lemmas 3.1 to 3.3. The the following properties hold.*

- The dependency graph $\mathcal{H}_\Psi$ of Ψ has maximum degree $d \leq k\Delta$ and thus $d + 1 \leq \bar{D}$ by (9).
- For every $S \subseteq \mathcal{C}^{\text{good}}$ with $|S| \leq L$ where L is an integer $L \in [\log_2, r]$,

$$|\text{vbl}^{\text{good}}(S)| \geq (k - 12)|S|.$$

Proof. Observe that $\mathcal{V}^{\text{high}} \subseteq \mathcal{V}^{\text{bad}}$ and $\mathcal{V}^{\text{good}} = \mathcal{V} \setminus \mathcal{V}^{\text{bad}}$. Then for every variable $v \in \mathcal{V}^{\text{good}}$, it holds that $\deg(v) < \Delta$. Therefore the maximum degree of $\mathcal{H}_\Psi$ is at most $k\Delta$.

For every $S \subseteq \mathcal{C}^{\text{good}}$ with $|S| \leq L$, by Lemma 3.1,

$$\begin{aligned} |\text{vbl}^{\text{good}}(S)| &\geq |\text{vbl}(S)| - \sum_{i \in S} |\text{vbl}(i) \cap \mathcal{V}^{\text{bad}}| \\ &\geq (k - 4)|S| - 8|S| \\ &\geq (k - 12)|S| \end{aligned}$$

which concludes the second property. $\square$

The following lemma shows that every feasible assignment to the core formula can be extended to a satisfying assignment of the original formula Φ . For every subset $F \subseteq \mathcal{C}^{\text{good}}$, the deletion sub-formula of Ψ induced by F is defined as

$$\Psi[F] \triangleq \bigwedge_{a \in F} C_a^{\text{good}}.$$

Let ν^{good} be the uniform product measure on $\{0, 1\}^{\mathcal{V}^{\text{good}}}$. For $a \in \mathcal{C}^{\text{good}}$, let E_a be the event that the residual clause C_a^{good} is violated, and for $S \subseteq \mathcal{C}^{\text{good}}$ define

$$E_S = \bigcap_{a \in S} E_a, \quad q_S = \nu^{\text{good}}(E_S).$$

Lemma 3.9. *Assume the joint event of Lemmas 3.1 to 3.3. Put $p_* = 2^{-k+12}$ and $C_* = 10^4$. If*

$$p_* \bar{D} \leq \frac{1}{2C_*}, \tag{12}$$

then every deletion sub-formula $\Psi[F]$ of Ψ is satisfiable and it holds that for every $a \notin F$,

$$\mu_F(E_a) \leq ep_*. \tag{13}$$

Consequently for $J \subseteq F$, $Z_{F \setminus J} \leq (1 - ep_)^{-|J|} Z_F$.*

Moreover, for every $\sigma \in \Omega$, it extends to a satisfying assignment of Φ . Let D_σ be the collection of labels whose bad-variable literals are all false at σ . Then it holds that

$$\mu_\Phi(\cdot \mid \mathcal{V}^{\text{bad}} \leftarrow \sigma) = \mu_{\Psi[D_\sigma]}(\cdot).$$

Proof. Fix a deletion sub-formula $\Psi[F]$. Every clause in it has violating probability at most p_* , and has at most $d + 1 \leq \overline{D}$ dependencies by Proposition 3.8. Take $x_a = x \triangleq ep_*$ in Theorem 2.2 for every clause label $a \in \Psi[F]$. It holds that for every a , by $x\overline{D} \leq e/(2C_*) < 1/128$ and $x < 1/2$,

$$p_* = xe^{-1} \leq x(1 - x)^d.$$

Therefore there exists a satisfying assignment for $\Psi[F]$.

To show (13), we prove it by induction on $|F|$. First, since all deletion sub-formula $\Psi[F]$ is satisfiable, $Z_F > 0$. For $F = \emptyset$, it is clear to see $\mu_\emptyset(E_a) \leq p_* \leq ep_*$. Now fix F with $|F| = t > 1$ and $a \notin F$. Split F into the neighbors and non-neighbors of a :

$$S = F \cap N_{\mathcal{H}_\Psi}(a), \quad T = F \setminus S$$

and order $S = \{b_1, \dots, b_r\}$ where $r \leq d$.

The event E_a uses variables disjoint from every clause in T . Hence

$$\nu(E_a \mid \text{Good}(F)) = \frac{\nu(E_a \cap \text{Good}(S) \mid \text{Good}(T))}{\nu(\text{Good}(S) \mid \text{Good}(T))} \leq \frac{p_*}{\nu(\text{Good}(S) \mid \text{Good}(T))}$$

To bound the denominator, by the chain rule,

$$\nu(\text{Good}(S) \mid \text{Good}(T)) = \prod_{j=1}^r (1 - \nu(E_{b_j} \mid \text{Good}(T \cup \{b_1, \dots, b_{j-1}\}))).$$

By induction, each factor is at least $1 - ep_*$. Therefore

$$\nu(\text{Good}(S) \mid \text{Good}(T)) \geq (1 - ep_*)^r \geq (1 - ep_*)^d.$$

Thus we conclude that $\nu(E_a \mid \text{Good}(F)) \leq p_*/(1 - ep_*)^d \leq ep_*$.

To complete the proof, fix a partial assignment $\sigma \in \Omega$ on the core formula. A good formula either is already satisfied by σ and disappears, or retains exactly its clause of Ψ . Then it is trivial to see the satisfiability under σ and the identity of conditional law. $\square$

4. RECURSIVE INSERTION-DYNAMIC FOR THE GOOD FORMULA

In this section, we provide the sampling algorithm for the good formula Ψ . The algorithm adapts the framework of [JMP26] to resample a feasible assignment after inserting an additional clause, specialized to the random k -SAT setting. Briefly, we adapt the recursive insertion-chain in [JMP26] on ordinary connected sets of constraints, and generate its lists from a global product sample.

For this section only, we redefine notations as follows. Let $\mathcal{V}$ be a finite variable set, $\Omega = \prod_{v \in \mathcal{V}} \Omega_v$ be the collection of all assignments and $\nu = \otimes_{v \in \mathcal{V}} \nu_v$ be a product probability measure. For simplicity, restrict Ω_v to the support of ν_v for each $v \in \mathcal{V}$. For a subset $U \subseteq \mathcal{V}$, define $\Omega_U = \prod_{v \in U} \Omega_v$ and $\nu_U = \otimes_{v \in U} \nu_v$ as the subspaces of Ω and ν restricted on U . A finite family $\mathcal{F}$ of constraints has bad events $B_c \subseteq \Omega$ depending on variables $\text{vbl}(c)$. Let $\mathcal{H}$ be a graph on $\mathcal{F}$ in which a pair of constraints intersecting with each other are adjacent. For $F \subseteq \mathcal{F}$, let $\text{Good}(F)$ be the event that all bad events in F do not occur and write

$$Z_F \triangleq \nu(\text{Good}(F)), \quad \mu_F(\cdot) = \nu(\cdot \mid \text{Good}(F)).$$

To guarantee the existence of measures, assume that $Z_{\mathcal{F}} > 0$.

For a non-empty subset $F \subseteq \mathcal{F}$ and $a \in F$, let $\text{CS}_F(a)$ consist of all connected subsets of $\mathcal{H}[F]$ containing a . For every $S \in \text{CS}_F(a)$, define

$$\begin{aligned} \mathcal{V}_S &\triangleq \bigcup_{c \in S} \text{vbl}(c), \quad E_S \triangleq \bigcap_{c \in S} B_c, \quad q_S \triangleq \nu(E_S), \\ F_S &\triangleq F \setminus N_F[S], \quad R_S = \frac{Z_{F_S}}{Z_F}, \quad \ell(S) \triangleq |N_F[S] \setminus \{a\}| \end{aligned} \tag{14}$$

where $N_F[S]$ is the closed neighborhood of S in $\mathcal{H}[F]$.

Theorem 4.1. *Suppose that there exist $K_0, K_2 \geq 0$ and $\kappa \in [0, 1)$ satisfying that for every non-empty subset $F \subseteq \mathcal{F}$ and $a \in F$,*

$$\sum_{S \in \text{CS}_F(a)} q_S R_S^2 \leq K_0, \tag{15}$$

$$\sum_{S \in \text{CS}_F(a)} q_S R_S^2 \ell(S) \leq \kappa, \tag{16}$$

$$\sum_{S, T \in \text{CS}_F(a)} R_S^2 q_{S \cup T} \leq K_2. \tag{17}$$

For every $F \subseteq \mathcal{F}$ and $\varepsilon \in (0, 1)$, there is a sampler supported on $\text{Good}(F)$ such that it outputs a sample whose law is within ε in total variation distance of μ_F with expected running time $\text{poly}(I, \varepsilon^{-1}, K_0, K_2, (1 - \kappa)^{-1})$ with an absolute polynomial degree where I is the length of the full input.

4.1. Preparations for the recursive insertion-chain. Before describing the recursive insertion-chain, we make some necessary preparations here.

4.1.1. The connected-set identity. We establish the following identity for all connected sets analogues to [JMP26, Theorem 2.4].

Lemma 4.2. *The following identity holds point-wisely on Ω :*

$$\mathbf{1}_{B_a} \cdot \mathbf{1}_{\text{Good}(F \setminus \{a\})} = \sum_{S \in \text{CS}_F(a)} (-1)^{|S|-1} \mathbf{1}_{E_S} \cdot \mathbf{1}_{\text{Good}(F_S)}.$$

Writing $F^- \triangleq F \setminus \{a\}$, $Z_- \triangleq Z_{F^-}$ and $r_{F^-, a} \triangleq \mu_{F^-}(B_a)$, we have

$$\frac{r_{F^-, a}}{1 - r_{F^-, a}} = \sum_{S \in \text{CS}_F(a)} (-1)^{|S|-1} q_S R_S.$$

Furthermore, if (15) holds, then $Z_-/Z_F \leq 1 + K_0$.

Proof. To show the first identity, we follow the proof of [JMP26, Theorem 2.4] which comes from the inclusion-exclusion argument of [LWY⁺26]. Fix an assignment σ . By the inclusion-exclusion principle,

$$\begin{aligned} \mathbf{1}_{B_a}(\sigma) \prod_{c \in F^-} (1 - \mathbf{1}_{B_c}(\sigma)) &= \sum_{S \subseteq F^- : a \in S} (-1)^{|S|-1} \prod_{c \in S} \mathbf{1}_{B_c}(\sigma) \\ &= \sum_{S \subseteq \text{CS}_F(a)} \sum_{T \subseteq F_S} (-1)^{|S|+|T|-1} \prod_{c \in S} \mathbf{1}_{B_c}(\sigma) \prod_{c \in T} \mathbf{1}_{B_c}(\sigma) \\ &= \sum_{S \subseteq \text{CS}_F(a)} (-1)^{|S|-1} \mathbf{1}_{E_S}(\sigma) \cdot \mathbf{1}_{\text{Good}(F_S)}(\sigma). \end{aligned}$$

To derive the second identity, note that E_S depends only on variables used in constraints S . Then variables in $F_S = F \setminus N_G[S]$ are disjoint from those in E_S . Hence we obtain

$$\nu(E_S \cap \text{Good}(F_S)) = \nu(E_S) \cdot \nu(\text{Good}(F_S)) = q_S Z_{F_S}$$

since ν is a product measure, $q_S = \nu(E_S)$ by (14) and $Z_{F_S} = \nu(\text{Good}(F_S))$. Taking expectation for the first identity, we have

$$\begin{aligned} Z_{F^-} - Z_F &= \sum_{S \in \text{CS}_F(a)} (-1)^{|S|-1} \nu(E_S) \nu(\text{Good}(F_S)) \\ &= \sum_{S \in \text{CS}_F(a)} (-1)^{|S|-1} q_S Z_{F_S}. \end{aligned}$$

Dividing Z_F on two sides gives

$$\frac{r_{F^-,a}}{1 - r_{F^-,a}} = \sum_{S \in \text{CS}_F(a)} (-1)^{|S|-1} q_S R_S$$

by $R_S = Z_{F_S}/Z_F$ in (14).

For the ratio upper bound, since $F_S \subseteq F$ and thus $R_S \geq 1$, it holds that

$$\begin{aligned} \frac{Z_-}{Z_F} &= 1 + \sum_{S \in \text{CS}_F(a)} (-1)^{|S|-1} q_S R_S \\ &\leq 1 + \sum_{S \in \text{CS}_F(a)} (-1)^{|S|-1} q_S R_S^2 \\ &\leq 1 + K_0. \end{aligned}$$

Therefore, we conclude the lemma. $\square$

4.1.2. Parity pairing. To precisely define transitions of the dynamics, we introduce the following parity pairing lemma. Given a fixed total order on F , the exploration of $S \in \text{CS}_F(a)$ starts with the accepted set $\{a\}$ and unlabeled other vertices. Repeatedly explore and label the first unlabeled neighbor of the accepted set, accepting it when it is in S , otherwise rejecting it. When there is no candidate, stop the exploration. By connectivity of S , the final accepting set is exactly S and the labeled set is $N_F[S] \setminus \{a\}$. Moreover, for an assignment σ , define $Q(\sigma) \triangleq \{c \in F \mid \sigma \in B_c\}$ as the constraints σ violates.

Lemma 4.3. *Suppose that $\sigma \in E_S \cap \text{Good}(F_S)$. When $Q(\sigma) \neq \{a\}$, let $d_\sigma(S)$ be the last member of $Q(\sigma) \setminus \{a\}$ explored by S , and define $\iota_\sigma(S) \triangleq S \triangle \{d_\sigma(S)\}$. Then $\iota_\sigma(S) \in \text{CS}_F(a)$, $\sigma \in E_{\iota_\sigma(S)} \cap \text{Good}(F_{\iota_\sigma(S)})$ and $\iota_\sigma(\iota_\sigma(S)) = S$. Trivially S and $\iota_\sigma(S)$ have opposite parity.*

When $Q(\sigma) = \{a\}$, the only possible choice of S is the singleton set $\{a\}$.

Proof. The statement $\sigma \in E_S \cap \text{Good}(F_S)$ is equivalent to

$$S \subseteq Q(\sigma) \subseteq N_F[S].$$

Therefore all constraints in $Q(\sigma)$ are labeled in the exploration by S . When $Q(\sigma) \neq \{a\}$, write $S' = S \triangle \{d_\sigma(S)\}$. We verify the following properties of S' .

- (1) $S' \in \text{CS}_F(a)$. Let A be the accepted set immediately before $d_\sigma(S)$ is explored. Note that there is no accepted vertex after $d_\sigma(S)$ is explored (since $S \subseteq Q(\sigma)$). Therefore, it must be $S' = A$ or $S' = A \cup \{d_\sigma(S)\}$ and $d_\sigma(S)$ must be adjacent to A . It is clear that $a \in A$ by the definition of $d_\sigma(S)$. Hence we conclude that $a \in S'$ and S' is connected, leading to the property $S' \in \text{CS}_F(a)$.

(2) $\sigma \in E_{S'} \cap \text{Good}(F_{S'})$. It is equivalent to show that

$$S' \subseteq Q(\sigma) \subseteq N_F[S'].$$

Since $A \subseteq S \subseteq Q(\sigma)$ and $d_\sigma(S) \subseteq Q(\sigma)$, we directly obtain that $S' \subseteq Q(\sigma)$. To prove $Q(\sigma) \subseteq N_F[S']$, recall the exploration of S and the definition that $d_\sigma(S)$ is the last member of $Q(\sigma) \setminus \{a\}$ in the exploration. Then all clauses in $Q(\sigma) \setminus \{a\}$ were explored and thus labeled by the time $d_\sigma(S)$ was explored. Hence $Q \subseteq N_F[A] \subseteq N_F[S']$ by $A \subseteq S'$.

(3) $\iota_\sigma(\iota_\sigma(S)) = S$. Compare the explorations of S and S' . Their membership decisions differ only at $d_\sigma(S)$. Then until $d_\sigma(S)$ is explored, they share the same accepted set and label set. Moreover, under the fixed total order on F , their exploration sequences keep identical through the selection of $d_\sigma(S)$.

At $d_\sigma(S)$, the accept/reject decision is reversed. Since either $S' = A$ or $S' = A \cup \{d_\sigma(S)\}$ holds, any subsequent candidate is rejected. Therefore, we obtain that $d_\sigma(S') = d_\sigma(S)$. Hence we conclude the property that $\iota_\sigma(\iota_\sigma(S)) = S$.

When $Q(\sigma) = \{a\}$, there must hold $S \subseteq \{a\} \subseteq N_F[S]$, meaning that S must be $\{a\}$. $\square$

4.1.3. *Random lists.* Draw one assignment $\sigma \sim \nu$ and define

$$\mathbf{L} = \{(S, \sigma(\mathcal{V}_S)) \mid S \in \text{CS}_F(a), \sigma \in E_S\}.$$

A *corridor label* is $e = (S, u)$ with $u \in E_S$ and $u \in \Omega_{\mathcal{V}_S}$. We omit labels of zero product probability and put $q_e = \nu_{\mathcal{V}_S}(u) > 0$.

Lemma 4.4. *For every corridor label $e = (S, u)$, it holds that*

$$\mathbb{P}(e \in \mathbf{L}) = q_e, \quad \text{Law}(\mathbf{L} \mid e \in \mathbf{L}) = \text{Law}(\mathbf{L} \text{ with } \sigma(\mathcal{V}_S) = u \text{ fixed}).$$

A realized random list $\mathbf{L}$ is enumerable in polynomial time of $|\mathbf{L}|$ and the input length I . Moreover, if (17) holds,

$$\mathbb{E}[|\mathbf{L}|^2] \leq K_2, \quad \sum_e q_e \mathbb{E}[1 + |\mathbf{L}| \mid e \in \mathbf{L}] \leq K_2 + \sqrt{K_2}.$$

Proof. The argument $\mathbb{P}(e \in \mathbf{L}) = q_e$ holds trivially by the definition of q_e . Since $u \in E_S$, the event $\{u \in \mathbf{L}\}$ is exactly the event $\{\sigma(\mathcal{V}_S) = u\}$. By the product independence of ν , for every possible list L ,

$$\mathbb{P}(\mathbf{L} = L \mid e \in \mathbf{L}) = \mathbb{P}(\mathbf{L} = L \wedge \sigma(\mathcal{V}_S) = u).$$

Thus we conclude the second argument for the law of random lists.

To enumerate the entries in $\mathbf{L}$, firstly test all constraints under the assignment σ and form the violating set $Q(\sigma)$ in time $\text{poly}(I)$ where I is the input length. If $a \notin Q(\sigma)$, $\mathbf{L}$ is the empty list. Otherwise,

$$\mathbf{L} = \{(S, \sigma(\mathcal{V}_S)) \mid S \in \text{CS}_F(a), \sigma \in E_S\}.$$

It suffices to enumerate every connected subset containing a in $\mathcal{H}[Q(\sigma)]$. Using a depth-first search, it takes at most $2|\mathbf{L}|$ time to enumerate all entries in $\mathbf{L}$ and therefore $\mathbf{L}$ is enumerable in $\text{poly}(I)(1 + |\mathbf{L}|)$.

When (17) holds, by the fact that $R_S \geq 1$,

$$\mathbb{E}[|\mathbf{L}|^2] = \sum_{S, T \in \text{CS}_F(a)} q_{S \cup T} \leq \sum_{S, T \in \text{CS}_F(a)} R_S^2 q_{S \cup T} \leq K_2.$$

For the last inequality, by law of total expectation,

$$\begin{aligned} \sum_e q_e \mathbb{E}[1 + |\mathbf{L}| \mid e \in \mathbf{L}] &= \mathbb{E}[|\mathbf{L}| + |\mathbf{L}|^2] \\ &\leq \mathbb{E}[|\mathbf{L}|^2] + \sqrt{\mathbb{E}[|\mathbf{L}|^2]} \\ &\leq K_2 + \sqrt{K_2} \end{aligned}$$

using $\mathbb{E}[|\mathbf{L}|] \leq \sqrt{\mathbb{E}[|\mathbf{L}|^2]}$. $\square$

Only for analysis, define the quantity $\omega_e = \mathbb{E}[\mathbb{1}[e \in \mathbf{L}] / |\mathbf{L}|]$ with the zero fraction for empty lists and therefore $\omega_e / q_e = \mathbb{E}[|\mathbf{L}|^{-1} \mid e \in \mathbf{L}]$.

4.2. The recursive state space and transition rules. Now we introduce the state space and the full transition rules of the recursive insertion dynamics, which is similar in [JMP26].

Definition 4.5 (State space). We define the state space $\mathcal{S}(F, a)$ inductively on $|F|$.

- **Input and output states:**

$$\forall \sigma \in \text{Good}(F^-), \text{I}(\sigma), \quad \text{and} \quad \forall \sigma \in \text{Good}(F), \text{O}(\sigma).$$

Furthermore, define $\mu_- (\text{I}(\sigma)) = \mu_{F^-}(\sigma)$ on the input space and $\mu_+ (\text{O}(\sigma)) = \mu_F(\sigma)$ on the output space.

- **Set states:** for every connected subset $S \in \text{CS}_F(a)$, there are states $\text{T}_S(\sigma)$ for $\sigma \in E_S \cap \text{Good}(F_S)$.
- **Layers:** order $N_F[S] \setminus \{a\}$ as $b_{S,1}, \dots, b_{S,\ell(S)}$. Set

$$\forall i = 0, \dots, \ell(S), \quad F_{S,i} = F_S \cup \{b_{S,1}, \dots, b_{S,i}\}.$$

For each label $e = (S, u)$ we introduce layers

$$\forall i = 0, \dots, \ell(S), \sigma \in \text{Good}(F_{S,i}), \quad \text{L}_i^e(\sigma).$$

For every $i = 1, \dots, \ell(S)$, we embed a distinct copy of $\mathcal{S}(F_{S,i}, b_{S,i})$ by a map $\psi_{e,i}$ that identifies its input $\text{I}(\sigma)$ with $\text{L}_{i-1}^e(\sigma)$ and its output $\text{O}(\sigma)$ with $\text{L}_i^e(\sigma)$. Every other state X becomes a distinct state (e, i, X) .

The following transition rules are described in [JMP26] to demonstrate an update of the insertion dynamics. For completeness, we state here.

Definition 4.6 (Transition rules [JMP26]). Consider the state space $\mathcal{S}(F, i)$. There are three kinds of transition rules **F**, **B** and **R** (forward, backward and reset):

- (1) At $\text{I}(\sigma)$, Rule **F** moves to $\text{O}(\sigma)$ if $\sigma \notin B_a$ and otherwise to $\text{T}_{\{a\}}(\sigma)$. Rule **R** generates a fresh list $\mathbf{L}$ and if it is non-empty, choose e uniformly in $\mathbf{L}$ and move to $\text{L}_{\ell(S)}^e(\sigma)$.
- (2) At $\text{O}(\sigma)$, Rule **B** moves to $\text{I}(\sigma)$.
- (3) At $\text{T}_S(\sigma)$, write $\sigma = (y, z)$ where y is on $\mathcal{V} \setminus \mathcal{V}_S$ and z on $\mathcal{V}_S$. Rule **F** samples $u \sim \nu_{\mathcal{V}_S}$ and moves to $\text{L}_0^{(S,z)}(y, u)$. Rule **B** moves to $\text{I}(\sigma)$ if $Q(\sigma) = \{a\}$ and to $\text{T}_{\iota_\sigma(S)}(\sigma)$ otherwise.
- (4) At $\text{L}_i^e(\sigma)$ where $e = (S, u)$:
 - (a) Rule **B** moves to $\text{L}_{i-1}^e(\sigma)$ if $i > 0$. If $i = 0$, write $\sigma = (y, z)$ and restore the saved value, moving to $\text{T}_S(y, u)$.
 - (b) If $i < \ell(S)$ and the rule is **F** or **R**, apply the same rule to the input states $\text{I}(\sigma)$ of $\mathcal{S}(F_{S,i+1}, b_{S,i+1})$ and embed the result by $\psi_{e,i+1}$.

- (c) If $i = \ell(S)$ and the rule is **F**, generate $\mathbf{L}^e \sim \text{Law}(\mathbf{L} \mid e \in \mathbf{L})$. Pick a uniform label in $\mathbf{L}^e$ and move to $\mathbf{l}(\sigma)$ if and only if the chosen label is e .
- (5) At an internal state (e, i, X) , apply the rule to X in its smaller copy and embed the result by $\psi_{e,i}$.

The transition kernel $P_{F,a}$ stays with probability $1/2$ and otherwise choose uniformly a rule from **F**, **B** and **R**. For $0 < \beta < 1$, we define

$$K_\beta(\mathbf{O}(\sigma), \mathbf{l}(\sigma)) = \frac{\beta}{6}, \quad K_\beta(\mathbf{O}(\sigma), \mathbf{O}(\sigma)) = 1 - \frac{\beta}{6}.$$

At every other state, set $K_\beta = P_{F,a}$.

Recursively define weights on the state space as

$$\begin{aligned} w(\mathbf{l}(\sigma)) &= w(\mathbf{O}(\sigma)) = w(\mathbf{T}_S(\sigma)) = \nu(\sigma), \\ w(\mathbf{L}_i^e(\sigma)) &= q_e \nu(\sigma), \\ w((e, i, X)) &= q_e w(X) \end{aligned}$$

and

$$\tilde{w}(X) = \begin{cases} w(X)/\beta & X = \mathbf{O}(\sigma) \text{ is an outer output state,} \\ w(X) & \text{otherwise.} \end{cases}$$

The normalizing factor of $\tilde{w}(\cdot)$ is denoted by $M = \sum_X \tilde{w}(X)$.

Lemma 4.7. *The kernel $P_{F,a}$ is reversible with respect to the weight w and K_β is reversible with respect to $\pi = \tilde{w}/M$. Furthermore, conditional on outer output states, π gives the distribution μ_+ .*

Proof. We adapt the proof of [JMP26, Lemma 6.4] to our setting by verifying the detailed balance equation for $P_{F,a}$ with respect to w :

$$\forall X, Y \in \mathcal{S}(F, a), \quad w(X)P_{F,a}(X, Y) = w(Y)P_{F,a}(Y, X) \quad (18)$$

by induction on the cardinality of F .

- **Direct, singleton and pairing transitions.** There are three kinds of transitions:

$$\begin{aligned} \mathbf{l}(\sigma) &\longleftrightarrow \mathbf{O}(\sigma), \\ \mathbf{l}(\sigma) &\longleftrightarrow \mathbf{T}_{\{a\}}(\sigma), \\ \mathbf{T}_S(\sigma) &\longleftrightarrow \mathbf{T}_{\iota(S)}(\sigma). \end{aligned}$$

In fact, by Definition 4.6, when the transition $\mathbf{l}(\sigma) \rightarrow \mathbf{O}(\sigma)$ is legal, σ satisfies all of F and thus $\mathbf{O}(\sigma)$ can move to $\mathbf{l}(\sigma)$ as well. Otherwise, the second transition occurs under the rule **F**. Under the rule **B**, $\mathbf{T}_{\{a\}}(\sigma)$ can move to $\mathbf{l}(\sigma)$.

The third transition occurs under the rule **B** and by Lemma 4.3, the reverse transition occurs.

By direct calculation, (18) holds for all of three kinds of transitions.

- **Resampling transitions.** Fix a corridor label $e = (S, u)$ and write the current assignment as $\sigma = (y, u)$ where y is on $\mathcal{V} \setminus \mathcal{V}_S$ and u is on $\mathcal{V}_S$. The forward rule **F** resamples the coordinates in $\mathcal{V}_S$, produces $z \sim \nu_{\mathcal{V}_S}$ and forms the transition

$$\mathbf{T}_S(y, u) \rightarrow \mathbf{L}_0^e(y, z)$$

with transition probability

$$P_{F,a}(\mathbf{T}_S(y, u), \mathbf{L}_0^e(y, z)) = \frac{\nu_{\mathcal{V}_S}(z)}{6}.$$

The reverse transition restores the saved value u , so

$$P_{F,a}(\mathbf{L}_0^e(y, z), \mathbf{T}_S(y, u)) = \frac{1}{6}.$$

Their corresponding weights are $w(\mathbf{T}_S(y, u)) = \nu(y, u)$ and $w(\mathbf{L}_0^e(y, z)) = q_e \nu(y, z)$. Since ν is a product measure and $q_e = \nu_{\mathcal{V}_S}(u)$, it holds that

$$\nu(y, u) \nu_{\mathcal{V}_S}(z) = \nu_{\mathcal{V} \setminus \mathcal{V}_S}(y) \nu_{\mathcal{V}_S}(u) \nu_{\mathcal{V}_S}(z) = q_e \nu(y, z).$$

Therefore (18) holds for all resampling transitions.

- **Random-list reset and return transitions.** Fix a corridor label $e = (S, u)$ and consider

$$\mathbf{l}(\sigma) \longleftrightarrow \mathbf{L}_{\ell(S)}^e(\sigma).$$

Recall the definition of ω_e . The reset rule **R** applied to the state $\mathbf{l}(\sigma)$ gives that

$$P_{F,a}(\mathbf{l}(\sigma), \mathbf{L}_{\ell(S)}^e(\sigma)) = \frac{\omega_e}{6}.$$

For the reverse direction, the forward rule **F** generates a fresh list $\mathbf{L}^e$ and choose a uniform entry. When the chosen entry is exactly e , **F** makes a move from $\mathbf{L}_{\ell(S)}^e(\sigma)$ to $\mathbf{l}(\sigma)$. Then the transition probability is

$$P_{F,a}(\mathbf{L}_{\ell(S)}^e(\sigma), \mathbf{l}(\sigma)) = \frac{1}{6} \cdot \mathbb{E} [|\mathbf{L}|^{-1} \mid e \in \mathbf{L}] = \frac{\mathbb{E} [\mathbb{1}[e \in \mathbf{L}] \cdot |\mathbf{L}|^{-1}]}{\mathbb{P}(e \in \mathbf{L})} = \frac{q_e}{\omega_e}.$$

Using weights $w(\mathbf{l}(\sigma)) = \nu(\sigma)$ and $w(\mathbf{L}_{\ell(S)}^e(\sigma)) = q_e \nu(\sigma)$, (18) holds for all random-list reset and return transitions.

- **Transitions inside smaller copies.** By induction, since a smaller copy $\mathcal{S}(F_{S,i}, b_{S,i})$ has a smaller cardinality $|F_{S,i}| \leq |F| - 1 < |F|$, it is easy to verify that all detail balanced equations still hold under the mapping $\psi_{e,i}$.

Therefore, we conclude that $P_{F,a}$ is reversible with respect to w . The reversibility of K_β with respect to π can be verified in a similar way and for simplicity of the proof, we omit here.

Conditional on the outer output states, for every assignment $\sigma \in \text{Good}(F)$,

$$\pi(\mathbf{O}(\sigma)) = \frac{\nu(\sigma)}{\beta M}$$

and therefore $\sum_{\tau \in \text{Good}(F)} \pi(\mathbf{O}(\tau)) = Z_F / (\beta M)$. Hence

$$\pi(\mathbf{O}(\sigma) \mid \text{the outer output states}) = \frac{\pi(\mathbf{O}(\sigma))}{\sum_{\tau \in \text{Good}(F)} \pi(\mathbf{O}(\tau))} = \frac{\nu(\sigma)}{Z_F} = \mu_F(\sigma)$$

which is exactly μ_+ . □

Now we bound the distance between π and μ_+ , which is an analogues to [JMP26, Lemma 6.5].

Lemma 4.8. *Assume that (15) and (16) hold. Then it holds that*

$$\sum_{X \in \mathcal{S}(F,a) \setminus \{\mathbf{O}(\sigma) : \sigma \in \text{Good}(F)\}} w(X) \leq \frac{1 + 2K_0}{1 - \kappa} \cdot Z_-.$$

Consequently $M \leq (\beta^{-1} + (1 + 2K_0) / (1 - \kappa)) Z_-$ and therefore

$$\|\pi - \mu_+\|_{\text{TV}} \leq \frac{\beta(1 + 2K_0)(1 + K_0)}{1 - \kappa}.$$

Proof. Fix $F \subseteq \mathcal{F}$ and $a \in F$. We show by induction on $|F|$,

$$\sum_{X \in \mathcal{S}(F, a) \setminus \{\mathbf{O}(\sigma) : \sigma \in \text{Good}(F)\}} w(X) \leq \frac{1 + 2K_0}{1 - \kappa} \cdot Z_-.$$

By Definition 4.5, the counted sets can be partitioned into the following groups:

- (1) The input states $\mathbf{l}$: by definition, the total weight of input states is

$$\sum_{\sigma \in \text{Good}(F^-)} w(\mathbf{l}(\sigma)) = \sum_{\sigma \in \text{Good}(F^-)} \nu(\sigma) = Z_-.$$

- (2) The set states $\mathbf{T}_S$ for every $S \in \text{CS}_F(a)$: for a fixed connected subset $S \in \text{CS}_F(a)$, the set states have assignments $\sigma \in E_S \cap \text{Good}(F_S)$ with weight $\nu(\sigma)$. The total weight of $\mathbf{T}_S$ is

$$\sum_{\sigma \in E_S \cap \text{Good}(F_S)} \nu(\sigma) = \nu(E_S \cap \text{Good}(F_S)) = q_S Z_{F_S}$$

since ν is a product measure.

- (3) The last layer states $\mathbf{L}_{\ell(S)}^e$ for every corridor label e : fix a label $e = (S, u)$. Observe that $F_{S, \ell(S)} = F \setminus \{a\} = F^-$. Then its layer has total weight

$$\sum_{\sigma \in \text{Good}(F^-)} w(\mathbf{L}_{\ell(S)}^e(\sigma)) = \sum_{\sigma \in \text{Good}(F^-)} q_e \nu(\sigma) = q_e Z_-.$$

Summing over all possible corridor labels,

$$\sum_{e=(S, u)} q_e Z_- = Z_- \sum_{u \in E_S} \nu_{\mathcal{V}_S}(u) = q_S Z_-.$$

- (4) The states of each smaller embed copy excluding that copy's output states: copy (e, i) is a scaled copy of the insertion construction for $(F_{S, i}, b_{S, i})$. By induction, its weight outside its own output is at most $(1 + 2K_0) Z_{F_{S, i-1}} / (1 - \kappa)$. Therefore, this copy contributes to the original states at most

$$q_e \frac{(1 + 2K_0)}{1 - \kappa} \cdot Z_{F_{S, i-1}}.$$

Put all of four cases together, and we obtain that

$$\begin{aligned} \sum_{X \in \mathcal{S}(F, a) \setminus \{\mathbf{O}(\sigma) : \sigma \in \text{Good}(F)\}} w(X) &\leq Z_- + \sum_{S \in \text{CS}_F(a)} \left(q_S Z_{F_S} + q_S Z_- + \sum_{e=(S, u)} \sum_{i=1}^{\ell(S)} q_e \frac{1 + 2K_0}{1 - \kappa} Z_{F_{S, i-1}} \right) \\ &\leq Z_- + \sum_{S \in \text{CS}_F(a)} q_S Z_{F_S} \left(2 + \frac{1 + 2K_0}{1 - \kappa} \ell(S) \right). \end{aligned}$$

The last inequality holds since $F_S \subseteq F_{S, i-1} \subseteq F_-$ and therefore $Z_- \leq Z_{F_{S, i-1}} \leq Z_{F_S}$. By the fact $Z_F \leq Z_-$ and $R_S \geq 1$,

$$\frac{q_S Z_{F_S}}{Z_-} \leq \frac{q_S Z_{F_S}}{Z_F} = q_S R_S \leq q_S R_S^2.$$

Therefore, by (15) and (16),

$$\begin{aligned}
\sum_{X \in \mathcal{S}(F,a) \setminus \{O(\sigma) : \sigma \in \text{Good}(F)\}} w(X) &\leq Z_- \left(1 + \sum_{S \in \text{CS}_F(a)} q_S R_S^2 \left(2 + \frac{1+2K_0}{1-\kappa} \ell(S) \right) \right) \\
&\leq Z_- \left(1 + 2K_0 + \kappa \cdot \frac{1+2K_0}{1-\kappa} \right) \\
&= \frac{1+2K_0}{1-\kappa} Z_-.
\end{aligned}$$

To show the upper bound of M , we observe

$$\sum_{\sigma \in \text{Good}(F)} \tilde{w}(O(\sigma)) = \sum_{\sigma \in \text{Good}(F)} \frac{\nu(\sigma)}{\beta} = \beta^{-1} Z_F.$$

Therefore, by the fact $Z_F \leq Z_-$,

$$M \leq \frac{1+2K_0}{1-\kappa} Z_- + \beta^{-1} Z_F \leq \left(\beta^{-1} + \frac{1+2K_0}{1-\kappa} \right) Z_-.$$

By Lemma 4.7, a coupling argument provides that the total variation distance is at most the probability of being outside the outputs. Hence, by the inequality $Z_-/Z_F \leq 1 + K_0$ in Lemma 4.2,

$$\|\pi - \mu_+\|_{\text{TV}} \leq \frac{\beta(1+2K_0)/(1-\kappa)Z_-}{Z_F} \leq \frac{\beta(1+2K_0)(1+K_0)}{1-\kappa}.$$

Thus we conclude the lemma. $\square$

4.3. Flow method. We apply the flow argument in [JMP26] to bound the running time of K_β . For a non-loop oriented transition $X \rightarrow Y$, define its capacity by

$$\text{Cap}(X, Y) \triangleq w(X) P_{F,a}(X, Y) = \tilde{w}(X) K_\beta(X, Y).$$

By detailed balanced equation, $\text{Cap}(X, Y) = \text{Cap}(Y, X)$ and they are independent of β . Recall that a signed flow φ assigns values on all transitions satisfying that $\varphi(X, Y) = -\varphi(Y, X)$ and its divergence $\partial\varphi$ is defined as

$$\partial\varphi(X) \triangleq \sum_Y \varphi(X, Y).$$

The energy of φ is the quantity

$$\mathcal{E}_{\text{flow}}(\varphi) \triangleq \sum_{\{X,Y\}} \frac{(\varphi(X, Y))^2}{\text{Cap}(X, Y)}$$

where each edge X, Y is counted only once.

Lemma 4.9. *Assume that (15), (16) and (17) hold. There is a signed flow φ satisfying that $\partial\varphi = \mu_- - \mu_+$ and its energy is at most*

$$\mathcal{E}_{\text{flow}}(\varphi) \leq \frac{6 + 18K_0 + 6K_2}{(1-\kappa)Z_-}.$$

Proof. We employ the induction on $|F|$. For a corridor label $e = (S, u)$, put $\gamma_e \triangleq (-1)^{|S|-1} R_S q_e$. Assign the following values:

$$\begin{aligned}\varphi(\mathbf{l}(\sigma), \mathbf{O}(\sigma)) &= \frac{\nu(\sigma)}{Z_F}, \\ \varphi(\mathbf{l}(\sigma), \mathbf{T}_{\{a\}}(\sigma)) &= \frac{\nu(\sigma)}{Z_F}, \\ \varphi(\mathbf{T}_S(\sigma), \mathbf{T}_{\iota(S)}(\sigma)) &= (-1)^{|S|} \frac{\nu(\sigma)}{Z_F}, \\ \varphi(\mathbf{T}_S(y, u), \mathbf{L}_0^e(y, z)) &= \gamma_e \mu_{F_S}(y, z), \\ \varphi(\mathbf{L}_{\ell(S)}^e(\sigma), \mathbf{l}(\sigma)) &= \gamma_e \mu_{-}(\sigma), \\ \varphi(\psi_{e,i}(X), \psi_{e,i}(Y)) &= \gamma_e \varphi_{F_{S,i}, b_{S,i}}(X, Y)\end{aligned}$$

where $\varphi_{F_{S,i}, b_{S,i}}$ is the flow from the induction.

The divergence identity holds by a similar argument as [JMP26, Lemma 6.6]. For simplicity, we omit here. To bound the energy, for the direct input-output transitions, their total energy contributes to $\mathcal{E}_{\text{flow}}(\varphi)$ at most

$$\sum_{\sigma \in \text{Good}(F)} \frac{(\nu(\sigma)/Z_F)^2}{\nu(\sigma)/6} = \frac{6}{Z_F}.$$

A singleton or pairing edge incident to $\mathbf{T}_S(\sigma)$ also has capacity $\nu(\sigma)/6$ and therefore energy $6\nu(\sigma)/Z_F^2$. Summing over all set states, they contributes at most

$$\sum_{S \in \text{CS}_F(a)} \frac{6q_S Z_{F_S}}{Z_F^2} = 6 \sum_{S \in \text{CS}_F(a)} \frac{q_S R_S^2}{Z_{F_S}}.$$

For a resampling edge, the capacity is $q_e \nu(y, z)/6$. Then for a fixed corridor label $e = (S, u)$, the total contribution of e is

$$\sum_{(y,z) \in \text{Good}(F_S)} \frac{\gamma_e^2 (\nu(y, z))^2 / Z_{F_S}^2}{q_e \nu(y, z)/6} = \frac{6q_e R_S^2}{Z_{F_S}}$$

The total contribution to $\mathcal{E}_{\text{flow}}(\varphi)$ of all three kinds of edges is at most

$$\frac{6}{Z_F} + 6 \sum_{S \in \text{CS}_F(a)} \frac{q_S R_S^2}{Z_{F_S}} + 6 \sum_{S \in \text{CS}_F(a)} \frac{q_S R_S^2}{Z_{F_S}} \leq \frac{6 + 18K_0}{Z_-}$$

using $Z_{F_S} \geq Z_-$, $\sum_S q_S R_S^2 \leq K_0$ and $Z_-/Z_F \leq 1 + K_0$.

For return edges, its capacity is $\omega_e \nu(\sigma)/6$. Then summing over all σ gives that the total contribution is

$$\sum_{\sigma \in \text{Good}(F^-)} \frac{q_e^2 R_S^2 (\nu(\sigma))^2 / Z_-^2}{\omega_e \nu(\sigma)/6} = \frac{6q_e^2 R_S^2}{\omega_e Z_-}.$$

By the Cauchy-Schwarz inequality,

$$\begin{aligned}q_e^2 &= \left(\mathbb{E} \left[\frac{\mathbb{1}[e \in \mathbf{L}]}{\sqrt{|\mathbf{L}|}} \sqrt{|\mathbf{L}|} \right] \right)^2 \\ &\leq \omega_e \mathbb{E} [\mathbb{1}[e \in \mathbf{L}] \cdot |\mathbf{L}|].\end{aligned}$$

Summing over all retained corridor labels,

$$\begin{aligned} \sum_{e=(S,u)} \frac{q_e^2 R_S^2}{\omega_e} &\leq \mathbb{E} \left[|\mathbf{L}| \cdot \sum_{(S,u) \in \mathbf{L}} R_S^2 \right] \\ &= \sum_{S,T \in \text{CS}_F(a)} q_{S \cup T} R_S^2 \\ &\leq K_2. \end{aligned}$$

Finally, to bound the recursive smaller copies, fix a copy (e, i) . By induction and multiplication factors from $\psi_{e,i}$, the total energy is at most

$$R_S^2 q_e \cdot \frac{6 + 18K_0 + 6K_2}{(1 - \kappa) Z_{F_{S,i-1}}}$$

Since $Z_{F_{S,i-1}} \geq Z_-$, summing over all labels and positions,

$$\begin{aligned} \sum_{e=(S,u)} \sum_{i=1}^{\ell(S)} R_S^2 q_e \cdot \frac{6 + 18K_0 + 6K_2}{(1 - \kappa) Z_{F_{S,i-1}}} &\leq \frac{6 + 18K_0 + 6K_2}{(1 - \kappa) Z_-} \sum_{S \in \text{CS}_F(a)} q_S R_S^2 \ell(S) \\ &\leq \kappa \cdot \frac{6 + 18K_0 + 6K_2}{(1 - \kappa) Z_-}. \end{aligned}$$

Putting all together,

$$Z_- \mathcal{E}_{\text{flow}}(\varphi) \leq 6 + 18K_0 + 6K_2 + \kappa \cdot \frac{6 + 18K_0 + 6K_2}{1 - \kappa} = \frac{6 + 18K_0 + 6K_2}{1 - \kappa}.$$

Thus we conclude the lemma. $\square$

To analyze $\left\| (\mu_- - \mu_+) K_\beta^N \right\|_{\text{TV}}$, the following lemma to bound the convergence rate by the energy of a signed flow generalizes the proof of [JMP26, Proposition 6.1].

Proposition 4.10. *Let K be a finite lazy reversible kernel with positive stationary distribution π . Suppose that a signed flow φ has energy at most A with capacities $\pi(X)K(X, Y)$. Then*

$$\forall N \in \mathbb{N}, \quad \|(\partial\varphi)K^N\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\frac{A}{2N+1}}.$$

In particular, if $\partial\varphi = \mu_- - \mu_+$,

$$\|\mu_- K^N - \mu_+\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\frac{A}{2N+1}} + 2\|\pi - \mu_+\|_{\text{TV}}.$$

Proof. For a test function f , write $\partial\varphi(f) = \sum_X \partial\varphi(X) f(X) = \sum_{\{X,Y\}} \varphi(X, Y) (f(X) - f(Y))$. By the Cauchy-Schwarz inequality,

$$\begin{aligned} |\partial\varphi(f)|^2 &= \left(\sum_{\{X,Y\}} \varphi(X, Y) (f(X) - f(Y)) \right)^2 \\ &= \left(\sum_{\{X,Y\}} \frac{\varphi(X, Y)}{\sqrt{\text{Cap}(X, Y)}} \sqrt{\text{Cap}(X, Y)} (f(X) - f(Y)) \right)^2 \\ &\leq A \sum_{\{X,Y\}} \text{Cap}(X, Y) (f(X) - f(Y))^2. \end{aligned}$$

The Dirichlet form on $g = f$ with respect to K is

$$\begin{aligned}\mathcal{E}_K(f, f) &= \frac{1}{2} \sum_{X, Y} \pi(X) K(X, Y) (f(X) - f(Y))^2 \\ &= \sum_{\{X, Y\}} \text{Cap}(X, Y) (f(X) - f(Y))^2.\end{aligned}$$

It provides that

$$|\partial\varphi(f)|^2 \leq A\mathcal{E}_K(f, f).$$

Applying the argument on $\mathcal{E}_K$ in the proof of [JMP26, Proposition 6.1], we have

$$\mathcal{E}_K(K^N g, K^N g) \leq \frac{1}{2N+1}$$

for all observable functions g with $\|g\|_\infty \leq 1$. Therefore,

$$\begin{aligned}\|(\partial\varphi)K^N\|_{\text{TV}} &= \frac{1}{2} \sup_{g: \|g\|_\infty \leq 1} |(\partial\varphi)K^N g| \\ &\leq \frac{1}{2} \sqrt{\frac{A}{2N+1}}.\end{aligned}$$

When $\partial\varphi = \mu_- - \mu_+$,

$$\begin{aligned}\mu_- K^N - \mu_+ &= (\mu_- - \mu_+) K^N + (\mu_+ K^N - \mu_+) \\ &= (\partial\varphi)K^N + (\mu_+ - \pi) K^N + (\pi - \mu_+)\end{aligned}$$

by stationarity of K with respect to π . Then it gives the bound of total variation distance by the triangle inequality as

$$\|\mu_- K^N - \mu_+\|_{\text{TV}} \leq \|(\partial\varphi)K^N\|_{\text{TV}} + 2\|\mu_+ - \pi\|_{\text{TV}}.$$

Here we use the fact that Markov kernels contract total variation distance. Using the above bound, we conclude the lemma. $\square$

4.4. The insertion routine. Now we provide the insertion routine in our sampling algorithm. Assume the hypothesis of Theorem 4.1. Given a tolerance $\delta \in (0, 1)$, set

$$\beta \triangleq \frac{(1 - \kappa)\delta}{8(1 + K_0)(1 + 2K_0)}, \quad N \triangleq \left\lceil 128 \frac{(1 + K_0)(1 + 2K_0)(6 + 18K_0 + 6K_2)}{(1 - \kappa)^2} \delta^{-3} \right\rceil. \quad (19)$$

By Lemma 4.9 and Proposition 4.10, it holds that

$$\|\mu_- K_\beta^N - \mu_+\|_{\text{TV}} \leq \frac{3\delta}{32} + \frac{\delta}{4} < \frac{\delta}{2}. \quad (20)$$

Let T_{step} bound the expectation cost of each step which will be specified later. We adapt the insertion routine in [JMP26] to our settings as follows.

Definition 4.11 (The insertion routine). The insertion kernel $\widehat{K}_{F,a,\delta}$ is the following procedure.

- (1) On a valid input $\sigma \in \text{Good}(F^-)$, initialize $X = \mathbf{l}(\sigma)$ and simulate N steps of K_β .
- (2) If the running cost exceeds $\lceil 2NT_{\text{step}}/\delta \rceil$, interrupt the simulation and return a failure symbol $\perp$.
- (3) Finally, return τ if the state is $\text{O}(\tau)$ otherwise return $\perp$.

By the Markov inequality,

$$\mathbb{P}_{\mu_-}(\text{the simulation is interrupted}) \leq \frac{NT_{\text{step}}}{2NT_{\text{step}}/\delta} \leq \frac{\delta}{2}.$$

Together with (20),

$$\left\| \mu_F - \widehat{K}_{F,a,\delta} - \mu_F \right\|_{\text{TV}} \leq \delta \quad (21)$$

with convention that μ_F gives zero mass to $\perp$.

Expected one-step cost. Now we bound the expected cost T_{step} of one step as in [JMP26]. A similar argument from [JMP26] provides that the total cost of state updates, pairing, constraint test and resampling is at most polynomial in I . For the operation to generalize a fresh random list, by Lemma 4.4, the expected cost is at most

$$p(I) (1 + \mathbb{E}[|\mathbf{L}|]) \leq p(I) (1 + \sqrt{K_2})$$

where p is a fixed polynomial.

Now we consider the cost to generate a list conditional on containing a particular label. Firstly observe that

$$\mu_- \leq \frac{\tilde{w}}{Z_-} \quad \text{and therefore} \quad \mu_- K_\beta^j \leq \frac{\tilde{w}}{Z_-}. \quad (22)$$

For any embedded copy with parameters (E, b) , suppose that its weight are multiplied by c_{cost} in the full construction. Fresh lists are generated at its input states. Then the total expected cost to generate these fresh lists is at most

$$c_{\text{cost}} Z_{E \setminus \{b\}} p(I) (1 + \sqrt{K_2})$$

The lists conditional on a particular label are generated at the copy's last corridor layers. For a label e , the layer has a combined weight $q_e c_{\text{cost}} Z_{E \setminus \{b\}}$. Consequently, by Lemma 4.4, this kind of cost is at most

$$c_{\text{cost}} Z_{E \setminus \{b\}} p(I) \sum_e q_e \mathbb{E}[1 + |\mathbf{L}| \mid e \in \mathbf{L}] \leq c_{\text{cost}} Z_{E \setminus \{b\}} p(I) (\sqrt{K_2} + K_2).$$

in expectation. Therefore, all list operations associated with this copy take at most

$$c_{\text{cost}} Z_{E \setminus \{b\}} p(I) (1 + 2\sqrt{K_2} + K_2).$$

A transition may pass through several active smaller copies. We bound its list cost by summing over all copies at which a list generation could occur. By Lemma 4.8, the input sets of these copies are disjoint and have a total weight at most

$$\sum_{(E,b)} c_{\text{cost}} Z_{E \setminus \{b\}} \leq \frac{1 + 2K_0}{1 - \kappa} \cdot Z_-.$$

All operations to generalize lists are at a state outside the outer outputs with weight $w = \tilde{w}$. By (22), the total expected cost is bounded by

$$T_{\text{step}} \leq p'(I, 1/\delta) \left(1 + \frac{1 + 2K_0}{1 - \kappa} (1 + 2\sqrt{K_2} + K_2) \right),$$

where p' is an absolute polynomial.

4.5. The sampling algorithm. Here we provide our sampling algorithm in Theorem 4.1.

Proof of Theorem 4.1. To sample from $\mathcal{F}$, order $\mathcal{F} = \{a_1, \dots, a_m\}$ and let $F_i = \{a_1, \dots, a_i\}$. Set $\delta = \varepsilon/(4(m+1))$. We repeatedly try the following procedures until a trial returns an assignment $\sigma \neq \perp$.

- (1) Draw $\sigma \sim \nu$.
- (2) Successively apply $\widehat{K}_{F_i, a_i, \delta}$ for $i = 1, \dots, m$; if $\perp$ is returned, output $\perp$ and abort this trial.
- (3) Return the final assignment with success.

By (21) and the union bound, the failure probability of one trial is at most $m\delta \leq \varepsilon/4$ and every successful assignment lies in $\text{Good}(\mathcal{F})$ and has a desired accuracy. Therefore the sampler terminates almost surely and the expected number of trials is at most $4/3$. Assume that one trial has a bounded expected cost τ with failure probability at most f and set $\mathcal{T}$ as the expected running cost until success. It holds that

$$\mathcal{T} \leq \tau + f\mathcal{T} \implies \mathcal{T} \leq \frac{\tau}{1-f}.$$

It remains to bound τ . By the above argument with the choice of β and N , the expected cost of one insertion is at most

$$\text{poly}(I, \delta^{-1}, K_0, K_2, (1-\kappa)^{-1})$$

with an absolute degree. By $\delta = \varepsilon/(4(m+1))$ and the fact that there are at most m insertions in a trial,

$$\tau \leq m \cdot \text{poly}(I, m, \varepsilon^{-1}, K_0, K_2, (1-\kappa)^{-1})$$

which is still polynomial in $I, \varepsilon^{-1}, K_0, K_2$ and $(1-\kappa)^{-1}$. Putting everything together, the sampler outputs an satisfying assignment of $\mathcal{F}$ within TV-distance ε to $\mu_{\mathcal{F}}$ with expected running time $\text{poly}(I, \varepsilon^{-1}, K_0, K_2, (1-\kappa)^{-1})$. $\square$

5. FAST SAMPLING FOR GOOD VARIABLES

In this section, we verify the hypothesis of Theorem 4.1 for efficiently sampling good variables of a random k -SAT formula. Recall the setting of parameters in Section 3:

$$p_* = 2^{-k+12}, \quad C_* = 10^4, \quad x = ep_*.$$

Furthermore, put

$$\tau = \ln \frac{1}{C_* p_* \overline{D}}, \quad \rho = \frac{4}{C_* \cdot \overline{D}}.$$

Throughout the remaining part of this paper, conditioned on the high-probability structural event established in Section 3, we use the following deterministic properties of the fixed formula Φ :

- (1) For every nonempty $J \subseteq [s]$, the variable set $W_J \triangleq \bigcup_{i \in J} B_i$ satisfies

$$|W_J| \leq \frac{k}{2} |W_J \cap \mathcal{V}^{\text{high}}|, \quad \forall v \in \mathcal{V}^{\text{high}}, \quad \deg(v) \geq \Delta. \quad (23)$$

- (2) Every connected variable set U in the original primal graph $\mathcal{G}_{\Phi}$ satisfies

$$I(U) \leq C_{\text{INC}} |U| + 4 \ln n. \quad (24)$$

- (3) The residual dependency graph $\mathcal{H}_{\Psi}$ satisfies

$$\forall a \in \mathcal{C}^{\text{good}}, \quad |N_{\mathcal{H}_{\Psi}}[a]| \leq \overline{D}. \quad (25)$$

Let $\text{CS}_\Psi(a)$ denote the nonempty connected sets in $\mathcal{H}_\Psi$ containing a .

For $k \geq 2^{20}$ and $m/n \leq 2^k/k^{16}$, the parameters p_* and $\bar{D}$ satisfy (12) and therefore Lemma 3.9 holds. Also direct calculations give that $\tau \geq \ln 2$, $x < 1/2$ and $x\bar{D} \leq e/(2C_*)$.

We need the following fact to bound the number of connected sets with given size.

Lemma 5.1 ([BCKL13, Lemma 2.1]). *In a graph of maximum degree at most Δ , the number of connected sets of size ℓ containing a certain vertex is at most $(e\Delta)^{\ell-1}$.*

We view the generation of Φ as the following three-stage process.

- **The structure generation:** sample all variable-occurrences without signs and separate variables into the bad core part $\mathcal{V}^{\text{bad}}$ and the good part $\mathcal{V}^{\text{good}}$.
- **Bad-variable signs:** sample bad-variable occurrence signs independently at random.
- **Good-variable signs:** sample good-variable occurrence signs independently at random.

We remark here that the dependency graph $\mathcal{H}_\Phi$ and the primal graph $\mathcal{G}_\Phi$ depend only on the structure generation.

Proposition 5.2. *Fix $\zeta \in (0, 1)$ such that $L \leq r = \lfloor q_0 n \rfloor$ for $L = \lceil \log_2(32m/\zeta) \rceil$. Conditional on the structure generation and every choice of bad-variable signs, with probability at least $1 - \zeta$ over the good-variable signs,*

$$\sup_{a \in \mathcal{C}^{\text{good}}} \sum_{\gamma \in \text{CS}_\Psi(a)} q_\gamma e^{(1+\tau)|\gamma|} \leq \rho. \quad (26)$$

Proof. Write

$$A_a \triangleq \sum_{\gamma \in \text{CS}_\Psi(a), |\gamma| \leq L} q_\gamma e^{(1+\tau)|\gamma|}, \quad T_a \triangleq \sum_{\gamma \in \text{CS}_\Psi(a), |\gamma| > L} q_\gamma e^{(1+\tau)|\gamma|}.$$

Put $z = e^2/C_* < 10^{-3}$. By Lemma 3.1 and Proposition 3.8, for every $S \subseteq \mathcal{C}^{\text{good}}$ with $|S| \leq L$, the violating probability is at most

$$q_S \leq 2^{-(k-12)|S|} = p_*^{|S|}.$$

Then with Lemma 5.1, the contribution from $|\gamma| \leq L$ is at most

$$A_a \leq \sum_{\ell \geq 1} (e\bar{D})^{\ell-1} p_*^\ell e^{(1+\tau)\ell} = \frac{e}{C_* \cdot \bar{D}(1-z)}.$$

For the tail T_a with $|\gamma| > L$, note that for every $S \subseteq \mathcal{C}^{\text{good}}$,

$$\mathbb{E}_{\text{good signs}} [q_S] \leq 2^{-\sum_{c \in S} |\text{vbl}^{\text{good}}(c)|} \leq p_*^{|S|}.$$

Therefore

$$\mathbb{E}[T_a] \leq \sum_{\ell=L+1}^{\infty} (e\bar{D})^{\ell-1} p_*^\ell e^{(1+\tau)\ell} = \frac{ez^L}{C_* \cdot \bar{D}(1-z)}.$$

The Markov inequality gives that

$$\mathbb{P}\left(\max_{a \in \mathcal{C}^{\text{good}}} T_a > \frac{1}{C_* \cdot \bar{D}}\right) \leq 3mz^L < \zeta.$$

Consequently, with probability at least $1 - \zeta$, for every $a \in \mathcal{C}^{\text{good}}$,

$$A_a + T_a \leq \frac{e}{C_* \cdot \bar{D}(1-z)} + \frac{1}{C_* \cdot \bar{D}} \leq \rho,$$

which concludes the proposition. $\square$

The verification of the moments provides an efficient sampler for the good formula Ψ by Theorem 4.1.

Corollary 5.3. *On the structural events and (26), there is a callable procedure $\text{GoodSampler}(F, \delta)$ for every $F \subseteq \mathcal{C}^{\text{good}}$ and $\delta \in (0, 1)$ that returns a satisfying assignment of $\Psi[F]$ with total variation error at most δ . Its expected running cost is bounded by a polynomial of universal degree in the input size I and δ^{-1} .*

Proof. For every $F \subseteq \mathcal{C}^{\text{good}}$ and $a \in F$, for convenience, we use the notation of Theorem 4.1. By Lemma 3.9,

$$R_S^2 \leq (1-x)^{-2|N_F[S]|} \leq e^{b_0|S|}, \quad \ell(S) \leq \bar{D}|S|$$

where $b_0 = -2\bar{D} \ln(1-x) \leq 2e/C_*$. On the event (26),

$$\sum_S q_S R_S^2 \leq \rho, \quad \sum_S q_S R_S^2 \ell(S) \leq \bar{D}\rho = \frac{4}{C_*}.$$

For the pairing moment, $S \cup T$ is connected since S and T are connected subsets containing a . A fixed union U has at most $3^{|U|-1}$ ordered representations by such a pair: each clause label in $U \setminus \{a\}$ can belong to only S , only T and both of them. Therefore, by (26),

$$\sum_{S,T} q_{S \cup T} R_S^2 \leq \sum_{\text{connected } U \ni a} 3^{|U|-1} e^{b_0|U|} q_U \leq \frac{\rho}{3}.$$

Consequently, by the choice $K_0 = K_2 = 1$ and $\kappa = 1/4$, the hypothesis of Theorem 4.1 holds. Thus there is a sampler $\text{GoodSampler}(F, \delta)$ on every input $F \subseteq \mathcal{C}^{\text{good}}$ and $\delta \in (0, 1)$ such that its output law is within TV-distance at most δ of the uniform distribution over all satisfying assignments of $\Psi[F]$ and its expected running cost is at most $\text{poly}(I, \delta^{-1})$ with an absolute polynomial degree. $\square$

6. RAPID MIXING OF HEAT-BATH DYNAMICS FOR THE BAD CORE

We prove a polynomial spectral gap for the heat-bath dynamics on the bad core. We return to the notation of the structural section: $B_1, \dots, B_s$ are the connected components of $\mathcal{G}_\Phi[\mathcal{V}^{\text{bad}}]$, Ω_i is the set of assignments to B_i satisfying its core clauses, and $\Omega = \prod_{i=1}^s \Omega_i$. A coordinate of Ω is an assignment to an entire component B_i .

Write μ^{bad} for the marginal of μ_Φ on $\mathcal{V}^{\text{bad}}$. We assume the core feasibility and extension condition

$$\forall i \in [s], \quad \Omega_i \neq \emptyset, \quad \forall \sigma \in \Omega, \quad \exists \eta \in \{0, 1\}^{\mathcal{V}} \text{ such that } \eta|_{\mathcal{V}^{\text{bad}}} = \sigma \text{ and } \eta \models \Phi. \quad (27)$$

Thus μ^{bad} is strictly positive on Ω . Condition (27) is established in the structural section, with n sufficiently large for its cutoff conditions.

For a random core configuration $X = (X_1, \dots, X_s)$ and a fixed configuration $\sigma = (\sigma_1, \dots, \sigma_s) \in \Omega$, each coordinate $\sigma_i = \sigma|_{B_i} \in \Omega_i$ denotes the assignment to the entire core component B_i . For $j \in [s]$, define

$$X_{-j} = (X_i)_{i \in [s] \setminus \{j\}}, \quad \sigma_{-j} = (\sigma_i)_{i \in [s] \setminus \{j\}}.$$

Thus the event $X_{-j} = \sigma_{-j}$ means that $X_i = \sigma_i$ for every $i \neq j$, or equivalently that the assignments on $\mathcal{V}^{\text{bad}} \setminus B_j$ agree with σ . For $A \subseteq [s]$, write $\sigma_A = (\sigma_i)_{i \in A}$ and $\sigma_{-A} = \sigma|_{[s] \setminus A}$, and use the same convention for random configurations.

Let $\mathcal{K}$ denote the transition kernel of the random-scan block heat-bath chain on Ω . From a configuration $\sigma \in \Omega$, the chain chooses $j \in [s]$ uniformly and replaces the assignment on B_j by a sample from $\mu^{\text{bad}}(\cdot | X_{-j} = \sigma_{-j})$, leaving all other core assignments unchanged. For a probability measure π on Ω and a real-valued function $f : \Omega \rightarrow \mathbb{R}$, let $X \sim \pi$ and set

$$V_j^\pi(f) = \mathbb{E}_\pi[\text{Var}_\pi(f(X) | X_{-j})].$$

For $s \geq 1$, the Dirichlet form of $\mathcal{K}$ with respect to μ^{bad} , following the definition in Section 2, is

$$\mathcal{E}_{\mathcal{K}}(f, f) = \frac{1}{2} \sum_{\sigma, \omega \in \Omega} \mu^{\text{bad}}(\sigma) \mathcal{K}(\sigma, \omega) (f(\omega) - f(\sigma))^2.$$

Viewing each block B_j as one coordinate, the standard heat-bath Dirichlet-form identity (see [CLV23, Definition 3.3(a)]) gives

$$\mathcal{E}_{\mathcal{K}}(f, f) = \frac{1}{s} \sum_{j=1}^s V_j^{\mu^{\text{bad}}}(f). \quad (28)$$

Each block update contributes its expected conditional variance, and the factor $1/s$ comes from the uniform choice of the updated block. To prove the spectral-gap bound $\text{gap}(\mathcal{K}) \geq 1/(sn)$ for the heat-bath chain, it therefore suffices to establish

$$\text{Var}_{\mu^{\text{bad}}}(f) \leq n \sum_{j=1}^s V_j^{\mu^{\text{bad}}}(f) \quad (29)$$

for every $f : \Omega \rightarrow \mathbb{R}$. Indeed, by (28), this is the Poincaré inequality $\text{Var}_{\mu^{\text{bad}}}(f) \leq sn \mathcal{E}_{\mathcal{K}}(f, f)$. Since each block heat-bath update satisfies detailed balance with respect to μ^{bad} , their average $\mathcal{K}$ is reversible with respect to this measure. For $|\Omega| > 1$, recall the Rayleigh–Rits variational characterization of the spectral in Section 2:

$$\text{gap}(\mathcal{K}) \triangleq 1 - \lambda_2(\mathcal{K}) = \inf_{\substack{f: \Omega \rightarrow \mathbb{R} \\ \text{Var}_{\mu^{\text{bad}}}(f) > 0}} \frac{\mathcal{E}_{\mathcal{K}}(f, f)}{\text{Var}_{\mu^{\text{bad}}}(f)},$$

Thus, if (29) holds for every f , then

$$\frac{\mathcal{E}_{\mathcal{K}}(f, f)}{\text{Var}_{\mu^{\text{bad}}}(f)} \geq \frac{1}{sn} \quad \text{whenever } \text{Var}_{\mu^{\text{bad}}}(f) > 0.$$

Taking the infimum over all such functions proves $\text{gap}(\mathcal{K}) \geq 1/(sn)$. If $|\Omega| = 1$, meaning that there is only one feasible core configuration, every function on Ω is constant, so the variance inequality holds automatically; we adopt the convention $\text{gap}(\mathcal{K}) = 1$.

6.1. The core marginal and its polymer representation. We use the parameters of the structural section:

$$\lambda = 1 + \alpha, \quad \Delta = 6k^7\lambda, \quad C_{\text{INC}} = 100k^2\lambda, \quad \overline{D} = 6k^8\lambda + 1, \quad \rho = \frac{4}{10^4\overline{D}}. \quad (30)$$

Like Section 5, we assume the structural event established in Section 3 and the deterministic properties of Φ . Throughout this section, polymer connectivity is in $\mathcal{H}_{\Psi}$, whose edges correspond to shared *good* variables, not in the original dependency graph $\mathcal{H}_{\Phi}$. Recall that $\text{CS}_{\Psi}(a)$ denote the nonempty connected sets in $\mathcal{H}_{\Psi}$ containing a . Furthermore, we assume the following *polymer-root estimate*:

$$\exists \tau \geq \ln 2 \text{ such that } \forall a \in \mathcal{C}^{\text{good}}, \quad \sum_{S \in \text{CS}_{\Psi}(a)} q_S \exp((1 + \tau)|S|) \leq \rho. \quad (31)$$

By Proposition 5.2, taking the failure parameter there to be n^{-2} shows that (31) holds with probability at least $1 - o_n(1)$. Fix such a value of τ throughout the remainder of this section.

Then we have the following theorem.

Theorem 6.1 (Spectral gap of the core heat bath). *Suppose $n \geq 2$ and $k \geq 30$, with the parameters defined in (30). Assume (27), (23), (24), (25), and (31). If $s \geq 1$, then, for every $f : \Omega \rightarrow \mathbb{R}$,*

$$\text{Var}_{\mu^{\text{bad}}}(f) \leq n \sum_{j=1}^s V_j^{\mu^{\text{bad}}}(f), \quad \text{gap}(\mathcal{K}) \geq \frac{1}{sn} \geq n^{-2}. \quad (32)$$

Consequently, for every $\sigma_0 \in \Omega$ and integer $t \geq 0$,

$$\left\| \mathcal{K}^t(\sigma_0, \cdot) - \mu^{\text{bad}} \right\|_{\text{TV}} \leq \frac{1}{2} 2^{n/2} \exp\left(-\frac{t}{sn}\right). \quad (33)$$

We first derive precisely the interaction expansion needed for the proof. For $\sigma \in \Omega$ put

$$\chi_a(\sigma) = \mathbb{1}[\text{all bad-variable literals of } C_a \text{ are false at } \sigma], \quad D_\sigma = \{a \in \mathcal{C}^{\text{good}} : \chi_a(\sigma) = 1\}.$$

If C_a contains no bad-variable literals, we set $\chi_a(\sigma) = 1$, since its satisfaction depends entirely on the good variables. All bad variables in an original clause lie in one component B_i , so a nonconstant χ_a depends on one core coordinate. The probability of a good-variable extension is

$$Z(\sigma) = \mathbb{E}_{\nu^{\text{good}}} \prod_{a \in \mathcal{C}^{\text{good}}} (1 - \chi_a(\sigma) \mathbf{1}_{E_a}), \quad \mu^{\text{bad}}(\sigma) = \frac{Z(\sigma)}{\sum_{\omega \in \Omega} Z(\omega)}. \quad (34)$$

In particular, this law need not be uniform on Ω .

A polymer is a nonempty set of clause labels $\gamma \subseteq \mathcal{C}^{\text{good}}$ such that the induced graph $\mathcal{H}_\Psi[\gamma]$ is connected; it need not be a maximal connected component of $\mathcal{H}_\Psi$. Two polymers are incompatible if they intersect or an edge of $\mathcal{H}_\Psi$ joins them; a polymer is incompatible with itself. We assign to each polymer the following signed weight, called its *activity*:

$$z_\gamma(\sigma) = (-1)^{|\gamma|} q_\gamma \prod_{a \in \gamma} \chi_a(\sigma). \quad (35)$$

The factor q_γ is the joint probability of the violation events indexed by γ . The product of the $\chi_a(\sigma)$ ensures that this activity is zero if any clause in γ is already satisfied by its bad-variable literals.

For completeness, we derive the polymer representation directly from (34). For fixed σ , expanding the product and using $\mathbb{E}_{\nu^{\text{good}}} \prod_{a \in S} \mathbf{1}_{E_a} = q_S$ gives

$$Z(\sigma) = \sum_{S \subseteq \mathcal{C}^{\text{good}}} (-1)^{|S|} q_S \prod_{a \in S} \chi_a(\sigma). \quad (36)$$

For $S = \emptyset$, we use $q_\emptyset = 1$ and the empty product is one.

For each nonempty S , let $\mathcal{F}(S)$ be the collection of vertex sets of the connected components of the induced graph $\mathcal{H}_\Psi[S]$. These components are polymers and are pairwise compatible. Distinct components share no good variable, since such a variable would give an edge of $\mathcal{H}_\Psi$ between them. Hence their joint violation events are independent under the product measure ν^{good} , and

$$q_S = \prod_{\gamma \in \mathcal{F}(S)} q_\gamma.$$

No independence is assumed among the violation events within a single polymer. Since $\mathcal{F}(S)$ partitions S , the corresponding summand in (36) factors as

$$(-1)^{|S|} q_S \prod_{a \in S} \chi_a(\sigma) = \prod_{\gamma \in \mathcal{F}(S)} \left((-1)^{|\gamma|} q_\gamma \prod_{a \in \gamma} \chi_a(\sigma) \right) = \prod_{\gamma \in \mathcal{F}(S)} z_\gamma(\sigma).$$

Conversely, for any set $\mathcal{F}$ of pairwise compatible polymers, its members are exactly the connected components of $\mathcal{H}_\Psi[\bigcup_{\gamma \in \mathcal{F}} \gamma]$. Thus $S \mapsto \mathcal{F}(S)$, with $\mathcal{F}(\emptyset) = \emptyset$, is a bijection between clause subsets and sets of pairwise compatible polymers. Reindexing (36) yields

$$Z(\sigma) = \sum_{\mathcal{F} \text{ pairwise compatible}} \prod_{\gamma \in \mathcal{F}} z_\gamma(\sigma). \quad (37)$$

The sum includes the empty set of polymers, whose contribution is one.

For an ordered tuple $\Gamma = (\gamma_1, \dots, \gamma_\ell)$ of polymers, not necessarily pairwise compatible and with repetitions allowed, let $I(\Gamma)$ and $\varphi(\Gamma)$ denote its incompatibility graph and Ursell coefficient, respectively, as defined in Section 2. Define

$$U_\Gamma = \bigcup_{i=1}^{\ell} \gamma_i, \quad K_\Gamma = \sum_{i=1}^{\ell} |\gamma_i|, \quad a_\Gamma = \frac{\varphi(\Gamma)}{\ell!} \prod_{i=1}^{\ell} (-1)^{|\gamma_i|} q_{\gamma_i}, \quad Y_\Gamma(\sigma) = \prod_{a \in U_\Gamma} \chi_a(\sigma). \quad (38)$$

We index the cluster-expansion terms $a_\Gamma Y_\Gamma(\sigma)$ by the ordered polymer tuples Γ .

By (38), we have the following lemma.

Lemma 6.2. *If $a_\Gamma \neq 0$, then $\mathcal{H}_\Psi[U_\Gamma]$ is connected.*

Proof. Suppose that $a_\Gamma \neq 0$. By (38), this implies $\varphi(\Gamma) \neq 0$. If the incompatibility graph $I(\Gamma)$ were disconnected, there would be no connected spanning subgraph in the sum defining $\varphi(\Gamma)$ in (4), and hence $\varphi(\Gamma) = 0$, a contradiction. Thus $I(\Gamma)$ is connected.

Each polymer γ_i is a connected set of clause labels in $\mathcal{H}_\Psi$. If i and j are adjacent in $I(\Gamma)$, then γ_i and γ_j are incompatible: they either intersect, or an edge of $\mathcal{H}_\Psi$ joins a clause in γ_i to a clause in γ_j . In either case, $\gamma_i \cup \gamma_j$ is connected in $\mathcal{H}_\Psi$. Since $I(\Gamma)$ is connected, following paths in $I(\Gamma)$ shows that the full union $U_\Gamma = \bigcup_{i=1}^{\ell} \gamma_i$ is connected in $\mathcal{H}_\Psi$. $\square$

The next lemma is the key bridge from the polymer representation of $Z(\sigma)$ to a controlled interaction expansion of the core marginal. It writes $\ln Z(\sigma)$ as a sum of cluster terms $a_\Gamma Y_\Gamma(\sigma)$ and bounds the total contribution of terms touching any fixed clause. This absolute-summability bound will let us regroup the interactions by core components and pass to finite truncations.

Lemma 6.3. *Under (31),*

$$\ln Z(\sigma) = \sum_{\Gamma} a_\Gamma Y_\Gamma(\sigma); \quad \forall a \in \mathcal{C}^{\text{good}}, \quad \sum_{\Gamma: a \in U_\Gamma} |a_\Gamma| e^{\tau K_\Gamma} \leq \rho. \quad (39)$$

Furthermore, we have

$$\sum_{\Gamma} \sup_{\sigma \in \Omega} |a_\Gamma Y_\Gamma(\sigma)| \leq m\rho < \infty.$$

Hence, the series $\sum_{\Gamma} a_\Gamma Y_\Gamma(\sigma)$ converges absolutely and uniformly for $\sigma \in \Omega$.

Proof. Apply Theorem 2.1 with

$$u_\gamma = q_\gamma e^{\tau |\gamma|}, \quad A(\gamma) = |\gamma|.$$

For every polymer γ and every $\sigma \in \Omega$, each $\chi_a(\sigma)$ belongs to $\{0, 1\}$. Hence, by (35),

$$|z_\gamma(\sigma)| e^{\tau |\gamma|} = q_\gamma e^{\tau |\gamma|} \prod_{a \in \gamma} \chi_a(\sigma) \leq q_\gamma e^{\tau |\gamma|} = u_\gamma.$$

Thus the upper bound u_γ is independent of the core configuration σ . Every polymer γ' incompatible with γ contains at least one clause in $N_{\mathcal{H}_\Psi}[\gamma]$. Therefore, using (31), (25) and (30),

$$\sum_{\gamma' \not\sim \gamma} u_{\gamma'} e^{|\gamma'|} = \sum_{\gamma' \not\sim \gamma} q_{\gamma'} e^{(1+\tau)|\gamma'|} \leq \sum_{a \in N_{\mathcal{H}_\Psi}[\gamma]} \sum_{\gamma' \in \text{CS}_\Psi(a)} q_{\gamma'} e^{(1+\tau)|\gamma'|} \leq |N_{\mathcal{H}_\Psi}[\gamma]| \rho \leq \overline{D} \rho |\gamma| \leq |\gamma|.$$

Thus (6) holds.

Applying Theorem 2.1 to the partition function (37), with $w_\gamma = z_\gamma(\sigma)$, gives the absolutely convergent series (5). Thus, for every $\sigma \in \Omega$,

$$\ln Z(\sigma) = \sum_{\ell \geq 1} \frac{1}{\ell!} \sum_{\gamma_1, \dots, \gamma_\ell} \varphi(\gamma_1, \dots, \gamma_\ell) \prod_{i=1}^{\ell} z_{\gamma_i}(\sigma), \quad (40)$$

where the inner sum ranges over all ordered tuples of polymers, with repetitions allowed.

For a tuple $\Gamma = (\gamma_1, \dots, \gamma_\ell)$ and $a \in U_\Gamma$, let $r_a(\Gamma) \triangleq |\{i \in [\ell] : a \in \gamma_i\}| \geq 1$. Since $\chi_a(\sigma) \in \{0, 1\}$, the definition of Y_Γ in (38) gives

$$\prod_{i=1}^{\ell} \prod_{a \in \gamma_i} \chi_a(\sigma) = \prod_{a \in U_\Gamma} \chi_a(\sigma)^{r_a(\Gamma)} = \prod_{a \in U_\Gamma} \chi_a(\sigma) = Y_\Gamma(\sigma).$$

Substituting the activities from (35) and using the definition of a_Γ in (38), we obtain

$$\frac{\varphi(\Gamma)}{\ell!} \prod_{i=1}^{\ell} z_{\gamma_i}(\sigma) = \frac{\varphi(\Gamma)}{\ell!} \left(\prod_{i=1}^{\ell} (-1)^{|\gamma_i|} q_{\gamma_i} \right) \left(\prod_{i=1}^{\ell} \prod_{a \in \gamma_i} \chi_a(\sigma) \right) = a_\Gamma Y_\Gamma(\sigma).$$

Consequently, (40) becomes

$$\ln Z(\sigma) = \sum_{\ell \geq 1} \sum_{\Gamma = (\gamma_1, \dots, \gamma_\ell)} a_\Gamma Y_\Gamma(\sigma) = \sum_{\Gamma} a_\Gamma Y_\Gamma(\sigma),$$

which proves the expansion asserted in (39).

For every fixed polymer γ_0 , the estimate (7), with $A(\gamma_0) = |\gamma_0|$, gives

$$\sum_{\ell \geq 1} \frac{1}{(\ell-1)!} \sum_{\gamma_2, \dots, \gamma_\ell} |\varphi(\gamma_0, \gamma_2, \dots, \gamma_\ell)| \prod_{i=2}^{\ell} u_{\gamma_i} \leq e^{|\gamma_0|}. \quad (41)$$

Fix a clause $a \in \mathcal{C}^{\text{good}}$ and put

$$S_a \triangleq \sum_{\Gamma: a \in U_\Gamma} |a_\Gamma| e^{\tau K_\Gamma}. \quad (42)$$

By (38) and the definition of u_γ , every tuple $\Gamma = (\gamma_1, \dots, \gamma_\ell)$ satisfies

$$|a_\Gamma| e^{\tau K_\Gamma} = \frac{|\varphi(\Gamma)|}{\ell!} \prod_{i=1}^{\ell} u_{\gamma_i}.$$

Also, $a \in U_\Gamma$ if and only if $a \in \gamma_i$ for at least one position i . Thus,

$$\mathbb{1}[a \in U_\Gamma] \leq \sum_{i=1}^{\ell} \mathbb{1}[a \in \gamma_i].$$

The right-hand side counts every position whose polymer contains a , including distinct occurrences of the same polymer. Since all terms below are nonnegative, this gives

$$S_a = \sum_{\ell \geq 1} \frac{1}{\ell!} \sum_{\gamma_1, \dots, \gamma_\ell} \mathbb{1}[a \in U_\Gamma] |\varphi(\Gamma)| \prod_{i=1}^{\ell} u_{\gamma_i} \leq \sum_{\ell \geq 1} \frac{1}{\ell!} \sum_{\gamma_1, \dots, \gamma_\ell} \left(\sum_{i=1}^{\ell} \mathbb{1}[a \in \gamma_i] \right) |\varphi(\Gamma)| \prod_{i=1}^{\ell} u_{\gamma_i}. \quad (43)$$

For $\ell \geq 1$ and $i \in [\ell]$, define the sum associated with position i by

$$T_{\ell, i}(a) \triangleq \sum_{\gamma_1, \dots, \gamma_\ell} \mathbb{1}[a \in \gamma_i] |\varphi(\gamma_1, \dots, \gamma_\ell)| \prod_{j=1}^{\ell} u_{\gamma_j}.$$

Swapping positions i and 1 bijectively maps the tuples with $a \in \gamma_i$ to those with $a \in \gamma_1$, while preserving the Ursell coefficient and the product $\prod_j u_{\gamma_j}$. Hence,

$$T_{\ell,i}(a) = T_{\ell,1}(a) \quad \text{for every } \ell \geq 1 \text{ and } i \in [\ell].$$

Writing the right-hand side of (43) in terms of these sums therefore gives

$$S_a \leq \sum_{\ell \geq 1} \frac{1}{\ell!} \sum_{i=1}^{\ell} T_{\ell,i}(a) = \sum_{\ell \geq 1} \frac{\ell}{\ell!} T_{\ell,1}(a) = \sum_{\ell \geq 1} \frac{1}{(\ell-1)!} T_{\ell,1}(a). \quad (44)$$

In $T_{\ell,1}(a)$, rename the first polymer γ_1 as γ_0 and factor out u_{γ_0} . This yields

$$T_{\ell,1}(a) = \sum_{\gamma_0: a \in \gamma_0} u_{\gamma_0} \sum_{\gamma_2, \dots, \gamma_\ell} |\varphi(\gamma_0, \gamma_2, \dots, \gamma_\ell)| \prod_{j=2}^{\ell} u_{\gamma_j}.$$

Substituting this identity into (44) and interchanging the nonnegative sums over ℓ and γ_0 gives

$$S_a \leq \sum_{\gamma_0: a \in \gamma_0} u_{\gamma_0} \left[\sum_{\ell \geq 1} \frac{1}{(\ell-1)!} \sum_{\gamma_2, \dots, \gamma_\ell} |\varphi(\gamma_0, \gamma_2, \dots, \gamma_\ell)| \prod_{i=2}^{\ell} u_{\gamma_i} \right].$$

Applying (41) to the bracketed sum and then (31) yields

$$S_a \leq \sum_{\gamma_0: a \in \gamma_0} u_{\gamma_0} e^{|\gamma_0|} = \sum_{\gamma_0 \in \text{CS}_\Psi(a)} q_{\gamma_0} e^{(1+\tau)|\gamma_0|} \leq \rho. \quad (45)$$

Since $a \in \mathcal{C}^{\text{good}}$ was arbitrary, this proves $\sum_{\Gamma: a \in U_\Gamma} |a_\Gamma| e^{\tau K_\Gamma} \leq \rho$ for every such a , as asserted in (39).

Finally, we show that the series $\sum_\Gamma a_\Gamma Y_\Gamma(\sigma)$ converges absolutely and uniformly on Ω . Since every polymer is a nonempty subset of $\mathcal{C}^{\text{good}}$, the definition of U_Γ in (38) gives $\emptyset \neq U_\Gamma \subseteq \mathcal{C}^{\text{good}}$ for every tuple. By (42),

$$\begin{aligned} \sum_{a \in \mathcal{C}^{\text{good}}} S_a &= \sum_{a \in \mathcal{C}^{\text{good}}} \sum_{\Gamma: a \in U_\Gamma} |a_\Gamma| e^{\tau K_\Gamma} = \sum_{a \in \mathcal{C}^{\text{good}}} \sum_{\Gamma} \mathbb{1}[a \in U_\Gamma] |a_\Gamma| e^{\tau K_\Gamma} \\ &= \sum_{\Gamma} \left(\sum_{a \in \mathcal{C}^{\text{good}}} \mathbb{1}[a \in U_\Gamma] \right) |a_\Gamma| e^{\tau K_\Gamma} = \sum_{\Gamma} |U_\Gamma| |a_\Gamma| e^{\tau K_\Gamma}. \end{aligned} \quad (46)$$

Since (31) implies $\tau \geq 0$ and (38) gives $K_\Gamma \geq 1$, we have $e^{\tau K_\Gamma} \geq 1$. Using $|U_\Gamma| \geq 1$, (46), and the bound $S_a \leq \rho$ just proved in (45), we obtain

$$\sum_{\Gamma} |a_\Gamma| \leq \sum_{\Gamma} |a_\Gamma| e^{\tau K_\Gamma} \leq \sum_{\Gamma} |U_\Gamma| |a_\Gamma| e^{\tau K_\Gamma} = \sum_{a \in \mathcal{C}^{\text{good}}} S_a \leq |\mathcal{C}^{\text{good}}| \rho \leq m\rho.$$

The last inequality uses $\mathcal{C}^{\text{good}} \subseteq \mathcal{C}$ and $|\mathcal{C}| = m$. By (38), $|Y_\Gamma(\sigma)| \leq 1$ for every $\sigma \in \Omega$. Consequently,

$$\sum_{\Gamma} \sup_{\sigma \in \Omega} |a_\Gamma Y_\Gamma(\sigma)| \leq \sum_{\Gamma} |a_\Gamma| \leq m\rho < \infty.$$

The Weierstrass M -test therefore gives absolute and uniform convergence of $\sum_\Gamma a_\Gamma Y_\Gamma(\sigma)$ on Ω . $\square$

6.2. Variance tensorization for finite-interaction measures. The cluster expansion above represents $\log Z(\sigma)$ as a sum of polymer-cluster interactions. In any finite truncation, terms involving no core component contribute only a constant, terms involving one component are absorbed into its local weight, and terms coupling multiple components become labelled hyperedges. The following lemma gives a variance bound for measures with finitely many such interactions. It controls the global variance by the sum of the coordinatewise conditional variances, with a prefactor determined by the expected sizes and incident interaction strengths of connected components in an associated random hypergraph. We will apply it to finite truncations of the cluster expansion and then pass to the limit.

Lemma 6.4 (Component-swap variance comparison). *Let $s \geq 1$ be an integer, let $\Omega_1, \dots, \Omega_s$ be nonempty finite sets, and put $\Omega = \prod_{i=1}^s \Omega_i$. For each $i \in [s]$, let $\ell_i : \Omega_i \rightarrow (0, \infty)$ be a local weight function. Let $\mathcal{J}$ be a finite set of labelled hyperedges on vertex set $[s]$. Each $e \in \mathcal{J}$ has a nonempty vertex set $V_e \subseteq [s]$; distinct hyperedges may have the same vertex set, but are distinguished by their labels. For each $e \in \mathcal{J}$, fix a real coefficient $a_e \in \mathbb{R}$ and an indicator function $Y_e : \prod_{i \in V_e} \Omega_i \rightarrow \{0, 1\}$. Define the probability measure π on Ω by*

$$\pi(\sigma) = \frac{1}{Z_\pi} \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) \exp \left(\sum_{e \in \mathcal{J}} a_e Y_e(\sigma_{V_e}) \right), \quad \sigma \in \Omega, \quad (47)$$

where Z_π is the normalizing constant. This measure is well defined and strictly positive. For $A \subseteq [s]$, define

$$R(A) = \sum_{\substack{e \in \mathcal{J} \\ V_e \cap A \neq \emptyset}} |a_e|. \quad (48)$$

Let ζ be a random labelled hypergraph on vertex set $[s]$, and write $\zeta_e = \mathbb{1}[e \in \zeta]$ for its edge indicators. These indicators are independent and satisfy

$$\forall e \in \mathcal{J}, \quad \mathbb{P}_\zeta(\zeta_e = 1) = 1 - e^{-2|a_e|}. \quad (49)$$

For $j \in [s]$, let $\mathcal{A}_j$ be the vertex set of the connected component of ζ containing j ; in particular, $\mathcal{A}_j = \{j\}$ when j is isolated. Then, for every $f : \Omega \rightarrow \mathbb{R}$,

$$\text{Var}_\pi(f) \leq \left(\max_{j \in [s]} \mathbb{E}_\zeta \left[2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \right] \right) \sum_{j=1}^s V_j^\pi(f). \quad (50)$$

If $a_e = 0$, then the label e contributes neither to π nor to $R(A)$, and its hyperedge is absent from ζ almost surely because $1 - e^{-2|a_e|} = 0$. We therefore remove all such labels. For every remaining $e \in \mathcal{J}$, define w_e and the function $h_e : \prod_{i \in V_e} \Omega_i \rightarrow \{0, 1\}$ by

$$w_e = e^{|a_e|} - 1, \quad h_e(\sigma_{V_e}) = \begin{cases} Y_e(\sigma_{V_e}), & a_e > 0, \\ 1 - Y_e(\sigma_{V_e}), & a_e < 0, \end{cases} \quad \sigma \in \Omega. \quad (51)$$

Then we have the following lemma.

Lemma 6.5. *For all $\sigma \in \Omega$, we have $\pi(\sigma) \propto (\prod_{i=1}^s \ell_i(\sigma_i)) \prod_{e \in \mathcal{J}} (1 + w_e h_e(\sigma_{V_e}))$.*

Let $\eta = (\eta_e)_{e \in \mathcal{J}} \in \{0, 1\}^\mathcal{J}$ and define

$$\hat{\pi}(\sigma, \eta) \propto \prod_{i=1}^s \ell_i(\sigma_i) \prod_{e: \eta_e=1} w_e h_e(\sigma_{V_e}). \quad (52)$$

Then we have the following result.

Lemma 6.6. *For each $\sigma \in \Omega$,*

$$\sum_{\eta \in \{0,1\}^{\mathcal{J}}} \widehat{\pi}(\sigma, \eta) = \pi(\sigma). \quad (53)$$

Moreover, for each $\sigma \in \Omega$ and $\eta \in \{0,1\}^{\mathcal{J}}$,

$$\widehat{\pi}(\eta \mid X = \sigma) = \prod_{e: \eta_e=1} \frac{w_e h_e(\sigma_{V_e})}{1 + w_e h_e(\sigma_{V_e})} \prod_{e: \eta_e=0} \frac{1}{1 + w_e h_e(\sigma_{V_e})}. \quad (54)$$

Draw independent copies $(X, \eta), (X', \xi) \sim \widehat{\pi}$. Let g be the labelled hypergraph on $[s]$ whose edge e is present exactly when $\eta_e \vee \xi_e = 1$. For $e \in \mathcal{J}$, we call the hyperedge labelled e *open* in g if $\eta_e \vee \xi_e = 1$, and *closed* if $\eta_e = \xi_e = 0$. For a hypergraph H on $[s]$, let $\text{Comp}(H)$ denote the collection of vertex sets of its connected components, including singleton sets for isolated vertices. These sets form a partition of $[s]$. Then we have the following lemma.

Lemma 6.7. *For every $C \in \text{Comp}(g)$, let ε_C be an independent fair bit. Let X^ε be the configuration obtained by swapping exactly those components C for which $\varepsilon_C = 1$, and let ε^C denote the vector obtained by flipping only the bit indexed by C . Then, for every $f: \Omega \rightarrow \mathbb{R}$,*

$$\text{Var}_\pi(f) \leq \frac{1}{2} \mathbb{E} \left[\sum_{C \in \text{Comp}(g)} \mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f(X^{\varepsilon^C}) \right)^2 \right] \right]. \quad (55)$$

The proofs of Lemmas 6.5–6.7 are provided in the appendix. Now we can prove the main result of this subsection.

Proof of Lemma 6.4. For each nonempty $C \subseteq [s]$, define

$$\theta_C = \sup_{\sigma \in \Omega} \mathbb{P}(C \in \text{Comp}(g) \mid X = \sigma), \quad \widetilde{f}_C(\sigma) = f(\sigma) - \mathbb{E}_\pi[f(X) \mid X_{-C} = \sigma_{-C}]. \quad (56)$$

The probability defining θ_C conditions only on the first configuration $X = \sigma$, not on η or the second sample (X', ξ) . The remaining randomness comes from η with its conditional law given $X = \sigma$ and the independent second sample (X', ξ) with law $\widehat{\pi}$. For $C \in \text{Comp}(g)$, passing from ε to ε^C flips only the swap bit for C and leaves the bits for all other components unchanged. Since the components are disjoint, every coordinate outside C still takes its value from the same configuration, X or X' . Thus, pointwise,

$$(X^{\varepsilon^C})_{-C} = (X^\varepsilon)_{-C}.$$

Consequently,

$$\mathbb{E}_\pi[f(X) \mid X_{-C} = (X^\varepsilon)_{-C}] = \mathbb{E}_\pi[f(X) \mid X_{-C} = (X^{\varepsilon^C})_{-C}].$$

Substituting into the definition of $\widetilde{f}_C$ in (56) yields

$$\widetilde{f}_C(X^\varepsilon) - \widetilde{f}_C(X^{\varepsilon^C}) = f(X^\varepsilon) - f(X^{\varepsilon^C}).$$

For a fixed sample pair with $C \in \text{Comp}(g)$, the uniform law of ε is invariant under $\varepsilon \mapsto \varepsilon^C$. Thus

$$\begin{aligned} \mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f(X^{\varepsilon^C}) \right)^2 \right] &= \mathbb{E}_\varepsilon \left[\left(\widetilde{f}_C(X^\varepsilon) - \widetilde{f}_C(X^{\varepsilon^C}) \right)^2 \right] \\ &\leq 2\mathbb{E}_\varepsilon \left[\left(\widetilde{f}_C(X^\varepsilon) \right)^2 + \left(\widetilde{f}_C(X^{\varepsilon^C}) \right)^2 \right] = 4\mathbb{E}_\varepsilon \left[\left(\widetilde{f}_C(X^\varepsilon) \right)^2 \right]. \end{aligned} \quad (57)$$

Hence, (55) gives

$$\text{Var}_\pi(f) \leq 2\mathbb{E}\left[\sum_{C \in \text{Comp}(g)} \mathbb{E}_\varepsilon\left[\left(\tilde{f}_C(X^\varepsilon)\right)^2\right]\right]. \quad (58)$$

Conditional on g , the collection $\text{Comp}(g)$ is fixed. Fix g and ε . For each $\sigma \in \Omega$, by (142),

$$\begin{aligned} \mathbb{P}(X^\varepsilon = \sigma \mid g, \varepsilon) &= \sum_{\omega \in \Omega} \mathbb{P}(X^\varepsilon = \sigma, (X')^\varepsilon = \omega \mid g, \varepsilon) = \sum_{\omega \in \Omega} \mathbb{P}(X = \sigma, X' = \omega \mid g) \\ &= \mathbb{P}(X = \sigma \mid g). \end{aligned}$$

By the law of total probability conditional on g ,

$$\forall \sigma \in \Omega, \quad \mathbb{P}(X^\varepsilon = \sigma \mid g) = \mathbb{P}(X = \sigma \mid g). \quad (59)$$

Condition on g , so that $\text{Comp}(g)$ is fixed. By (58) and (59),

$$\text{Var}_\pi(f) \leq 2\mathbb{E}\left[\sum_{C \in \text{Comp}(g)} \left(\tilde{f}_C(X)\right)^2\right]. \quad (60)$$

Since the marginal law of X is π , the law of total expectation gives

$$\mathbb{E}\left[\sum_{C \in \text{Comp}(g)} \left(\tilde{f}_C(X)\right)^2\right] = \sum_{\emptyset \neq C \subseteq [s]} \mathbb{E}_\pi\left[\mathbb{P}(C \in \text{Comp}(g) \mid X) \left(\tilde{f}_C(X)\right)^2\right]. \quad (61)$$

Substituting (61) into (60) and using the definition of θ_C in (56) yields

$$\text{Var}_\pi(f) \leq 2 \sum_{\emptyset \neq C \subseteq [s]} \theta_C \mathbb{E}_\pi\left[\left(\tilde{f}_C(X)\right)^2\right]. \quad (62)$$

By the definition of $\tilde{f}_C$ in (56) and the definition of conditional variance, for every nonempty $C \subseteq [s]$,

$$\mathbb{E}_\pi\left[\left(\tilde{f}_C(X)\right)^2 \mid X_{-C}\right] = \mathbb{E}_\pi\left[(f(X) - \mathbb{E}_\pi[f(X) \mid X_{-C}])^2 \mid X_{-C}\right] = \text{Var}_\pi(f(X) \mid X_{-C}). \quad (63)$$

The law of total expectation, followed by (63), gives

$$\mathbb{E}_\pi\left[\left(\tilde{f}_C(X)\right)^2\right] = \mathbb{E}_\pi\left[\mathbb{E}_\pi\left[\left(\tilde{f}_C(X)\right)^2 \mid X_{-C}\right]\right] = \mathbb{E}_\pi[\text{Var}_\pi(f(X) \mid X_{-C})]. \quad (64)$$

Substituting (64) into (62) gives

$$\text{Var}_\pi(f) \leq 2 \sum_{\emptyset \neq C \subseteq [s]} \theta_C \mathbb{E}_\pi[\text{Var}_\pi(f(X) \mid X_{-C})]. \quad (65)$$

Fix a nonempty $C \subseteq [s]$ and an outside configuration $y \in \prod_{i \notin C} \Omega_i$. Let π_C^y be the law of X_C given $X_{-C} = y$. For $x_C \in \prod_{i \in C} \Omega_i$, let $f_y(x_C)$ denote the value of f on the unique assignment $\sigma \in \Omega$ satisfying $\sigma_C = x_C$ and $\sigma_{-C} = y$. We write (x_C, y) for this assignment. Define the product reference measure ν_C by

$$\nu_C(x_C) \propto \prod_{i \in C} \ell_i(x_i). \quad (66)$$

For each $x_C \in \prod_{i \in C} \Omega_i$, define

$$q_y(x_C) = \frac{\pi_C^y(x_C)}{\nu_C(x_C)}, \quad L_y = \min_{x_C} q_y(x_C) > 0, \quad U_y = \max_{x_C} q_y(x_C). \quad (67)$$

Since $\nu_C(x_C)$ is strictly positive, q_y is well defined.

By the definition of conditional probability,

$$\pi_C^y(x_C) = \frac{\pi((x_C, y))}{\sum_{z_C \in \prod_{i \in C} \Omega_i} \pi((z_C, y))}. \quad (68)$$

Substitute (47) into the numerator and denominator of (68). The outside local weight $\prod_{i \notin C} \ell_i(y_i)$ has the same value in every term of the denominator and in the numerator, so it cancels from their ratio. Similarly, if $V_e \cap C = \emptyset$, then $Y_e((z_C, y)_{V_e}) = Y_e(y_{V_e})$ for every z_C . Hence the corresponding exponential factor is also common to the numerator and every term of the denominator and cancels. The remaining factors are precisely those that may depend on x_C , giving

$$\pi_C^y(x_C) \propto \left(\prod_{i \in C} \ell_i(x_i) \right) \exp \left(\sum_{\substack{e \in \mathcal{J} \\ V_e \cap C \neq \emptyset}} a_e Y_e((x_C, y)_{V_e}) \right). \quad (69)$$

Here and below, the proportionality constants may depend on C and y but not on x_C . Substituting (69) and (66) into (67) gives

$$q_y(x_C) \propto \exp \left(\sum_{\substack{e \in \mathcal{J} \\ V_e \cap C \neq \emptyset}} a_e Y_e((x_C, y)_{V_e}) \right). \quad (70)$$

Consequently, by (70), for any x_C, x'_C ,

$$\ln \frac{q_y(x_C)}{q_y(x'_C)} = \sum_{\substack{e \in \mathcal{J} \\ V_e \cap C \neq \emptyset}} a_e \left(Y_e((x_C, y)_{V_e}) - Y_e((x'_C, y)_{V_e}) \right) \leq \sum_{\substack{e \in \mathcal{J} \\ V_e \cap C \neq \emptyset}} |a_e| = R(C), \quad (71)$$

where the inequality uses $Y_e \in \{0, 1\}$, and the final equality is (48). Choose

$$x_C^{\max} \in \arg \max_{x_C} q_y(x_C), \quad x_C^{\min} \in \arg \min_{x_C} q_y(x_C).$$

Then (67) and (71) give

$$\ln \frac{U_y}{L_y} = \ln \frac{q_y(x_C^{\max})}{q_y(x_C^{\min})} \leq R(C). \quad (72)$$

Hence,

$$\frac{U_y}{L_y} \leq e^{R(C)}. \quad (73)$$

Let λ be a probability measure on $\prod_{i \in C} \Omega_i$, and let $Z_C = (Z_i)_{i \in C} \sim \lambda$. For $j \in C$ and every $z_{C \setminus \{j\}}$ having positive marginal probability, define the conditional mean

$$\begin{aligned} m_{j,\lambda}(z_{C \setminus \{j\}}) &\triangleq \mathbb{E}_\lambda[f_y(Z_C) \mid Z_{C \setminus \{j\}} = z_{C \setminus \{j\}}] \\ &= \sum_{\substack{u_C \in \prod_{i \in C} \Omega_i \\ u_{C \setminus \{j\}} = z_{C \setminus \{j\}}}} f_y(u_C) \mathbb{P}_\lambda(Z_C = u_C \mid Z_{C \setminus \{j\}} = z_{C \setminus \{j\}}). \end{aligned} \quad (74)$$

Then

$$\begin{aligned} \forall j \in C, \quad V_j^\lambda(f_y) &\triangleq \mathbb{E}_\lambda[\text{Var}_\lambda(f_y(Z_C) \mid Z_{C \setminus \{j\}})] \\ &= \sum_{z_C \in \prod_{i \in C} \Omega_i} \mathbb{P}_\lambda(Z_C = z_C) (f_y(z_C) - m_{j,\lambda}(z_{C \setminus \{j\}}))^2. \end{aligned} \quad (75)$$

Equivalently, minimizing separately for each value of $Z_{C \setminus \{j\}}$ gives the variational representation

$$\forall j \in C, \quad V_j^\lambda(f_y) = \inf_{\psi: \prod_{i \in C \setminus \{j\}} \Omega_i \rightarrow \mathbb{R}} \mathbb{E}_\lambda \left[\left(f_y(Z_C) - \psi(Z_{C \setminus \{j\}}) \right)^2 \right], \quad (76)$$

whose minimizer is $\psi = m_{j,\lambda}$ from (74).

For every function h on $\prod_{i \in C} \Omega_i$, the density $q_y = \pi_C^y / \nu_C$ gives

$$\mathbb{E}_{\pi_C^y}[h(Z_C)] = \sum_{z_C} \pi_C^y(z_C) h(z_C) = \sum_{z_C} \nu_C(z_C) q_y(z_C) h(z_C) = \mathbb{E}_{\nu_C}[q_y(Z_C) h(Z_C)]. \quad (77)$$

Applying this identity with $h = (f_y - c)^2$, and then using the definition of U_y in (67), we obtain

$$\begin{aligned} \text{Var}_{\pi_C^y}(f_y) &= \inf_{c \in \mathbb{R}} \mathbb{E}_{\pi_C^y} \left[(f_y(Z_C) - c)^2 \right] \\ &= \inf_{c \in \mathbb{R}} \mathbb{E}_{\nu_C} \left[q_y(Z_C) (f_y(Z_C) - c)^2 \right] \\ &\leq \inf_{c \in \mathbb{R}} U_y \mathbb{E}_{\nu_C} \left[(f_y(Z_C) - c)^2 \right] \\ &= U_y \text{Var}_{\nu_C}(f_y). \end{aligned} \quad (78)$$

Combining (76) with (77), and the definition of L_y in (67), we obtain

$$\begin{aligned} \forall j \in C, \quad V_j^{\pi_C^y}(f_y) &= \inf_{\psi} \mathbb{E}_{\pi_C^y} \left[\left(f_y(Z_C) - \psi(Z_{C \setminus \{j\}}) \right)^2 \right] \\ &= \inf_{\psi} \mathbb{E}_{\nu_C} \left[q_y(Z_C) \left(f_y(Z_C) - \psi(Z_{C \setminus \{j\}}) \right)^2 \right] \\ &\geq \inf_{\psi} L_y \mathbb{E}_{\nu_C} \left[\left(f_y(Z_C) - \psi(Z_{C \setminus \{j\}}) \right)^2 \right] \\ &= L_y V_j^{\nu_C}(f_y), \end{aligned} \quad (79)$$

where each infimum is over $\psi: \prod_{i \in C \setminus \{j\}} \Omega_i \rightarrow \mathbb{R}$. By (66), the coordinates under ν_C are independent. Applying Lemma 2.3 to f_y under this product measure gives

$$\text{Var}_{\nu_C}(f_y) \leq \sum_{j \in C} V_j^{\nu_C}(f_y). \quad (80)$$

By (78), (80), (79) and (73), we have

$$\text{Var}_{\pi_C^y}(f_y) \leq U_y \text{Var}_{\nu_C}(f_y) \leq U_y \sum_{j \in C} V_j^{\nu_C}(f_y) \leq \frac{U_y}{L_y} \sum_{j \in C} V_j^{\pi_C^y}(f_y) \leq e^{R(C)} \sum_{j \in C} V_j^{\pi_C^y}(f_y). \quad (81)$$

For $j \in C$, the coordinates outside j decompose as

$$X_{-j} = (X_{-C}, X_{C \setminus \{j\}}).$$

Hence, for every y with $\mathbb{P}_\pi(X_{-C} = y) > 0$, the definition of $V_j^{\pi_C^y}(f_y)$ gives

$$V_j^{\pi_C^y}(f_y) = \mathbb{E}_\pi \left[\text{Var}_\pi(f(X) \mid X_{-j}) \mid X_{-C} = y \right].$$

The law of total expectation therefore yields

$$\forall j \in C, \quad \sum_y \mathbb{P}_\pi(X_{-C} = y) V_j^{\pi_C^y}(f_y) = \mathbb{E}_\pi[\text{Var}_\pi(f(X) \mid X_{-j})] = V_j^\pi(f). \quad (82)$$

By (81) and (82),

$$\begin{aligned}\mathbb{E}_\pi[\text{Var}_\pi(f(X) \mid X_{-C})] &= \sum_y \mathbb{P}_\pi(X_{-C} = y) \text{Var}_{\pi_C^y}(f_y) \\ &\leq e^{R(C)} \sum_{j \in C} \sum_y \mathbb{P}_\pi(X_{-C} = y) V_j^{\pi_C^y}(f_y) = e^{R(C)} \sum_{j \in C} V_j^\pi(f).\end{aligned}\tag{83}$$

To bound θ_C defined in (56), put $p_e = 1 - e^{-|a_e|}$. By (54), for every $e \in \mathcal{J}$,

$$\begin{aligned}\mathbb{P}(\eta_e = 1 \mid X = \sigma) &= \sum_{\eta_{-e} \in \{0,1\}^{\mathcal{J} \setminus \{e\}}} \hat{\pi}(\eta_e = 1, \eta_{-e} \mid X = \sigma) \\ &= \frac{w_e h_e(\sigma_{V_e})}{1 + w_e h_e(\sigma_{V_e})} \prod_{d \in \mathcal{J} \setminus \{e\}} \left(\frac{1}{1 + w_d h_d(\sigma_{V_d})} + \frac{w_d h_d(\sigma_{V_d})}{1 + w_d h_d(\sigma_{V_d})} \right) \\ &= \frac{w_e h_e(\sigma_{V_e})}{1 + w_e h_e(\sigma_{V_e})}.\end{aligned}\tag{84}$$

Since $h_e(\sigma_{V_e}) \in \{0, 1\}$, we have

$$\mathbb{P}(\eta_e = 1 \mid X = \sigma) = \frac{w_e h_e(\sigma_{V_e})}{1 + w_e h_e(\sigma_{V_e})} = \begin{cases} 0, & h_e(\sigma_{V_e}) = 0, \\ \frac{w_e}{1 + w_e}, & h_e(\sigma_{V_e}) = 1. \end{cases}\tag{85}$$

Using $w_e = e^{|a_e|} - 1$ from (51) and $p_e = 1 - e^{-|a_e|}$, we have

$$\frac{w_e}{1 + w_e} = \frac{e^{|a_e|} - 1}{e^{|a_e|}} = 1 - e^{-|a_e|} = p_e.\tag{86}$$

Substitution into (85) therefore yields

$$\mathbb{P}(\eta_e = 1 \mid X = \sigma) = p_e h_e(\sigma_{V_e}).\tag{87}$$

Likewise, for every $\omega \in \Omega$,

$$\mathbb{P}(\xi_e = 1 \mid X' = \omega) = p_e h_e(\omega_{V_e}).\tag{88}$$

To compare g with the random hypergraph ζ defined in (49), we place all relevant random objects on a common probability space. In this construction, (X, η) and (X', ξ) are independent, each has distribution $\hat{\pi}$, and the auxiliary random hypergraph has the distribution of ζ .

First draw X, X' independently from π . For every label $e \in \mathcal{J}$, introduce two auxiliary random variables U_e, U'_e , each uniformly distributed on $[0, 1]$. All $2|\mathcal{J}|$ variables are mutually independent and are independent of (X, X') . Define

$$\forall e \in \mathcal{J}, \quad \eta_e = \mathbb{1}[U_e \leq p_e h_e(X_{V_e})], \quad \xi_e = \mathbb{1}[U'_e \leq p_e h_e(X'_{V_e})].\tag{89}$$

Indeed, $0 \leq p_e h_e(\sigma_{V_e}) \leq p_e < 1$, so independence of U_e from X and uniformity of U_e give

$$\mathbb{P}(\eta_e = 1 \mid X = \sigma) = \mathbb{P}(U_e \leq p_e h_e(\sigma_{V_e})) = p_e h_e(\sigma_{V_e}).$$

Thus, (87) is satisfied. Moreover, conditional on $X = \sigma$, each quantity $p_e h_e(\sigma_{V_e})$ in the comparison $U_e \leq p_e h_e(\sigma_{V_e})$ is deterministic. Since the variables $(U_e)_{e \in \mathcal{J}}$ are independent, the bits $(\eta_e)_{e \in \mathcal{J}}$ are therefore conditionally independent. Consequently, for every $\varepsilon \in \{0, 1\}^{\mathcal{J}}$,

$$\mathbb{P}(\eta = \varepsilon \mid X = \sigma) = \prod_{e: \varepsilon_e = 1} p_e h_e(\sigma_{V_e}) \prod_{e: \varepsilon_e = 0} (1 - p_e h_e(\sigma_{V_e})).\tag{90}$$

Since $h_e(\sigma_{V_e}) \in \{0, 1\}$, using (86), the two Bernoulli probabilities can be rewritten as

$$p_e h_e(\sigma_{V_e}) = \frac{w_e h_e(\sigma_{V_e})}{1 + w_e h_e(\sigma_{V_e})}, \quad 1 - p_e h_e(\sigma_{V_e}) = \frac{1}{1 + w_e h_e(\sigma_{V_e})}.$$

Substituting these identities into (90) gives

$$\mathbb{P}(\eta = \varepsilon \mid X = \sigma) = \prod_{e: \varepsilon_e = 1} \frac{w_e h_e(\sigma_{V_e})}{1 + w_e h_e(\sigma_{V_e})} \prod_{e: \varepsilon_e = 0} \frac{1}{1 + w_e h_e(\sigma_{V_e})},$$

which is exactly the conditional distribution in (54).

By (53), the X -marginal of $\hat{\pi}$ is π . Therefore, the conditional probability formula gives, for every $\sigma \in \Omega$ and $\varepsilon \in \{0, 1\}^{\mathcal{J}}$,

$$\hat{\pi}(\sigma, \varepsilon) = \pi(\sigma) \hat{\pi}(\varepsilon \mid X = \sigma). \quad (91)$$

In our construction, $X \sim \pi$, and the conditional law of η given $X = \sigma$ is the same as the conditional law under $\hat{\pi}$. Consequently,

$$\mathbb{P}(X = \sigma, \eta = \varepsilon) = \pi(\sigma) \hat{\pi}(\varepsilon \mid X = \sigma) = \hat{\pi}(\sigma, \varepsilon), \quad (92)$$

where the last equality is (91). The same argument shows that $(X', \xi) \sim \hat{\pi}$. Moreover, $(X, (U_e)_{e \in \mathcal{J}})$ and $(X', (U'_e)_{e \in \mathcal{J}})$ are independent, so (X, η) and (X', ξ) are independent random pairs, each with joint distribution $\hat{\pi}$. Thus they have exactly the distribution assumed when g was defined. Define g from these bits as before, with label e open precisely when $\eta_e \vee \xi_e = 1$.

Next, define

$$\zeta_e = \mathbb{1} [U_e \leq p_e \text{ or } U'_e \leq p_e], \quad e \in \mathcal{J}, \quad (93)$$

and let ζ be the labelled hypergraph that includes e exactly when $\zeta_e = 1$. By independence and uniformity of U_e, U'_e ,

$$\mathbb{P}(\zeta_e = 1) = 1 - \mathbb{P}(U_e > p_e, U'_e > p_e) = 1 - (1 - p_e)^2 = 1 - e^{-2|a_e|}. \quad (94)$$

Different indicators ζ_e depend on disjoint pairs (U_e, U'_e) , so they are mutually independent. They are also independent of (X, X') , since their definitions use only the auxiliary variables. Consequently, this ζ has exactly the independent-hypergraph law specified in the lemma.

Recall $0 \leq p_e h_e(\sigma_{V_e}) \leq p_e < 1$. By (89) and (93),

$$\begin{aligned} \eta_e = 1 &\implies U_e \leq p_e h_e(X_{V_e}) \leq p_e \implies \zeta_e = 1, \\ \xi_e = 1 &\implies U'_e \leq p_e h_e(X'_{V_e}) \leq p_e \implies \zeta_e = 1. \end{aligned}$$

It follows that the set of labelled hyperedges of g is a subset of the set of labelled hyperedges of ζ . Recall that $\mathcal{A}_j$ is the vertex set of the connected component of ζ containing j . Fix a nonempty deterministic set $C \subseteq [s]$ and a vertex $j \in C$. On the event $\{C \in \text{Comp}(g)\}$, the set C is the vertex set of a connected component of g . Hence every vertex of C is connected to j in g , and therefore also in ζ . In other words, $C \in \text{Comp}(g)$ only if $C \subseteq \mathcal{A}_j$. Consequently, for every $\sigma \in \Omega$,

$$\mathbb{P}(C \in \text{Comp}(g) \mid X = \sigma) \leq \mathbb{P}(C \subseteq \mathcal{A}_j \mid X = \sigma) = \mathbb{P}_{\zeta}(C \subseteq \mathcal{A}_j),$$

where the equality uses independence of ζ from X . Using the definition of θ_C in (56), for every $j \in C$,

$$\theta_C = \sup_{\sigma \in \Omega} \mathbb{P}(C \in \text{Comp}(g) \mid X = \sigma) \leq \sup_{\sigma \in \Omega} \mathbb{P}_{\zeta}(C \subseteq \mathcal{A}_j) = \mathbb{P}_{\zeta}(C \subseteq \mathcal{A}_j). \quad (95)$$

Finally, substituting (83) into (65) gives

$$\begin{aligned} \text{Var}_{\pi}(f) &\leq 2 \sum_{\emptyset \neq C \subseteq [s]} \theta_C \mathbb{E}_{\pi}[\text{Var}_{\pi}(f(X) \mid X_{-C})] \leq 2 \sum_{\emptyset \neq C \subseteq [s]} \theta_C e^{R(C)} \sum_{j \in C} V_j^{\pi}(f) \\ &= 2 \sum_{j=1}^s V_j^{\pi}(f) \sum_{\substack{\emptyset \neq C \subseteq [s] \\ j \in C}} \theta_C e^{R(C)} = \sum_{j=1}^s \left(2 \sum_{\substack{\emptyset \neq C \subseteq [s] \\ j \in C}} \theta_C e^{R(C)} \right) V_j^{\pi}(f). \end{aligned} \quad (96)$$

Fix $j \in [s]$. By (95), we have

$$\begin{aligned}
2 \sum_{\substack{\emptyset \neq C \subseteq [s] \\ j \in C}} \theta_C e^{R(C)} &\leq 2 \sum_{\substack{\emptyset \neq C \subseteq [s] \\ j \in C}} \mathbb{P}_\zeta(C \subseteq \mathcal{A}_j) e^{R(C)} = 2 \mathbb{E}_\zeta \left[\sum_{\substack{\emptyset \neq C \subseteq [s] \\ j \in C}} \mathbb{1}[C \subseteq \mathcal{A}_j] e^{R(C)} \right] \\
&= 2 \mathbb{E}_\zeta \left[\sum_{\substack{C \subseteq \mathcal{A}_j \\ j \in C}} e^{R(C)} \right] \leq 2 \mathbb{E}_\zeta \left[2^{|\mathcal{A}_j|-1} e^{R(\mathcal{A}_j)} \right] = \mathbb{E}_\zeta \left[2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \right] \leq \max_{i \in [s]} \mathbb{E}_\zeta \left[2^{|\mathcal{A}_i|} e^{R(\mathcal{A}_i)} \right]. \quad (97)
\end{aligned}$$

Indeed, for each realization of $\mathcal{A}_j$, exactly $2^{|\mathcal{A}_j|-1}$ subsets of $\mathcal{A}_j$ contain j , and the definition of R in (48) implies $R(C) \leq R(\mathcal{A}_j)$ whenever $C \subseteq \mathcal{A}_j$. Substitution into (96) proves (50). $\square$

6.3. The exploration bound and the spectral gap. Recall that $\Gamma = (\gamma_1, \dots, \gamma_\ell)$ is the ordered polymer tuple whose associated data $U_\Gamma, K_\Gamma, a_\Gamma, Y_\Gamma$ are defined in (38). For each tuple Γ , let $\text{vbl}(U_\Gamma)$ denote the set of all variables appearing in the original clauses C_a with $a \in U_\Gamma$, including both good and bad variables. Define its *geometric scope* to be the set of core-component indices touched by these clauses:

$$J_\Gamma = \{i \in [s] : B_i \cap \text{vbl}(U_\Gamma) \neq \emptyset\}. \quad (98)$$

Here $\sigma_{J_\Gamma} = (\sigma_i)_{i \in J_\Gamma}$ denotes the tuple of assignments to the core components indexed by J_Γ . Equivalently, it is the restriction of σ to $\bigcup_{i \in J_\Gamma} B_i$. The value of $Y_\Gamma(\sigma)$ defined in (38) is determined by this restricted assignment. Indeed, each factor $\chi_a(\sigma)$ in $Y_\Gamma(\sigma) = \prod_{a \in U_\Gamma} \chi_a(\sigma)$ depends only on the bad variables in C_a . By (98), all these variables belong to $\bigcup_{i \in J_\Gamma} B_i$. We write $Y_\Gamma(\sigma_{J_\Gamma})$ for that value.

Fix an arbitrary finite truncation set $\Lambda \subseteq \{\Gamma : |J_\Gamma| \geq 2\}$ with $|\Lambda| < \infty$. The finite set Λ indexes the candidate labelled hyperedges on vertex set $[s]$: the candidate with label $\Gamma \in \Lambda$ has vertex set J_Γ . The random hypergraph ζ from Lemma 6.4 is obtained by retaining these candidates independently. Equivalently, we identify ζ with the random subset $\{\Gamma \in \Lambda : \zeta_\Gamma = 1\}$, where ζ_Γ is the indicator that the candidate labelled Γ is retained, with $\mathbb{P}(\zeta_\Gamma = 1) = 1 - e^{-2|a_\Gamma|}$ by (49). Define

$$R(A) = \sum_{\substack{\Gamma \in \Lambda \\ J_\Gamma \cap A \neq \emptyset}} |a_\Gamma|.$$

For any $j \in [s]$, recall that $\mathcal{A}_j$ is the vertex set of the connected component of ζ containing j . The main result of this subsection is the following lemma.

Lemma 6.8. $\max_j \mathbb{E}_\zeta \left[2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \right] \leq n.$

We identify $\mathcal{A}_j$ by an ordinary component search, using good clauses to locate the hyperedges that touch the part already found. Start with the single block index j . Choose a good clause a that contains a variable from one of the blocks found so far and has not been used before. For every label Γ not checked earlier with $a \in U_\Gamma$, check whether $\zeta_\Gamma = 1$. If it is, the hyperedge J_Γ is present in ζ , so add all indices in J_Γ to the set found so far; if $\zeta_\Gamma = 0$, add nothing. Repeat this until every good clause incident to the blocks found so far has been used. Thus the procedure checks only hyperedges that touch the growing component, although in the worst case this may include all labels in Λ .

We use a fixed deterministic order whenever several clauses are available. Let a_t be the clause chosen at the t -th step, $\mathcal{R}_t$ the block indices found before that step, and $\mathcal{E}_t$ the interaction labels checked before that step. Put

$$\mathcal{R}_0 = \{j\}, \quad \mathcal{E}_0 = \emptyset.$$

The labels checked for the first time at step t are

$$\mathcal{D}_t = \{\Gamma \in \Lambda \setminus \mathcal{E}_t : a_t \in U_\Gamma\}. \quad (99)$$

After checking their indicators, set

$$\mathcal{E}_{t+1} = \mathcal{E}_t \cup \mathcal{D}_t, \quad \mathcal{R}_{t+1} = \mathcal{R}_t \cup \bigcup_{\substack{\Gamma \in \mathcal{D}_t \\ \zeta_\Gamma = 1}} J_\Gamma.$$

Let Q be the number of steps before the procedure stops, and define

$$\mathcal{P}_Q \triangleq \{a_0, \dots, a_{Q-1}\}. \quad (100)$$

No clause is chosen twice, so $Q = |\mathcal{P}_Q| \leq |\mathcal{C}^{\text{good}}| \leq m$. Actually, the final set $\mathcal{R}_Q$ is exactly $\mathcal{A}_j$.

Lemma 6.9. $\mathcal{R}_Q = \mathcal{A}_j$.

Recall the definition of K_Γ in (38). For $0 \leq t \leq Q$, define

$$T_t = \sum_{\Gamma \in \mathcal{E}_t} K_\Gamma \zeta_\Gamma, \quad T = T_Q.$$

We have the following bound on T .

Lemma 6.10. $\mathbb{E}e^{T/2-2\rho Q} \leq 1$.

For $A \subseteq [s]$, define the union of the corresponding core blocks by

$$W_A = \bigcup_{i \in A} B_i. \quad (101)$$

For simplicity, let $W = W_{\mathcal{A}_j}$. Recall the definitions of Δ and $I(W)$ in (30) and (2), respectively. Then we have the following lemmas.

Lemma 6.11. For $A \subseteq [s]$, we have $R(A) \leq \rho I(W_A)$.

Lemma 6.12. $|W| \leq \frac{k^2}{2\Delta} I(W)$.

Lemma 6.13. There is a connected variable set in $\mathcal{G}_\Phi$ containing W and having at most $|W| + T$ variables.

The proofs of Lemmas 6.9–6.13 are provided in the appendix. Now we can prove Lemma 6.8.

Proof of Lemma 6.8. Recall that a_t is the previously unused good clause selected at step t in the fixed deterministic order from those incident to the blocks indexed by $\mathcal{R}_t$. For every $t < Q$, its choice gives $\text{vbl}(a_t) \cap W_{\mathcal{R}_t} \neq \emptyset$, because the exploration rule selects a_t only if it contains a variable from a block indexed by $\mathcal{R}_t$, and $W_{\mathcal{R}_t} = \bigcup_{i \in \mathcal{R}_t} B_i$ by (101). Since $\mathcal{R}_t \subseteq \mathcal{R}_Q = \mathcal{A}_j$, we have $W_{\mathcal{R}_t} \subseteq W_{\mathcal{A}_j} = W$. Hence, a_t contains a variable from W . By (100) we have

$$\mathcal{P}_Q \subseteq \left\{ a \in \mathcal{C}^{\text{good}} : \text{vbl}(a) \cap W \neq \emptyset \right\}.$$

Consequently, by the definition of $I(W)$ in (2),

$$Q = |\mathcal{P}_Q| \leq \left| \left\{ a \in \mathcal{C}^{\text{good}} : \text{vbl}(a) \cap W \neq \emptyset \right\} \right| \leq I(W). \quad (102)$$

Moreover, by Lemma 6.13, there is a connected set L in $\mathcal{G}_\Phi$ such that $W \subseteq L$ and $|L| \leq |W| + T$. By $W \subseteq L$ and (2), we have

$$I(W) \leq I(L).$$

By L is a connected set in $\mathcal{G}_\Phi$, $|L| \leq |W| + T$ and (24), we have

$$I(L) \leq C_{\text{INC}}|L| + 4 \ln n \leq C_{\text{INC}}(|W| + T) + 4 \ln n.$$

Hence,

$$I(W) \leq C_{\text{INC}}(|W| + T) + 4 \ln n.$$

By Lemma 6.12, $|W| \leq k^2 I(W)/(2\Delta)$, so

$$I(W) \leq C_{\text{INC}} \left(\frac{k^2}{2\Delta} I(W) + T \right) + 4 \ln n.$$

Moreover, for $k \geq 30$, the parameter definitions in (30) give

$$\frac{C_{\text{INC}} k^2}{2\Delta} = \frac{25}{3k^3} \leq \frac{1}{100}, \quad C_{\text{INC}} \rho = \frac{k^2 \lambda}{25(6k^8 \lambda + 1)} \leq \frac{1}{150k^6} \leq \frac{1}{100}, \quad C_{\text{INC}} \geq 1. \quad (103)$$

Therefore,

$$I(W) \leq \frac{I(W)}{100} + C_{\text{INC}} T + 4 \ln n.$$

Rearranging yields

$$\frac{99}{100} I(W) \leq C_{\text{INC}} T + 4 \ln n,$$

and hence

$$I(W) \leq \frac{100}{99} C_{\text{INC}} T + \frac{400}{99} \ln n \leq 2C_{\text{INC}} T + 8 \ln n. \quad (104)$$

We now expand the pointwise estimate one step at a time. Since the blocks B_i are nonempty and pairwise disjoint, by (101) we have $|\mathcal{A}_j| \leq |W_{\mathcal{A}_j}|$. Thus, $W = W_{\mathcal{A}_j}$ implies $|\mathcal{A}_j| \leq |W|$. Moreover, we have $R(\mathcal{A}_j) \leq \rho I(W)$ by Lemma 6.11, and $Q \leq I(W)$ by (102). Therefore,

$$\begin{aligned} \ln \left(2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \right) + 2\rho Q &= (\ln 2)|\mathcal{A}_j| + R(\mathcal{A}_j) + 2\rho Q \\ &\leq (\ln 2)|W| + \rho I(W) + 2\rho I(W) \\ &\leq |W| + 3\rho I(W). \end{aligned}$$

By Lemma 6.12 and (103),

$$\ln \left(2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \right) + 2\rho Q \leq \left(\frac{k^2}{2\Delta} + 3\rho \right) I(W) \leq \frac{4I(W)}{100C_{\text{INC}}}.$$

By (104) and (103),

$$\ln \left(2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \right) + 2\rho Q \leq \frac{4}{100C_{\text{INC}}} (2C_{\text{INC}} T + 8 \ln n) = \frac{2}{25} T + \frac{8}{25C_{\text{INC}}} \ln n \leq \frac{T}{2} + \ln n. \quad (105)$$

Exponentiating (105) and rearranging gives

$$2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \leq n e^{T/2 - 2\rho Q}.$$

Taking expectation and applying Lemma 6.10 now yields,

$$\max_j \mathbb{E}_\zeta \left[2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \right] \leq n. \quad (106)$$

The lemma is proved. $\square$

6.4. Proof of Theorem 6.1. Now we can prove the main theorem of this section.

Proof of Theorem 6.1. Assume $s \geq 1$. Recall that a polymer γ is a nonempty set of clause labels $\gamma \subseteq \mathcal{C}^{\text{good}}$ whose induced subgraph $\mathcal{H}_\Psi[\gamma]$ is connected. A cluster label Γ is an ordered tuple $\Gamma = (\gamma_1, \dots, \gamma_\ell)$ of polymers, with repetitions allowed and finite length $\ell \geq 1$. Let $\mathfrak{L}$ be the set of all cluster labels, and define

$$\mathfrak{L}_{\geq 2} = \{\Gamma \in \mathfrak{L} : |J_\Gamma| \geq 2\}.$$

The clause-label set $\mathcal{C}^{\text{good}}$ is finite, so it has only finitely many polymers (each polymer is a nonempty subset of $\mathcal{C}^{\text{good}}$). Consequently, for each fixed length ℓ there are only finitely many possible tuples $(\gamma_1, \dots, \gamma_\ell)$. Taking the union over the positive integer lengths shows that the set of all cluster labels, and hence its subfamily $\mathfrak{L}_{\geq 2}$, is countable. Choose an increasing sequence of finite sets $\Lambda_N \subseteq \mathfrak{L}_{\geq 2}$ such that $\bigcup_N \Lambda_N = \mathfrak{L}_{\geq 2}$.

We partition the terms of the expansion of $\ln Z(\sigma)$ in (39), indexed by $\Gamma \in \mathfrak{L}$, according to whether J_Γ is empty, a singleton, or contains at least two indices. The last class is indexed by $\mathfrak{L}_{\geq 2}$. Absolute convergence in Lemma 6.3 justifies this regrouping. If $J_\Gamma = \emptyset$, none of the original clauses indexed by U_Γ contains a bad variable. Thus $\chi_a(\sigma) = 1$ for every $a \in U_\Gamma$, so $Y_\Gamma(\sigma) = 1$ and $a_\Gamma Y_\Gamma(\sigma) = a_\Gamma$. If $J_\Gamma = \{i\}$, we abbreviate $Y_\Gamma(\sigma_{J_\Gamma})$ as $Y_\Gamma(\sigma_i)$ and define the local weight

$$\ell_i(\sigma_i) = \exp \left(\sum_{\Gamma \in \mathfrak{L}: J_\Gamma = \{i\}} a_\Gamma Y_\Gamma(\sigma_i) \right).$$

Absolute summability in Lemma 6.3 ensures that this exponent is finite, so $\ell_i(\sigma_i) > 0$. Exponentiating the regrouped expansion gives

$$Z(\sigma) = \exp \left(\sum_{\Gamma \in \mathfrak{L}: J_\Gamma = \emptyset} a_\Gamma \right) \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) \exp \left(\sum_{\Gamma \in \mathfrak{L}_{\geq 2}} a_\Gamma Y_\Gamma(\sigma_{J_\Gamma}) \right).$$

The first exponential factor is independent of σ and cancels in the ratio $Z(\sigma)/\sum_{\omega \in \Omega} Z(\omega)$ in (34). Hence

$$\mu^{\text{bad}}(\sigma) \propto \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) \exp \left(\sum_{\Gamma \in \mathfrak{L}_{\geq 2}} a_\Gamma Y_\Gamma(\sigma_{J_\Gamma}) \right). \quad (107)$$

Define

$$\pi_N(\sigma) = \frac{1}{Z_N} \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) \exp(H_N(\sigma)), \quad H_N(\sigma) = \sum_{\Gamma \in \Lambda_N} a_\Gamma Y_\Gamma(\sigma_{J_\Gamma}), \quad (108)$$

where Z_N is the sum of the numerator over Ω . Thus, for each N , π_N is a strictly positive probability measure on the same finite state space Ω .

For each N , apply the component-swap comparison Lemma 6.4 to π_N using the identification

$$\mathcal{J} = \Lambda_N, \quad e = \Gamma, \quad V_e = J_\Gamma, \quad a_e = a_\Gamma, \quad Y_e = Y_\Gamma.$$

Under this identification, the oscillation in (48) is exactly the quantity $R(A)$ used in Lemma 6.8, restricted to labels in Λ_N . The edge indicators have the same independent law $\mathbb{P}(\zeta_\Gamma = 1) = 1 - e^{-2|a_\Gamma|}$. Applying Lemma 6.8 to the finite label set Λ_N gives

$$\max_{j \in [s]} \mathbb{E}_\zeta \left[2^{|\mathcal{A}_j|} e^{R(\mathcal{A}_j)} \right] \leq n. \quad (109)$$

Applying Lemma 6.4 with this bound yields, for every $f : \Omega \rightarrow \mathbb{R}$,

$$\text{Var}_{\pi_N}(f) \leq n \sum_{j=1}^s V_j^{\pi_N}(f). \quad (110)$$

We now pass from the finite-truncation measures π_N , for which the Poincaré inequality above follows from Lemma 6.4 and Lemma 6.8, to the target measure μ^{bad} . The convergence argument uses the absolute-summability bound from Lemma 6.3 and the interaction representation (107). Define

$$H(\sigma) \triangleq \sum_{\Gamma \in \mathfrak{L}_{\geq 2}} a_{\Gamma} Y_{\Gamma}(\sigma_{J_{\Gamma}}). \quad (111)$$

The proof of Lemma 6.3 gives $\sum_{\Gamma} |a_{\Gamma}| \leq m\rho$. Since $|Y_{\Gamma}(\sigma_{J_{\Gamma}})| \leq 1$, we therefore have

$$\sum_{\Gamma \in \mathfrak{L}_{\geq 2}} \sup_{\sigma \in \Omega} |a_{\Gamma} Y_{\Gamma}(\sigma_{J_{\Gamma}})| \leq \sum_{\Gamma \in \mathfrak{L}_{\geq 2}} |a_{\Gamma}| \leq \sum_{\Gamma} |a_{\Gamma}| \leq m\rho < \infty. \quad (112)$$

Thus the series defining H is absolutely convergent, and, because $\Lambda_N \subseteq \mathfrak{L}_{\geq 2}$,

$$H(\sigma) - H_N(\sigma) = \sum_{\Gamma \in \mathfrak{L}_{\geq 2} \setminus \Lambda_N} a_{\Gamma} Y_{\Gamma}(\sigma_{J_{\Gamma}}).$$

Applying the triangle inequality and then $|Y_{\Gamma}| \leq 1$ gives

$$\sup_{\sigma \in \Omega} |H_N(\sigma) - H(\sigma)| \leq \sum_{\Gamma \in \mathfrak{L}_{\geq 2} \setminus \Lambda_N} \sup_{\sigma \in \Omega} |a_{\Gamma} Y_{\Gamma}(\sigma_{J_{\Gamma}})| \leq \sum_{\Gamma \in \mathfrak{L}_{\geq 2} \setminus \Lambda_N} |a_{\Gamma}|. \quad (113)$$

For any $\varepsilon > 0$, absolute summability lets us choose a finite $F \subseteq \mathfrak{L}_{\geq 2}$ such that $\sum_{\Gamma \in \mathfrak{L}_{\geq 2} \setminus F} |a_{\Gamma}| < \varepsilon$. Since Λ_N is increasing and $\bigcup_N \Lambda_N = \mathfrak{L}_{\geq 2}$, there is N_0 such that $F \subseteq \Lambda_N$ for all $N \geq N_0$. Hence, for such N ,

$$\sup_{\sigma \in \Omega} |H_N(\sigma) - H(\sigma)| < \varepsilon,$$

which proves $H_N \rightarrow H$ uniformly on Ω . Set

$$\begin{aligned} W_N(\sigma) &\triangleq \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) e^{H_N(\sigma)}, & W(\sigma) &\triangleq \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) e^{H(\sigma)}, \\ Z_N &= \sum_{\omega \in \Omega} W_N(\omega), & \mathcal{Z} &\triangleq \sum_{\omega \in \Omega} W(\omega). \end{aligned}$$

The uniform convergence $H_N \rightarrow H$ implies, by continuity of the exponential, that

$$\forall \sigma \in \Omega, \quad W_N(\sigma) \rightarrow W(\sigma). \quad (114)$$

Because Ω is finite, we have

$$|Z_N - \mathcal{Z}| = \left| \sum_{\omega \in \Omega} (W_N(\omega) - W(\omega)) \right| \leq \sum_{\omega \in \Omega} |W_N(\omega) - W(\omega)| \rightarrow 0. \quad (115)$$

Each $W(\omega) > 0$ since $\ell_i(\omega_i) > 0$ and $H(\omega) \in \mathbb{R}$; as $\Omega \neq \emptyset$, it follows that

$$\mathcal{Z} = \sum_{\omega \in \Omega} W(\omega) > 0.$$

By (107), $\mu^{\text{bad}}(\sigma)$ is proportional to $W(\sigma)$. Since W is positive on the nonempty finite set Ω , normalizing this proportionality gives

$$\mu^{\text{bad}}(\sigma) = \frac{W(\sigma)}{\mathcal{Z}}.$$

Recall that π_N was defined in (108) with weight W_N and normalizer Z_N . By (114) and (115), and since $\mathcal{Z} > 0$, we obtain

$$\forall \sigma \in \Omega, \quad \pi_N(\sigma) = \frac{W_N(\sigma)}{Z_N} \rightarrow \frac{W(\sigma)}{\mathcal{Z}} = \mu^{\text{bad}}(\sigma).$$

For completeness, fix $j \in [s]$ and $y \in \prod_{i \neq j} \Omega_i$. Since $\Omega_j \neq \emptyset$ and $\Omega = \prod_{i=1}^s \Omega_i$, every pair (x, y) with $x \in \Omega_j$ is a state in Ω . By (27), μ^{bad} assigns positive mass to each such state. Therefore

$$\mu^{\text{bad}}(X_{-j} = y) = \sum_{x \in \Omega_j} \mu^{\text{bad}}(X_j = x, X_{-j} = y) > 0.$$

The convergence above then implies

$$\forall x \in \Omega_j, \quad \pi_N(X_j = x \mid X_{-j} = y) \longrightarrow \mu^{\text{bad}}(X_j = x \mid X_{-j} = y).$$

Set $\Omega_{-j} := \prod_{i \neq j} \Omega_i$. For $\pi \in \{\pi_N, \mu^{\text{bad}}\}$, $x \in \Omega_j$, and $y \in \Omega_{-j}$, define

$$p_j^\pi(x, y) \triangleq \pi(X_j = x, X_{-j} = y), \quad p_{-j}^\pi(y) \triangleq \sum_{x \in \Omega_j} p_j^\pi(x, y),$$

and let $f(x, y)$ denote f evaluated at the configuration with j th coordinate x and remaining coordinates y . Then

$$V_j^\pi(f) = \sum_{y \in \Omega_{-j}} \left[\sum_{x \in \Omega_j} p_j^\pi(x, y) f(x, y)^2 - \frac{\left(\sum_{x \in \Omega_j} p_j^\pi(x, y) f(x, y) \right)^2}{p_{-j}^\pi(y)} \right]. \quad (116)$$

The pointwise convergence $\pi_N(\sigma) \rightarrow \mu^{\text{bad}}(\sigma)$ and the positivity established above give

$$p_j^{\pi_N}(x, y) \longrightarrow p_j^{\mu^{\text{bad}}}(x, y), \quad p_{-j}^{\pi_N}(y) \longrightarrow p_{-j}^{\mu^{\text{bad}}}(y) > 0 \quad \text{for all } x \in \Omega_j, y \in \Omega_{-j}.$$

Thus (116) yields

$$V_j^{\pi_N}(f) \longrightarrow V_j^{\mu^{\text{bad}}}(f).$$

Similarly, for any probability measure π on Ω ,

$$\text{Var}_\pi(f) = \sum_{\sigma \in \Omega} \pi(\sigma) f(\sigma)^2 - \left(\sum_{\sigma \in \Omega} \pi(\sigma) f(\sigma) \right)^2.$$

Thus pointwise convergence of the masses also gives

$$\text{Var}_{\pi_N}(f) \longrightarrow \text{Var}_{\mu^{\text{bad}}}(f). \quad (117)$$

Since s is finite and each summand converges as shown above, we may pass the limit through the sum:

$$\lim_{N \rightarrow \infty} \sum_{j=1}^s V_j^{\pi_N}(f) = \sum_{j=1}^s \lim_{N \rightarrow \infty} V_j^{\pi_N}(f) = \sum_{j=1}^s V_j^{\mu^{\text{bad}}}(f). \quad (118)$$

Taking $N \rightarrow \infty$ in (110) and using (117) and (118) yields:

$$\text{Var}_{\mu^{\text{bad}}}(f) = \lim_{N \rightarrow \infty} \text{Var}_{\pi_N}(f) \leq n \lim_{N \rightarrow \infty} \sum_{j=1}^s V_j^{\pi_N}(f) = n \sum_{j=1}^s V_j^{\mu^{\text{bad}}}(f). \quad (119)$$

By the heat-bath Dirichlet-form identity (28),

$$n \sum_{j=1}^s V_j^{\mu^{\text{bad}}}(f) = sn \mathcal{E}_\mathcal{K}(f, f).$$

Thus (119) gives $\text{Var}_{\mu^{\text{bad}}}(f) \leq sn \mathcal{E}_\mathcal{K}(f, f)$. If $|\Omega| > 1$, the Rayleigh–Ritz characterization in Section 2 then yields

$$\text{gap}(\mathcal{K}) = \inf_{\text{Var}_{\mu^{\text{bad}}}(f) > 0} \frac{\mathcal{E}_\mathcal{K}(f, f)}{\text{Var}_{\mu^{\text{bad}}}(f)} \geq \frac{1}{sn}.$$

Each of the s blocks is a nonempty, pairwise disjoint subset of the n variables, so $s \leq n$ and consequently $1/(sn) \geq n^{-2}$. If $|\Omega| = 1$, all functions on Ω are constant, so both sides of the variance inequality (119) vanish. We use the convention $\text{gap}(\mathcal{K}) = 1$.

It remains to prove the total-variation estimate (33) when $|\Omega| > 1$. For each j , let P_j be the conditional-resampling operator, with transition kernel

$$P_j(\sigma, \omega) = \mathbf{1}_{\{\omega_{-j} = \sigma_{-j}\}} \mu^{\text{bad}}(X_j = \omega_j \mid X_{-j} = \sigma_{-j}), \quad \sigma, \omega \in \Omega.$$

View each P_j as an operator on $L^2(\mu^{\text{bad}})$, equipped with the inner product $\langle f, g \rangle_{\mu^{\text{bad}}} = \mathbb{E}_{\mu^{\text{bad}}}[fg]$. For any $g : \Omega \rightarrow \mathbb{R}$, its action is

$$(P_j g)(\sigma) = \sum_{\omega \in \Omega} P_j(\sigma, \omega) g(\omega) = \mathbb{E}_{\mu^{\text{bad}}}[g(X) \mid X_{-j} = \sigma_{-j}].$$

Thus P_j is the orthogonal projection onto the subspace of functions depending only on X_{-j} . In particular, $P_j = P_j^* = P_j^2$, where P_j^* is the adjoint with respect to this inner product; hence P_j is positive semidefinite. The Markov operator corresponding to $\mathcal{K}$ is $P = s^{-1} \sum_{j=1}^s P_j$, so it too is self-adjoint and positive semidefinite. Since μ^{bad} has full support on the product space Ω , each block can be resampled to any of its values with positive probability. Thus $\mathcal{K}$ is irreducible, and the eigenvalue 1 is simple. The gap bound in (32) then implies that every eigenvalue of P on the mean-zero subspace lies in $[0, 1 - 1/(sn)]$. Consequently, for every mean-zero g ,

$$\|P^t g\|_{L^2(\mu^{\text{bad}})} \leq \left(1 - \frac{1}{sn}\right)^t \|g\|_{L^2(\mu^{\text{bad}})}. \quad (120)$$

Fix $\sigma_0 \in \Omega$ and define

$$g_0(\sigma) \triangleq \frac{\mathbf{1}_{\{\sigma_0\}}(\sigma)}{\mu^{\text{bad}}(\sigma_0)} - 1. \quad (121)$$

This is the initial density relative to μ^{bad} , minus its constant part. In particular, (121) gives $\mathbb{E}_{\mu^{\text{bad}}}[g_0] = 1 - 1 = 0$, so the contraction (120) applies to g_0 . For $t \geq 0$, write $h_t(\omega) = \mathcal{K}^t(\sigma_0, \omega) / \mu^{\text{bad}}(\omega)$ for the density at time t . Since $\mathcal{K}^0$ is the identity kernel,

$$h_0(\omega) - 1 = \frac{\mathcal{K}^0(\sigma_0, \omega)}{\mu^{\text{bad}}(\omega)} - 1 = \frac{\mathbf{1}_{\{\omega = \sigma_0\}}}{\mu^{\text{bad}}(\omega)} - 1 = \frac{\mathbf{1}_{\{\omega = \sigma_0\}}}{\mu^{\text{bad}}(\sigma_0)} - 1 = g_0(\omega).$$

The last equality holds because the indicator vanishes unless $\omega = \sigma_0$. Detailed balance gives

$$h_{t+1}(\omega) = \frac{1}{\mu^{\text{bad}}(\omega)} \sum_{\sigma \in \Omega} \mu^{\text{bad}}(\sigma) h_t(\sigma) \mathcal{K}(\sigma, \omega) = \sum_{\sigma \in \Omega} \mathcal{K}(\omega, \sigma) h_t(\sigma) = (Ph_t)(\omega).$$

Since P is a Markov operator, it preserves the constant function 1, so $P1 = 1$. Subtracting this identity from the recurrence above gives

$$h_{t+1} - 1 = Ph_t - P1 = P(h_t - 1).$$

Starting from $h_0 - 1 = g_0$ and iterating this recurrence, we obtain

$$h_t - 1 = P^t(h_0 - 1) = P^t g_0. \quad (122)$$

By (121), $g_0(\sigma_0) = 1/\mu^{\text{bad}}(\sigma_0) - 1$, whereas $g_0(\sigma) = -1$ for every $\sigma \neq \sigma_0$. Therefore,

$$\begin{aligned} \|g_0\|_2^2 &= \sum_{\sigma \in \Omega} \mu^{\text{bad}}(\sigma) g_0(\sigma)^2 = \mu^{\text{bad}}(\sigma_0) \left(\frac{1}{\mu^{\text{bad}}(\sigma_0)} - 1 \right)^2 + \sum_{\sigma \neq \sigma_0} \mu^{\text{bad}}(\sigma) (-1)^2 \\ &= \mu^{\text{bad}}(\sigma_0) \left(\frac{1}{\mu^{\text{bad}}(\sigma_0)} - 1 \right)^2 + 1 - \mu^{\text{bad}}(\sigma_0) = \frac{1}{\mu^{\text{bad}}(\sigma_0)} - 1. \end{aligned} \quad (123)$$

By the total-variation identity (1),

$$\left\| \mathcal{K}^t(\sigma_0, \cdot) - \mu^{\text{bad}} \right\|_{\text{TV}} = \frac{1}{2} \mathbb{E}_{\mu^{\text{bad}}} [|h_t - 1|].$$

Using (122), this becomes

$$\left\| \mathcal{K}^t(\sigma_0, \cdot) - \mu^{\text{bad}} \right\|_{\text{TV}} = \frac{1}{2} \mathbb{E}_{\mu^{\text{bad}}} [|P^t g_0|].$$

Since μ^{bad} is a probability measure, Cauchy–Schwarz gives

$$\frac{1}{2} \mathbb{E}_{\mu^{\text{bad}}} [|P^t g_0|] \leq \frac{1}{2} \|P^t g_0\|_2.$$

Applying the contraction (120) (which applies to g_0 because it is mean-zero by (121)), we get

$$\left\| \mathcal{K}^t(\sigma_0, \cdot) - \mu^{\text{bad}} \right\|_{\text{TV}} \leq \frac{1}{2} \|P^t g_0\|_2 \leq \frac{1}{2} \left(1 - \frac{1}{sn}\right)^t \|g_0\|_2.$$

Finally, substituting the norm calculation (123) yields

$$\left\| \mathcal{K}^t(\sigma_0, \cdot) - \mu^{\text{bad}} \right\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\frac{1}{\mu^{\text{bad}}(\sigma_0)} - 1} \left(1 - \frac{1}{sn}\right)^t. \quad (124)$$

By (27), every σ_0 has at least one satisfying full extension. Since μ^{bad} is the induced marginal described by (34) and there are at most 2^n full assignments in total, it follows that $\mu^{\text{bad}}(\sigma_0) \geq 2^{-n}$. Consequently, the prefactor is bounded as follows:

$$\sqrt{\frac{1}{\mu^{\text{bad}}(\sigma_0)} - 1} \leq 2^{n/2}. \quad (125)$$

Moreover, we also have

$$\left(1 - \frac{1}{sn}\right)^t \leq \exp\left(-\frac{t}{sn}\right). \quad (126)$$

Combining (125) with (126) and substituting both bounds into (124) proves (33). If $|\Omega| = 1$, the total-variation distance is identically zero. The theorem is proved. $\square$

7. PUTTING ALL TOGETHER: DESIGN OF THE COMPLETE SAMPLER

Now we are ready to show Theorem 1.1. The main obstacle is that, the analysis in Section 6 for $\text{gap}(\mathcal{K})$ defines the exact marginal updates while we can only ‘approximately’ simulate the updates of $\mathcal{K}$ in an executable sampler using Corollary 5.3. Here we show how to appropriately construct the sampler, provide the accuracy guarantee and the total running cost, and finally conclude a proof of Theorem 1.1.

Before proceeding, we recall the structural assumptions as in Section 5 and the polymer-root estimate as in Section 6 and assume them unless stated otherwise.

7.1. Approximate marginal heat-bath updates. To approximately simulate marginal updates of the heat-bath dynamics $\mathcal{K}$, we first provide the following proposition to approximately count sub-deletions of good formula Ψ .

Proposition 7.1. *Given $F \subseteq \mathcal{C}^{\text{good}}$, $\varepsilon \in (0, 1/2)$ and $\delta \in (0, 1)$, there is an estimator $\hat{Z}_F$ such that*

$$\mathbb{P}\left((1 - \varepsilon)Z_F \leq \hat{Z}_F \leq (1 + \varepsilon)Z_F\right) \geq 1 - \delta$$

and its expected running cost is at most $P_1(I, \varepsilon^{-1}, \ln(1/\delta))$ where P_1 is a fixed polynomial.

Proof. If $F = \emptyset$, $Z_F = 1$ and we return $\hat{Z}_F = 1$. Otherwise, order $F = \{a_1, \dots, a_\ell\}$ and set

$$\forall j = 0, \dots, \ell, \quad F_j \triangleq \{a_1, \dots, a_j\}.$$

Define

$$q_j \triangleq \mathbb{P}_{\mu_{\Psi[F_{j-1}]}}(a_j \text{ is satisfied}).$$

Therefore $Z_F = \prod_{j=1}^{\ell} q_j$. By Lemma 3.9, $q_j \geq 1 - \varepsilon p_* \geq 1/2$.

Set $h = \varepsilon/(16\ell)$ and

$$N_{\text{sample}} \triangleq \left\lceil 128\ell^2 \varepsilon^{-2} \ln \frac{2\ell}{\delta} \right\rceil.$$

For each j , make N_{sample} independent calls to $\text{GoodSampler}(F_{j-1}, h)$ in Corollary 5.3 and let the returned assignments be $X_{j,1}, \dots, X_{j,N_{\text{sample}}}$. Set

$$\forall t = 1, \dots, N_{\text{sample}}, \quad Y_{j,t} \triangleq \mathbb{1}[X_{j,t} \text{ satisfies } a_j].$$

and $\hat{q}_j \triangleq \sum_{t=1}^{N_{\text{sample}}} Y_{j,t} / N_{\text{sample}}$. Finally output $\hat{Z}_F = \prod_{j=1}^{\ell} \hat{q}_j$.

We first bound the error of $\hat{Z}_F$. For every $j = 1, \dots, \ell$, put

$$\theta_j \triangleq \mathbb{E}[\hat{q}_j].$$

By Corollary 5.3, the law of the output of each call is within h total variation distance of $\mu_{F_{j-1}}$. Therefore $|\theta_j - q_j| \leq h$. The random variables $Y_{j,1}, \dots, Y_{j,N_{\text{sample}}}$ are independent Bernoulli variables with mean θ_j . By the Hoeffding's inequality,

$$\mathbb{P}(|\hat{q}_j - \theta_j| > h) \leq 2e^{-2N_{\text{sample}}h^2} \leq \frac{\delta}{\ell}.$$

Then by the union bound, with probability at least $1 - \delta$, for every $j = 1, \dots, \ell$,

$$|\hat{q}_j - q_j| \leq 2h = \frac{\varepsilon}{8\ell}.$$

Using $q_j \geq 1/2$, we obtain that with probability at least $1 - \delta$,

$$\forall j = 1, \dots, \ell, \quad |\hat{q}_j/q_j - 1| \leq \varepsilon/(4\ell). \quad (127)$$

On the event (127),

$$\left(1 - \frac{\varepsilon}{4\ell}\right)^\ell \leq \frac{\hat{Z}_F}{Z_F} \leq \left(1 + \frac{\varepsilon}{4\ell}\right)^\ell.$$

For every $\varepsilon \in (0, 1)$, $(1 - \varepsilon/(4\ell))^\ell \geq 1 - \varepsilon/4$ and $(1 + \varepsilon/(4\ell))^\ell \leq e^{\varepsilon/4} \leq 1 + \varepsilon$.

To complete the proof, we bound the expected cost of the estimation. There are ℓN_{sample} calls for $\text{GoodSampler}(\cdot, \cdot)$. Then by Corollary 5.3, $\ell \leq \text{poly}(I)$ and the choice of $N_{\text{sample}} = \text{poly}(\ell, \varepsilon^{-1}, \ln(1/\delta))$, the total expected cost of these calls is

$$\ell N_{\text{sample}} \cdot P(I, h^{-1}) \leq P_1(I, \varepsilon^{-1}, \ln(1/\delta))$$

where P and P_1 are fixed polynomials. Computing $\hat{Z}_F$ takes time linear in ℓ . Hence the expected running cost is at most $\text{poly}(I, \varepsilon^{-1}, \ln(1/\delta))$ with a universal polynomial degree. $\square$

With Proposition 7.1, we use the following lemma to simulate an update of $\mathcal{K}$.

Lemma 7.2. *For every component B_i , every feasible assignment σ_{-i} on $\mathcal{V}^{\text{bad}} \setminus B_i$ and $\eta \in (0, 1)$, one can implement an update whose law is within η in total variation distance of $\mu^{\text{bad}}(\cdot \mid \sigma_{-i})$. Every returned core assignment is feasible and the update has a deterministic cost cap bounded by $P_2(I, \eta^{-1})$ for a fixed polynomial P_2 .*

Proof. Fix a component B_i and a feasible assignment σ_{-i} on $\mathcal{V}^{\text{bad}} \setminus B_i$ and $\eta \in (0, 1)$. Since $|B_i| < \log_2 n$ by Lemma 3.5, to generate Ω_i , we list all assignments on B_i and check the satisfiability by Lemma 3.9 in time $2^{|B_i|} \text{poly}(I) = \text{poly}(I)$. Hence Ω_i is enumerable in $\text{poly}(I)$ and $|\Omega_i| \leq 2^{|B_i|} < n$. For every $z \in \Omega_i$, (z, σ_{-i}) is feasible and set $w_z \triangleq Z_{D(z, \sigma_{-i})}$. It is clear that $w_z > 0$ for every assignment $z \in \Omega_i$ and

$$p_z \triangleq \mu^{\text{bad}}(z \mid \sigma_{-i}) = \frac{w_z}{W}$$

where $W \triangleq \sum_{z \in \Omega_i} w_z$.

Now we estimate w_z for every $z \in \Omega_i$. Apply Proposition 7.1 on $D_{(z, \sigma_{-i})}$ with choice of $\varepsilon = \eta/8$ and $\delta = \eta/(4|\Omega_i|)$. We obtain a non-negative estimate $\hat{w}_z$ such that

$$\mathbb{P}\left(\left(1 - \frac{\eta}{8}\right)w_z \leq \hat{w}_z \leq \left(1 + \frac{\eta}{8}\right)w_z\right) \geq 1 - \frac{\eta}{4|\Omega_i|}.$$

Let $\mathcal{E}$ be the event that every estimate satisfies the relative-error bound. By the union bound,

$$\mathbb{P}(\overline{\mathcal{E}}) \leq |\Omega_i|\delta = \frac{\eta}{4}.$$

Set $\widehat{W} \triangleq \sum_{z \in \Omega_i} \hat{w}_z$. Draw $\tau \in \Omega_i$ as follows: if $\widehat{W} = 0$, pick any feasible assignment in Ω_i ; otherwise, sample a candidate according to $\hat{p}_z = \hat{w}_z / \widehat{W}$.

Now we bound the error. On $\mathcal{E}$, by Proposition 7.1, for every $z \in \Omega_i$,

$$\left|\frac{\hat{w}_z}{w_z} - 1\right| \leq \frac{\eta}{8}$$

and

$$\left|\frac{\widehat{W}}{W} - 1\right| = \left|\sum_{z \in \Omega_i} p_z \left(\frac{\hat{w}_z}{w_z} - 1\right)\right| \leq \frac{\eta}{8}.$$

Therefore,

$$\begin{aligned} \left\|\text{Law}(\tau \mid \mathcal{E}) - \mu^{\text{bad}}(\cdot \mid \sigma_{-i})\right\|_{\text{TV}} &= \frac{1}{2} \sum_{z \in \Omega_i} |p_z - \hat{p}_z| \\ &\leq \frac{1}{2(1 - \eta/8)} \sum_{z \in \Omega_i} p_z \cdot \frac{\eta}{4} \\ &\leq \frac{\eta}{4}. \end{aligned}$$

By the convexity of total variation distance,

$$\left\|\text{Law}(\tau) - \mu^{\text{bad}}(\cdot \mid \sigma_{-i})\right\|_{\text{TV}} \leq \mathbb{P}(\mathcal{E}) \frac{\eta}{4} + \mathbb{P}(\overline{\mathcal{E}}) \leq \frac{\eta}{2}.$$

To bound the cost cap, Proposition 7.1 provides that the expect cost of each weight estimate is at most

$$P_1(I, 8/\eta, \ln(4|\Omega_i|/\eta)) = P(I, \eta^{-1})$$

for some fixed polynomial P . Therefore, there exists $H_{I, \eta} = \text{poly}(I, \eta^{-1})$ with a universal polynomial degree such that the total running cost is at most $H_{I, \eta}$. Set the cap budget $C_\eta = \lceil 4H_{I, \eta}/\eta \rceil$. If the budget is exceeded, we stop the update and return any feasible assignment in Ω_i . By the Markov inequality,

$$\mathbb{P}(\text{the budget is exceeded}) \leq \frac{H_{I, \eta}}{C_\eta} \leq \frac{\eta}{4}.$$

Then the capped output law is within $\eta/2 + \eta/4 < \eta$ in total variation distance with $\mu^{\text{bad}}(\cdot \mid \sigma_{-i})$ and the deterministic cap is bounded by $\text{poly}(I, \eta^{-1}) \cdot (12/\eta) = P_2(I, \eta^{-1})$ for some fixed polynomial P_2 . $\square$

7.2. Design and analysis of the main sampler. We present our algorithm as follows.

Definition 7.3 (The main sampler). On input a k -SAT formula Φ with n variables and m clause, and a tolerance error $\varepsilon \in (0, 1/2)$, we define the **FastSampler** (Φ, ε) as:

- (1) If $m = 0$, output n independent fair bits. Otherwise, compute λ and Δ as (9) and generate $\mathcal{V}^{\text{high}}, \mathcal{V}^{\text{bad}}, \mathcal{V}^{\text{good}}$, bad components $B_1, \dots, B_s$ and the good formula Ψ . If $s = 0$, call **GoodSampler** $(\Phi, \varepsilon/8)$ and output the result.
- (2) Check $|B_i| < \log_2 n$ and construct Ω_i for every $i = 1, \dots, s$. If there exists some $\Omega_i = \emptyset$, output $\perp$.
- (3) Initialize any feasible assignment $\sigma_0 \in \prod_{i=1}^s \Omega_i$.
- (4) Set

$$T^{\text{bad}} = n^4(n + \lceil \log_2(8/\varepsilon) \rceil + 1), \quad \eta = \frac{\varepsilon}{8(T^{\text{bad}} + 1)}.$$

For $t = 1, \dots, T^{\text{bad}}$, simulate T^{bad} steps of $\mathcal{K}$: choose i uniformly at random from the core components and use Lemma 7.2 with error budget η to update B_i with the assignment on other components fixed.

- (5) Let σ be the assignment on $\mathcal{V}^{\text{bad}}$ after T^{bad} updates. Call **GoodSampler** $(D_\sigma, \varepsilon/8)$ once to obtain τ on $\mathcal{V}^{\text{good}}$ and output the resulting pair (σ, τ) .

On the structural event and the polymer-root estimate, we provide the following accuracy guarantee and running cost of **GoodSampler** (Φ, ε) .

Proposition 7.4. *On the structural event and the polymer-root estimate, **FastSampler** (Φ, ε) returns a satisfying assignment within $5\varepsilon/16$ in total variation distance with μ_Φ . Its expected running cost is at most $(I/\varepsilon)^{C_2}$ where C_2 is a universal constant.*

Proof. When $s = 0$, the proposition holds directly from Corollary 5.3. Now we consider the case $s \geq 1$.

We firstly provide the running cost. There are $T^{\text{bad}} = \text{poly}(n, \ln(1/\varepsilon))$ calls for the capped subroutine with accuracy $\eta = \text{poly}(I, \varepsilon^{-1})$ in Lemma 7.2 to simulate an update for $\mathcal{K}$ and each simulator has a running time $(I/\varepsilon)^{C_1}$ for some universal constant $C_1 > 0$. For the final call **GoodSampler** $(D_\sigma, \varepsilon/8)$, its expected cost is at most $\text{poly}(I, \varepsilon^{-1})$. Therefore, the total expected running cost is at most $(I/\varepsilon)^{C_1} \cdot \text{poly}(n, \ln(1/\varepsilon)) + \text{poly}(I, \varepsilon^{-1}) = (I/\varepsilon)^{C_2}$ for a universal constant $C_2 > 0$.

Now we check the accuracy. By Theorem 6.1 with the choice of T^{bad} ,

$$\left\| \mathcal{K}^{T^{\text{bad}}}(\sigma_0, \cdot) - \mu^{\text{bad}} \right\|_{\text{TV}} \leq \frac{\varepsilon}{16}.$$

Let P_i be the exact update of the component B_i and $\hat{P}_i$ be its approximate implementation in Lemma 7.2. For every $\sigma \in \Omega_i$, by Lemma 7.2,

$$\left\| P_i(\sigma, \cdot) - \hat{P}_i(\sigma, \cdot) \right\|_{\text{TV}} \leq \eta.$$

Then for the actual implementation core chain $\hat{\mathcal{K}} = \frac{1}{s} \sum_{i=1}^s \hat{P}_i$, it holds that

$$\sup_{\sigma \in \Omega} \left\| \hat{\mathcal{K}}(\sigma, \cdot) - \mathcal{K}(\sigma, \cdot) \right\|_{\text{TV}} \leq \eta.$$

Let σ_t be the actual assignment on $\mathcal{V}^{\text{bad}}$ after t simulating updates. By the triangle inequality of the total variation distance, for every $t = 1, \dots, T^{\text{bad}}$,

$$\begin{aligned} \|\mathcal{K}^t(\sigma_0, \cdot) - \text{Law}(\sigma_t)\|_{\text{TV}} &\leq \|\mathcal{K}^{t-1}(\sigma_0, \cdot)\hat{\mathcal{K}} - \text{Law}(\sigma_{t-1})\hat{\mathcal{K}}\|_{\text{TV}} + \|\mathcal{K}^{t-1}(\sigma_0, \cdot)\hat{\mathcal{K}} - \mathcal{K}^t(\sigma_0, \cdot)\|_{\text{TV}} \\ &\leq \|\mathcal{K}^{t-1}(\sigma_0, \cdot) - \text{Law}(\sigma_{t-1})\|_{\text{TV}} + \eta. \end{aligned}$$

Then by induction,

$$\|\text{Law}(\sigma_{T^{\text{bad}}}) - \mathcal{K}^{T^{\text{bad}}}(\sigma_0, \cdot)\|_{\text{TV}} \leq \eta T^{\text{bad}} \leq \frac{\varepsilon}{8}.$$

By the triangle inequality of total variation distance,

$$\|\text{Law}(\sigma_{T^{\text{bad}}}) - \mu^{\text{bad}}\|_{\text{TV}} \leq \frac{\varepsilon}{8} + \frac{\varepsilon}{16} \leq \frac{3\varepsilon}{16}.$$

Finally, given $\sigma = \sigma_{T^{\text{bad}}}$, by Lemma 3.9 and Corollary 5.3, at least one feasible τ exists and $\|\text{Law}(\tau) - \mu_{\Psi[D_\sigma]}\|_{\text{TV}} \leq \varepsilon/8$. Put everything together, and we obtain that $\|\text{Law}(\sigma, \tau) - \mu_\Phi\|_{\text{TV}} \leq \varepsilon/8 + 3\varepsilon/16 = 5\varepsilon/16$, thus leading to the proposition. $\square$

Proof of Theorem 1.1. By Lemmas 3.1 to 3.3 and Proposition 7.1 with choice of $\zeta = n^{-2}$, the structural event and the polymer-root estimate hold with probability at least $1 - o_n(1)$. Since the input length I is upper bounded by $(n + nkm)^{C_1} \leq (nk(\alpha + 1))^{C_1}$ for some universal constant $C_1 > 0$, by Proposition 7.4, with probability at least $1 - o_n(1)$, **FastSampler** (Φ, ε) outputs a satisfying assignment of Φ within ε -total variation distance to μ_Φ and its expected running time is at most

$$\left(\frac{(nk(\alpha + 1))^{C_1}}{\varepsilon}\right)^{C_2} \leq \left(\frac{nk(\alpha + 1)}{\varepsilon}\right)^C$$

for a universal constant $C > 0$. Thus we conclude the main theorem. $\square$

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APPENDIX A. SUPPLEMENTARY PROOFS

A.1. Missing proofs in Section 6.

Proof of Lemma 6.5. For every $\sigma \in \Omega$, we have

$$e^{a_e Y_e(\sigma_{V_e})} = c_e (1 + w_e h_e(\sigma_{V_e})), \quad c_e = \begin{cases} 1, & a_e > 0, \\ e^{a_e}, & a_e < 0. \end{cases} \quad (128)$$

To verify (128), fix $e \in \mathcal{J}$ and $\sigma \in \Omega$. If $a_e > 0$, then $c_e = 1$, $w_e = e^{a_e} - 1$, and $h_e(\sigma_{V_e}) = Y_e(\sigma_{V_e})$, so

$$c_e (1 + w_e h_e(\sigma_{V_e})) = 1 + (e^{a_e} - 1) Y_e(\sigma_{V_e}) = \begin{cases} 1, & Y_e(\sigma_{V_e}) = 0, \\ e^{a_e}, & Y_e(\sigma_{V_e}) = 1. \end{cases}$$

If $a_e < 0$, then $c_e = e^{a_e}$, $w_e = e^{-a_e} - 1$, and $h_e(\sigma_{V_e}) = 1 - Y_e(\sigma_{V_e})$, so

$$\begin{aligned} c_e (1 + w_e h_e(\sigma_{V_e})) &= e^{a_e} (1 + (e^{-a_e} - 1) (1 - Y_e(\sigma_{V_e}))) \\ &= 1 + (e^{a_e} - 1) Y_e(\sigma_{V_e}) = \begin{cases} 1, & Y_e(\sigma_{V_e}) = 0, \\ e^{a_e}, & Y_e(\sigma_{V_e}) = 1. \end{cases} \end{aligned}$$

Since $Y_e(\sigma_{V_e}) \in \{0, 1\}$, the displayed values equal $e^{a_e Y_e(\sigma_{V_e})}$ in every case, proving (128).

By (128),

$$\begin{aligned} \forall \sigma \in \Omega, \quad \pi(\sigma) &\propto \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) \left(\prod_{e \in \mathcal{J}} c_e \right) \prod_{e \in \mathcal{J}} (1 + w_e h_e(\sigma_{V_e})) \\ &\propto \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) \prod_{e \in \mathcal{J}} (1 + w_e h_e(\sigma_{V_e})). \end{aligned} \quad (129)$$

The second proportionality uses that $\prod_{e \in \mathcal{J}} c_e$ is positive and independent of σ . The lemma is proved. $\square$

Proof of Lemma 6.6. Summing the full joint weight in (52) over η gives, for every $\sigma \in \Omega$,

$$\begin{aligned} \sum_{\eta \in \{0,1\}^{\mathcal{J}}} \hat{\pi}(\sigma, \eta) &\propto \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) \sum_{\eta \in \{0,1\}^{\mathcal{J}}} \prod_{e: \eta_e=1} w_e h_e(\sigma_{V_e}) \\ &= \left(\prod_{i=1}^s \ell_i(\sigma_i) \right) \prod_{e \in \mathcal{J}} (1 + w_e h_e(\sigma_{V_e})) \propto \pi(\sigma), \end{aligned} \quad (130)$$

where the last proportionality follows from Lemma 6.5. The proportionality constants are independent of σ ; normalizing over σ therefore shows that the σ -marginal of $\hat{\pi}$ is exactly π , namely,

$$\sum_{\eta \in \{0,1\}^{\mathcal{J}}} \hat{\pi}(\sigma, \eta) = \pi(\sigma), \quad \sigma \in \Omega. \quad (131)$$

Moreover, for fixed σ ,

$$\begin{aligned}\hat{\pi}(\eta \mid X = \sigma) &= \frac{\left(\prod_{i=1}^s \ell_i(\sigma_i)\right) \prod_{e:\eta_e=1} w_e h_e(\sigma_{V_e})}{\sum_{\eta' \in \{0,1\}^{\mathcal{J}}} \left(\prod_{i=1}^s \ell_i(\sigma_i)\right) \prod_{e:\eta'_e=1} w_e h_e(\sigma_{V_e})} = \frac{\prod_{e:\eta_e=1} w_e h_e(\sigma_{V_e})}{\sum_{\eta' \in \{0,1\}^{\mathcal{J}}} \prod_{e:\eta'_e=1} w_e h_e(\sigma_{V_e})} \\ &= \prod_{e:\eta_e=1} \frac{w_e h_e(\sigma_{V_e})}{1 + w_e h_e(\sigma_{V_e})} \prod_{e:\eta_e=0} \frac{1}{1 + w_e h_e(\sigma_{V_e})}.\end{aligned}\quad (132)$$

The lemma is proved. $\square$

Proof of Lemma 6.7. For any fixed set $D \subseteq [s]$, define the involution T_D on $(\Omega \times \{0,1\}^{\mathcal{J}})^2$ by

$$T_D((\sigma, \eta), (\omega, \xi)) = ((\tilde{\sigma}, \tilde{\eta}), (\tilde{\omega}, \tilde{\xi})), \quad (133)$$

where, for every $i \in [s]$ and $e \in \mathcal{J}$,

$$(\tilde{\sigma}_i, \tilde{\omega}_i) = \begin{cases} (\omega_i, \sigma_i), & i \in D, \\ (\sigma_i, \omega_i), & i \notin D, \end{cases} \quad (\tilde{\eta}_e, \tilde{\xi}_e) = \begin{cases} (\xi_e, \eta_e), & V_e \subseteq D, \\ (\eta_e, \xi_e), & V_e \not\subseteq D. \end{cases}$$

Whenever we apply T_D below to an input pair $((X, \eta), (X', \xi))$, we assume that D is a union of connected components of its associated hypergraph g , where e is present exactly when $\eta_e \vee \xi_e = 1$. Formally,

$$D = \bigcup_{C \in \mathcal{D}} C \subseteq [s] \quad \text{for some } \mathcal{D} \subseteq \text{Comp}(g). \quad (134)$$

The subcollection $\mathcal{D}$ may be empty or equal to $\text{Comp}(g)$.

Fix an input pair $((X, \eta), (X', \xi))$, and let g be its associated hypergraph and D be a union of components of g . Fix $e \in \mathcal{J}$ with $\eta_e \vee \xi_e = 1$. The hyperedge labelled e belongs to g , so any two vertices of V_e are connected by this hyperedge. Hence $V_e \subseteq C_e$ for a unique $C_e \in \text{Comp}(g)$. If $C_e \in \mathcal{D}$, then (134) gives $V_e \subseteq C_e \subseteq D$. If $C_e \notin \mathcal{D}$, then $C_e \cap C = \emptyset$ for every $C \in \mathcal{D}$, because distinct connected components have disjoint vertex sets. Thus (134) gives

$$C_e \cap D = C_e \cap \bigcup_{C \in \mathcal{D}} C = \bigcup_{C \in \mathcal{D}} (C_e \cap C) = \emptyset,$$

and therefore $V_e \cap D = \emptyset$. We have proved

$$\forall e \in \mathcal{J}, \quad \eta_e \vee \xi_e = 1 \implies V_e \subseteq D \text{ or } V_e \cap D = \emptyset. \quad (135)$$

Write

$$\left((\tilde{X}, \tilde{\eta}), (\tilde{X}', \tilde{\xi})\right) = T_D((X, \eta), (X', \xi)).$$

For each $e \in \mathcal{J}$, define

$$F_e(\sigma, \eta, \omega, \xi) = (w_e h_e(\sigma_{V_e}))^{\eta_e} (w_e h_e(\omega_{V_e}))^{\xi_e}, \quad \sigma, \omega \in \Omega, \quad \eta, \xi \in \{0,1\}^{\mathcal{J}},$$

where a factor with exponent zero equals one, including $0^0 = 1$. We verify the invariance of these factors in three cases.

(1) If $V_e \subseteq D$, the definition in (133) gives

$$(\tilde{X}_{V_e}, \tilde{X}'_{V_e}, \tilde{\eta}_e, \tilde{\xi}_e) = (X'_{V_e}, X_{V_e}, \xi_e, \eta_e).$$

Consequently,

$$\begin{aligned}
F_e \left(\tilde{X}, \tilde{\eta}, \tilde{X}', \tilde{\xi} \right) &= \left(w_e h_e \left(\tilde{X}_{V_e} \right) \right)^{\tilde{\eta}_e} \left(w_e h_e \left(\tilde{X}'_{V_e} \right) \right)^{\tilde{\xi}_e} \\
&= \left(w_e h_e \left(X'_{V_e} \right) \right)^{\xi_e} \left(w_e h_e \left(X_{V_e} \right) \right)^{\eta_e} \\
&= \left(w_e h_e \left(X_{V_e} \right) \right)^{\eta_e} \left(w_e h_e \left(X'_{V_e} \right) \right)^{\xi_e} = F_e \left(X, \eta, X', \xi \right).
\end{aligned}$$

(2) If $V_e \cap D = \emptyset$, then $V_e \not\subseteq D$ because $V_e \neq \emptyset$. Hence (133) gives

$$\left(\tilde{X}_{V_e}, \tilde{X}'_{V_e}, \tilde{\eta}_e, \tilde{\xi}_e \right) = \left(X_{V_e}, X'_{V_e}, \eta_e, \xi_e \right).$$

Then $F_e \left(\tilde{X}, \tilde{\eta}, \tilde{X}', \tilde{\xi} \right) = F_e \left(X, \eta, X', \xi \right)$.

(3) If $V_e \cap D \neq \emptyset$ and $V_e \not\subseteq D$, then (135) implies $\eta_e \vee \xi_e = 0$. Since $\eta_e, \xi_e \in \{0, 1\}$, we have $\eta_e = \xi_e = 0$. Then (133) yields

$$\tilde{\eta}_e = \eta_e = 0, \quad \tilde{\xi}_e = \xi_e = 0.$$

Therefore the zero-exponent convention gives

$$F_e \left(\tilde{X}, \tilde{\eta}, \tilde{X}', \tilde{\xi} \right) = 1 = F_e \left(X, \eta, X', \xi \right).$$

These cases exhaust all possibilities, so

$$\forall e \in \mathcal{J}, \quad F_e \left(\tilde{X}, \tilde{\eta}, \tilde{X}', \tilde{\xi} \right) = F_e \left(X, \eta, X', \xi \right). \quad (136)$$

For every $i \in [s]$, (133) also gives

$$\begin{aligned}
\ell_i \left(\tilde{X}_i \right) \ell_i \left(\tilde{X}'_i \right) &= \begin{cases} \ell_i \left(X'_i \right) \ell_i \left(X_i \right), & i \in D, \\ \ell_i \left(X_i \right) \ell_i \left(X'_i \right), & i \notin D, \end{cases} \\
&= \ell_i \left(X_i \right) \ell_i \left(X'_i \right),
\end{aligned}$$

Combined with (136) and (52), we obtain

$$\hat{\pi} \left(\tilde{X}, \tilde{\eta} \right) \hat{\pi} \left(\tilde{X}', \tilde{\xi} \right) = \hat{\pi} \left(X, \eta \right) \hat{\pi} \left(X', \xi \right). \quad (137)$$

Moreover, for every $e \in \mathcal{J}$,

$$\begin{aligned}
\tilde{\eta}_e \vee \tilde{\xi}_e &= \begin{cases} \xi_e \vee \eta_e, & V_e \subseteq D, \\ \eta_e \vee \xi_e, & V_e \not\subseteq D, \end{cases} \\
&= \eta_e \vee \xi_e.
\end{aligned}$$

Thus,

$$\{e \in \mathcal{J} : \tilde{\eta}_e \vee \tilde{\xi}_e = 1\} = \{e \in \mathcal{J} : \eta_e \vee \xi_e = 1\}.$$

Finally, for every $\sigma, \omega \in \Omega$ and $\eta, \xi \in \{0, 1\}^{\mathcal{J}}$, applying (133) twice gives

$$T_D \left(T_D \left((\sigma, \eta), (\omega, \xi) \right) \right) = ((\sigma, \eta), (\omega, \xi)).$$

Indeed, each selected coordinate and auxiliary bit is swapped twice, returning to its original value, while all others remain unchanged. Moreover, the edge indicators $\eta_e \vee \xi_e$ are unchanged by T_D for every input pair and every fixed D . Thus, for each fixed g and each D that is a union of components of g , the restriction of T_D to the fiber of pairs whose hypergraph is g is a bijection from that fiber to itself. By (137), this restricted bijection preserves the pair probabilities.

For each $C \in \text{Comp}(g)$, independently draw a fresh bit $\varepsilon_C \in \{0, 1\}$ with

$$\mathbb{P}_\varepsilon (\varepsilon_C = 0) = \mathbb{P}_\varepsilon (\varepsilon_C = 1) = \frac{1}{2}.$$

Collect these bits into the vector $\varepsilon = (\varepsilon_C)_{C \in \text{Comp}(g)} \in \{0, 1\}^{\text{Comp}(g)}$ and define its complementary bit vector by

$$\bar{\varepsilon} = (1 - \varepsilon_C)_{C \in \text{Comp}(g)} \in \{0, 1\}^{\text{Comp}(g)}.$$

Define

$$D(\varepsilon) = \bigcup_{\substack{C \in \text{Comp}(g) \\ \varepsilon_C = 1}} C$$

and apply $T_{D(\varepsilon)}$ to this sample pair. By definition, $D(\varepsilon)$ is a union of connected components of the associated hypergraph g , so it satisfies the condition in (134). Thus a component C is swapped in its entirety if $\varepsilon_C = 1$ and is left unchanged if $\varepsilon_C = 0$. Let $X^\varepsilon, (X')^\varepsilon$ be the two resulting configurations. Since every component is swapped in exactly one of ε and $\bar{\varepsilon}$, we have $X^{\bar{\varepsilon}} = (X')^\varepsilon$. The notation $\mathbb{E}_\varepsilon$ and Var_ε denotes expectation and variance over ε alone, with the sample pair $((X, \eta), (X', \xi))$ held fixed. The map $\varepsilon \mapsto \bar{\varepsilon}$ is a bijection of $\{0, 1\}^{\text{Comp}(g)}$, on which ε is uniform. Thus ε and $\bar{\varepsilon}$ have the same distribution, as do $f(X^\varepsilon)$ and $f(X^{\bar{\varepsilon}})$; in particular, they have the same mean and variance. Since $\mathbb{E}_\varepsilon[f(X^\varepsilon)] = \mathbb{E}_\varepsilon[f(X^{\bar{\varepsilon}})]$, we have

$$f(X^\varepsilon) - f(X^{\bar{\varepsilon}}) = (f(X^\varepsilon) - \mathbb{E}_\varepsilon[f(X^\varepsilon)]) - (f(X^{\bar{\varepsilon}}) - \mathbb{E}_\varepsilon[f(X^{\bar{\varepsilon}})]).$$

Applying $(u - v)^2 \leq 2u^2 + 2v^2$ to these two mean-subtracted terms and taking expectation over ε gives

$$\begin{aligned} \mathbb{E}_\varepsilon (f(X^\varepsilon) - f(X^{\bar{\varepsilon}}))^2 &\leq 2\mathbb{E}_\varepsilon \left[(f(X^\varepsilon) - \mathbb{E}_\varepsilon[f(X^\varepsilon)])^2 \right] + 2\mathbb{E}_\varepsilon \left[(f(X^{\bar{\varepsilon}}) - \mathbb{E}_\varepsilon[f(X^{\bar{\varepsilon}})])^2 \right] \\ &= 2\text{Var}_\varepsilon(f(X^\varepsilon)) + 2\text{Var}_\varepsilon(f(X^{\bar{\varepsilon}})) = 4\text{Var}_\varepsilon(f(X^\varepsilon)). \end{aligned} \quad (138)$$

For each $C \in \text{Comp}(g)$, define the bit vector $\varepsilon^C \in \{0, 1\}^{\text{Comp}(g)}$ by

$$\forall C' \in \text{Comp}(g), \quad (\varepsilon^C)_{C'} = \begin{cases} 1 - \varepsilon_C, & C' = C, \\ \varepsilon_{C'}, & C' \neq C. \end{cases}$$

Thus ε^C differs from ε only at the bit indexed by C . This changes whether the entire component C is swapped, while leaving the choices for all other components unchanged. Write $\varepsilon_{-C} = (\varepsilon_{C'})_{C' \in \text{Comp}(g) \setminus \{C\}}$ for all the other bits. Since the bits are independent, applying Lemma 2.3 in Section 2.5 to the function $\varepsilon \mapsto f(X^\varepsilon)$ gives

$$\text{Var}_\varepsilon(f(X^\varepsilon)) \leq \sum_{C \in \text{Comp}(g)} \mathbb{E}_\varepsilon[\text{Var}_\varepsilon(f(X^\varepsilon) \mid \varepsilon_{-C})]. \quad (139)$$

For each fixed value of ε_{-C} , let a and b denote the values of $f(X^\varepsilon)$ when $\varepsilon_C = 0$ and $\varepsilon_C = 1$, respectively. Since each choice of ε_C has probability $1/2$, the conditional mean of $f(X^\varepsilon)$ is $(a + b)/2$, and its conditional variance is

$$\text{Var}_\varepsilon(f(X^\varepsilon) \mid \varepsilon_{-C}) = \frac{1}{2} \left(a - \frac{a+b}{2} \right)^2 + \frac{1}{2} \left(b - \frac{a+b}{2} \right)^2 = \frac{(a-b)^2}{4}.$$

Flipping ε_C changes the function value from a to b if $\varepsilon_C = 0$, and from b to a if $\varepsilon_C = 1$. Thus $(f(X^\varepsilon) - f(X^{\varepsilon^C}))^2 = (a - b)^2$ for either choice of ε_C . It follows that

$$\text{Var}_\varepsilon(f(X^\varepsilon) \mid \varepsilon_{-C}) = \frac{1}{4} \mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f(X^{\varepsilon^C}) \right)^2 \mid \varepsilon_{-C} \right].$$

Taking expectations gives

$$\begin{aligned}
& \mathbb{E}_\varepsilon[\text{Var}_\varepsilon(f(X^\varepsilon) \mid \varepsilon_{-C})] \\
&= \mathbb{E}_{\varepsilon_{-C}}[\text{Var}_\varepsilon(f(X^\varepsilon) \mid \varepsilon_{-C})] \\
&= \frac{1}{4} \mathbb{E}_{\varepsilon_{-C}} \left[\mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f(X^{\varepsilon^C}) \right)^2 \mid \varepsilon_{-C} \right] \right] \\
&= \frac{1}{4} \mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f(X^{\varepsilon^C}) \right)^2 \right].
\end{aligned} \tag{140}$$

By (140) and (139),

$$\begin{aligned}
4 \text{Var}_\varepsilon(f(X^\varepsilon)) &\leq 4 \sum_{C \in \text{Comp}(g)} \mathbb{E}_\varepsilon[\text{Var}_\varepsilon(f(X^\varepsilon) \mid \varepsilon_{-C})] \\
&= \sum_{C \in \text{Comp}(g)} \mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f(X^{\varepsilon^C}) \right)^2 \right].
\end{aligned} \tag{141}$$

Let $Y = ((X, \eta), (X', \xi))$, and let G be its associated labelled hypergraph. For every g with $\mathbb{P}(G = g) > 0$ and every fixed $\varepsilon \in \{0, 1\}^{\text{Comp}(g)}$, the set $D(\varepsilon)$ is a union of components of g . Thus $T_{D(\varepsilon)}$ is a bijection on the fiber of sample pairs whose hypergraph is g , and (137) shows that this map leaves the product probabilities $\hat{\pi}(X, \eta) \hat{\pi}(X', \xi)$ unchanged. Since conditioning on $G = g$ amounts to normalizing these probabilities on that fiber, for every sample-pair value y in the fiber we have

$$\mathbb{P}(T_{D(\varepsilon)}(Y) = y \mid G = g) = \mathbb{P}(Y = y \mid G = g).$$

Conditional on $G = g$, the bit vector ε is sampled uniformly from $\{0, 1\}^{\text{Comp}(g)}$, independently of $((X, \eta), (X', \xi))$. Consequently, for every $\sigma, \omega \in \Omega$,

$$\mathbb{P}(X^\varepsilon = \sigma, (X')^\varepsilon = \omega \mid G = g, \varepsilon) = \mathbb{P}(X = \sigma, X' = \omega \mid G = g). \tag{142}$$

By the law of total probability and the independence of X and X' , for every $\sigma, \omega \in \Omega$,

$$\mathbb{P}(X^\varepsilon = \sigma, (X')^\varepsilon = \omega) = \mathbb{P}(X = \sigma, X' = \omega) = \pi(\sigma)\pi(\omega). \tag{143}$$

Using $X^\varepsilon = (X')^\varepsilon$, (143), and the independence of X and X' with common law π , we obtain

$$\begin{aligned}
& \mathbb{E} \left[\mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f(X^{\varepsilon^C}) \right)^2 \right] \right] = \mathbb{E} \left[\mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f((X')^\varepsilon) \right)^2 \right] \right] \\
&= \mathbb{E} \left[\left(f(X) - f(X') \right)^2 \right] = 2 \text{Var}_\pi(f).
\end{aligned} \tag{144}$$

We now average over the independent lifted samples $(X, \eta), (X', \xi) \sim \hat{\pi}$. In the following inequality, the outer $\mathbb{E}$ averages over X, η, X', ξ , whereas the inner $\mathbb{E}_\varepsilon$ averages only over the swap bits, holding the entire sample pair fixed. Combining (138) and (141), taking these expectations, and using (144) gives

$$\text{Var}_\pi(f) \leq \frac{1}{2} \mathbb{E} \left[\sum_{C \in \text{Comp}(g)} \mathbb{E}_\varepsilon \left[\left(f(X^\varepsilon) - f(X^{\varepsilon^C}) \right)^2 \right] \right].$$

□

Proof of Lemma 6.11. Fix $\Gamma \in \Lambda$ satisfying $J_\Gamma \cap A \neq \emptyset$, and choose $i \in J_\Gamma \cap A$. By (98), $B_i \cap \text{vbl}(U_\Gamma) \neq \emptyset$. Since $U_\Gamma = \bigcup_{r=1}^\ell \gamma_r$ by (38), there exists $a \in U_\Gamma$ such that $\text{vbl}(a) \cap B_i \neq \emptyset$. Each

polymer γ_r is a subset of $\mathcal{C}^{\text{good}}$, so $U_\Gamma \subseteq \mathcal{C}^{\text{good}}$ and hence $a \in \mathcal{C}^{\text{good}}$. Since $i \in A$ and $B_i \subseteq W_A$, this clause satisfies $\text{vbl}(a) \cap W_A \neq \emptyset$. Hence,

$$R(A) \leq \sum_{\substack{a \in \mathcal{C}^{\text{good}} \\ \text{vbl}(a) \cap W_A \neq \emptyset}} \sum_{\substack{\Gamma \in \Lambda \\ a \in U_\Gamma}} |a_\Gamma| e^{\tau K_\Gamma}.$$

The right-hand side may count the same Γ once for each clause in U_Γ incident to W_A , which only increases the sum. For each $a \in \mathcal{C}^{\text{good}}$, (39) gives

$$\sum_{\substack{\Gamma \in \Lambda \\ a \in U_\Gamma}} |a_\Gamma| e^{\tau K_\Gamma} \leq \sum_{\Gamma: a \in U_\Gamma} |a_\Gamma| e^{\tau K_\Gamma} \leq \rho.$$

Therefore,

$$R(A) \leq \rho \left| \left\{ a \in \mathcal{C}^{\text{good}} : \text{vbl}(a) \cap W_A \neq \emptyset \right\} \right|.$$

Since $\mathcal{C}^{\text{good}} \subseteq \mathcal{C}$, the definition (2) gives

$$\left| \left\{ a \in \mathcal{C}^{\text{good}} : \text{vbl}(a) \cap W_A \neq \emptyset \right\} \right| \leq I(W_A).$$

Hence, $R(A) \leq \rho I(W_A)$. The lemma is proved. $\square$

Proof of Lemma 6.9. Recall that ζ is a hypergraph on vertex set $[s]$, and that the hyperedge indexed by a retained label Γ has vertex set J_Γ . Suppose that $\Gamma \in \mathcal{D}_t$ and $\zeta_\Gamma = 1$. By the choice of a_t , there is some $i \in \mathcal{R}_t$ such that $B_i \cap \text{vbl}(a_t) \neq \emptyset$. Since $a_t \in U_\Gamma$, the definition of J_Γ in (98) gives $i \in J_\Gamma$. Thus the retained hyperedge J_Γ meets $\mathcal{R}_t$, and adding its vertices preserves connection to j . It follows inductively that every vertex in $\mathcal{R}_Q$ is connected to j in the hypergraph ζ , and hence $\mathcal{R}_Q \subseteq \mathcal{A}_j$.

Conversely, consider any hyperedge J_Γ of ζ meeting $\mathcal{R}_Q$. By the definition of J_Γ , some clause $a \in U_\Gamma$ contains a variable from a block B_i with $i \in \mathcal{R}_Q$. Before the procedure stops, every such good clause is used. When a is used, the label Γ is checked, unless it was checked earlier; since $\zeta_\Gamma = 1$, all vertices of J_Γ are therefore added to $\mathcal{R}_Q$. Thus every retained hyperedge that meets $\mathcal{R}_Q$ is entirely contained in $\mathcal{R}_Q$. Consequently, no path in ζ can connect a vertex of $\mathcal{R}_Q$ to a vertex outside $\mathcal{R}_Q$. Since $j \in \mathcal{R}_Q$, this proves $\mathcal{A}_j \subseteq \mathcal{R}_Q$, and therefore $\mathcal{R}_Q = \mathcal{A}_j$. $\square$

Proof of Lemma 6.10. For $t < Q$, let $\mathcal{F}_t$ denote the information revealed before step t . Then $\mathcal{D}_t$ is $\mathcal{F}_t$ -measurable and

$$\delta T_t \triangleq T_{t+1} - T_t = \sum_{\Gamma \in \mathcal{D}_t} K_\Gamma \zeta_\Gamma. \quad (145)$$

Because the indicators in (49) are independent and $\mathcal{D}_t \subseteq \Lambda \setminus \mathcal{E}_t$, conditionally on $\mathcal{F}_t$ the variables $(\zeta_\Gamma)_{\Gamma \in \mathcal{D}_t}$ are independent and have the same Bernoulli laws as in (49). Define $p_\Gamma \triangleq 1 - e^{-2|a_\Gamma|}$. By (49) and the conditional independence just noted,

$$\mathbb{E} \left[e^{\delta T_t/2} \mid \mathcal{F}_t \right] = \prod_{\Gamma \in \mathcal{D}_t} \mathbb{E} \left[e^{K_\Gamma \zeta_\Gamma/2} \mid \mathcal{F}_t \right] = \prod_{\Gamma \in \mathcal{D}_t} \left((1 - p_\Gamma) + p_\Gamma e^{K_\Gamma/2} \right) = \prod_{\Gamma \in \mathcal{D}_t} \left(1 + p_\Gamma (e^{K_\Gamma/2} - 1) \right).$$

Applying $1 + x \leq e^x$ and $e^{K_\Gamma/2} - 1 \leq e^{K_\Gamma/2}$, we obtain

$$\mathbb{E} \left[e^{\delta T_t/2} \mid \mathcal{F}_t \right] \leq \exp \left(\sum_{\Gamma \in \mathcal{D}_t} p_\Gamma (e^{K_\Gamma/2} - 1) \right) \leq \exp \left(\sum_{\Gamma \in \mathcal{D}_t} p_\Gamma e^{K_\Gamma/2} \right).$$

Moreover, $p_\Gamma = 1 - e^{-2|a_\Gamma|} \leq 2|a_\Gamma|$. Therefore,

$$\mathbb{E}\left[e^{\delta T_t/2} \mid \mathcal{F}_t\right] \leq \exp\left(2 \sum_{\Gamma \in \mathcal{D}_t} |a_\Gamma| e^{K_\Gamma/2}\right).$$

Finally, $\tau \geq \ln 2 > 1/2$ by (31), so $e^{K_\Gamma/2} \leq e^{\tau K_\Gamma}$. Moreover, $\mathcal{D}_t \subseteq \{\Gamma : a_t \in U_\Gamma\}$. Combining these two observations with (39), we obtain

$$\sum_{\Gamma \in \mathcal{D}_t} |a_\Gamma| e^{K_\Gamma/2} \leq \sum_{\Gamma : a_t \in U_\Gamma} |a_\Gamma| e^{\tau K_\Gamma} \leq \rho.$$

Therefore,

$$\mathbb{E}\left[e^{\delta T_t/2} \mid \mathcal{F}_t\right] \leq e^{2\rho}.$$

Equivalently,

$$\mathbb{E}\left[e^{\delta T_t/2 - 2\rho} \mid \mathcal{F}_t\right] \leq 1, \quad t < Q. \quad (146)$$

Here $t \wedge Q = \min\{t, Q\}$. Define the stopped process

$$M_t = \exp(T_{t \wedge Q}/2 - 2\rho(t \wedge Q)), \quad t \geq 0, \quad (147)$$

with respect to the exploration filtration. On the event $t < Q$, we have

$$M_{t+1} = M_t e^{\delta T_t/2 - 2\rho}.$$

Since M_t is determined by the information revealed before step t , it is $\mathcal{F}_t$ -measurable. We may therefore take it outside the conditional expectation. Using (146), on the event $t < Q$ we obtain

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t \mathbb{E}\left[e^{\delta T_t/2 - 2\rho} \mid \mathcal{F}_t\right] \leq M_t.$$

On the event $t \geq Q$, the process has already stopped, so $M_{t+1} = M_t$. Thus $(M_t)_{t \geq 0}$ is a nonnegative supermartingale.

Finally, each of the Q exploration steps uses a distinct clause $a_t \in \mathcal{C}^{\text{good}}$. Hence $Q \leq |\mathcal{C}^{\text{good}}| \leq |\mathcal{C}| = m$. It follows that $m \wedge Q = Q$, while $T_Q = T$. Thus (147) gives $M_m = e^{T/2 - 2\rho Q}$. Moreover, the supermartingale property gives $\mathbb{E}M_m \leq \mathbb{E}M_0$. Therefore

$$\mathbb{E}e^{T/2 - 2\rho Q} = \mathbb{E}M_m \leq \mathbb{E}M_0 = M_0 = 1.$$

The lemma is proved. $\square$

Proof of Lemma 6.12. By (23), every $v \in W \cap \mathcal{V}^{\text{high}}$ satisfies $\deg(v) \geq \Delta$. Hence,

$$\Delta|W \cap \mathcal{V}^{\text{high}}| \leq \sum_{v \in W \cap \mathcal{V}^{\text{high}}} \deg(v).$$

By the definition of $I(W)$ in (2), exactly $I(W)$ clauses contain at least one variable of W . Each such clause contains at most k variables and therefore contributes at most k to the degree sum. Hence

$$\sum_{v \in W \cap \mathcal{V}^{\text{high}}} \deg(v) \leq kI(W).$$

Therefore,

$$|W \cap \mathcal{V}^{\text{high}}| \leq \frac{k}{\Delta} I(W).$$

Combining this with (23) gives

$$|W| \leq \frac{k}{2} |W \cap \mathcal{V}^{\text{high}}| \leq \frac{k^2}{2\Delta} I(W).$$

The lemma is proved. $\square$

Proof of Lemma 6.13. Recall that $\mathcal{H}_\Psi$ is the dependency graph of the residual formula Ψ on the clause labels $\mathcal{C}^{\text{good}}$, where two clause labels are adjacent exactly when their residual clauses share a good variable. Recall also that $\mathcal{G}_\Phi$ is the primal graph of the original formula Φ : its vertices are the variables of Φ , and two variables are adjacent when they occur together in an original clause.

Fix $\Gamma \in \mathcal{E}_Q$ with $\zeta_\Gamma = 1$. Recall that $\Gamma = (\gamma_1, \dots, \gamma_\ell)$ is a tuple of polymers, each γ_i is a connected set of clause labels in $\mathcal{H}_\Psi$, and $U_\Gamma = \bigcup_{i=1}^\ell \gamma_i$ is itself a set of clause labels. Since $\zeta_\Gamma = 1$, (49) forces $a_\Gamma \neq 0$: indeed, if $a_\Gamma = 0$, then $\mathbb{P}_\zeta(\zeta_\Gamma = 1) = 1 - e^{-2|a_\Gamma|} = 0$. By Lemma 6.2, the induced clause graph $\mathcal{H}_\Psi[U_\Gamma]$ is connected.

Choose a spanning tree T_Γ of $\mathcal{H}_\Psi[U_\Gamma]$. For each edge $e = \{c, d\}$ of T_Γ , choose a good variable x_e appearing in both original clauses C_c and C_d ; such a variable exists by the definition of $\mathcal{H}_\Psi$. Define

$$L_\Gamma = \left(\bigcup_{c \in U_\Gamma} (\text{vbl}(c) \cap \mathcal{V}^{\text{bad}}) \right) \cup \{x_e : e \in E(T_\Gamma)\} \cup W_{J_\Gamma}. \quad (148)$$

For each $c \in U_\Gamma$, the connector variables x_e on tree edges incident to c , together with the bad variables of C_c , lie in the same clause C_c and hence are mutually adjacent in $\mathcal{G}_\Phi$. Since T_Γ is connected, the union of the first two sets in (148) is connected.

For every $i \in J_\Gamma$, (98) gives $B_i \cap \text{vbl}(U_\Gamma) \neq \emptyset$. Thus B_i intersects the first set in (148). Since B_i is itself connected in $\mathcal{G}_\Phi$, adjoining the blocks $\{B_i : i \in J_\Gamma\}$ to the union of the first two sets in (148) preserves connectivity. By (101), the resulted set is exactly L_Γ . Hence, L_Γ is connected. Moreover, the label Γ belongs to a unique $\mathcal{D}_t$. By (99) and the choice of a_t , we have $a_t \in U_\Gamma$. Furthermore, by the exploration rule, a_t intersects some currently discovered block B_i with $i \in \mathcal{R}_t$, and hence $\text{vbl}(a_t) \cap W_{\mathcal{R}_t} \neq \emptyset$ by (101). Since $a_t \in U_\Gamma$ and $W_{\mathcal{R}_t} \subseteq \mathcal{V}^{\text{bad}}$, we have

$$\emptyset \neq \text{vbl}(a_t) \cap W_{\mathcal{R}_t} \subseteq W_{\mathcal{R}_t} \cap \bigcup_{c \in U_\Gamma} (\text{vbl}(c) \cap \mathcal{V}^{\text{bad}}).$$

Combined with (148), we have

$$L_\Gamma \cap W_{\mathcal{R}_t} \neq \emptyset. \quad (149)$$

Set $L_0 = B_j$ and, for $0 \leq t < Q$, define

$$L_{t+1} = L_t \cup \bigcup_{\substack{\Gamma \in \mathcal{D}_t \\ \zeta_\Gamma = 1}} L_\Gamma. \quad (150)$$

We prove by induction that L_t is connected and contains $W_{\mathcal{R}_t}$. At $t = 0$, we have $L_0 = B_j = W_{\mathcal{R}_0}$ because $\mathcal{R}_0 = \{j\}$. Now suppose that L_t is connected and $W_{\mathcal{R}_t} \subseteq L_t$. Then every set adjoined in the definition of L_{t+1} intersects $W_{\mathcal{R}_t}$ by (149), so L_{t+1} is connected because $W_{\mathcal{R}_t} \subseteq L_t$. By (148), we have

$$\forall \Gamma, \quad W_{J_\Gamma} \subseteq L_\Gamma. \quad (151)$$

Moreover, by the update rule for $\mathcal{R}_{t+1}$,

$$W_{\mathcal{R}_{t+1}} = W_{\mathcal{R}_t} \cup \bigcup_{\substack{\Gamma \in \mathcal{D}_t \\ \zeta_\Gamma = 1}} W_{J_\Gamma}.$$

Combined with (150), (151), and $W_{\mathcal{R}_t} \subseteq L_t$, we obtain

$$W_{\mathcal{R}_{t+1}} = W_{\mathcal{R}_t} \cup \bigcup_{\substack{\Gamma \in \mathcal{D}_t \\ \zeta_\Gamma = 1}} W_{J_\Gamma} \subseteq L_t \cup \bigcup_{\substack{\Gamma \in \mathcal{D}_t \\ \zeta_\Gamma = 1}} L_\Gamma = L_{t+1}.$$

Thus L_Q is connected and contains $W_{\mathcal{R}_Q} = W_{\mathcal{A}_j} = W$.

Since T_Γ has $|U_\Gamma| - 1$ edges, it requires at most $|U_\Gamma| - 1$ connector variables. By (38), $U_\Gamma = \bigcup_{i=1}^\ell \gamma_i$ and $K_\Gamma = \sum_{i=1}^\ell |\gamma_i|$, and therefore $|U_\Gamma| \leq K_\Gamma$. Hence the total number of variables added to W is at most

$$\sum_{\Gamma \in \mathcal{E}_Q} K_\Gamma \zeta_\Gamma = T,$$

which proves the claimed bound $|W| + T$. If $|U_\Gamma| = 1$, no connector variable is needed. $\square$